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BACHELOR'S THESIS

**CAUSAL MACHINE LEARNING AND ITS
APPLICATION IN CREDIT RISK PREDICTION**

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Abstract

Amid the digital transformation of the finance and banking sector and the growth of online lending, building credit risk prediction systems that offer high accuracy, transparent interpretability, and practical decision-support value has become an urgent requirement. However, while most modern machine learning models achieve strong predictive performance, they remain limited in explaining the causes leading to default risk, which reduces their trustworthiness and applicability in financial risk management settings.

This study proposes a credit risk prediction framework based on causal machine learning combined with Bayesian networks, implemented in a logical and rigorous manner through five advanced integrated stages: balancing the data with the SMOTE algorithm to optimise sample quality (S_Pro), screening and selecting strongly discriminative features via Lasso regression (L_Pro), discretising continuous variables with the K-means algorithm (K_Pro), learning the network structure and parameters through the Tabu Search algorithm combined with Maximum Likelihood Estimation (MLE) (BN_Pro), and finally performing forward/backward inference together with sensitivity analysis to fully exploit the causal structure (In_Pro). In addition, the study integrates model explanation techniques such as SHAP, LIME, sensitivity analysis, forward–backward reasoning, classification threshold optimisation, and Decision Curve Analysis in order to comprehensively evaluate the model’s applied effectiveness.

The experimental results show that the model achieves superior discrimination of high-risk customers compared with many traditional methods, while also providing clear causal relationships between financial factors and the likelihood of default. Optimising the decision threshold helps to effectively balance profit against the cost of risk, whereas Decision Curve Analysis demonstrates the model’s practical economic value across a range of credit-granting scenarios. The study contributes a transparent, interpretable and decision-supporting credit risk assessment framework that helps financial institutions improve appraisal quality, reduce credit losses, and optimise risk management effectiveness in the modern business environment.

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Lists of symbols

AI Artificial Intelligence

BI Business Intelligence

BIC Bayesian Information Criterion

BN Bayesian Network

CML Causal Machine Learning

CNN Convolutional Neural Network

CPD Conditional Probability Distribution

CPT Conditional Probability Table

DBM Deep Boltzmann Machine

DBN Deep Belief Network

DAG Directed Acyclic Graph

DCA Decision Curve Analysis

DT Decision Tree

FN False Negative – the number of customers who actually default but are incorrectly predicted by the model as non-defaulting.

FP False Positive – the number of customers who do not actually default but are incorrectly predicted by the model as defaulting.

GDPR General Data Protection Regulation

LIME Local Interpretable Model-agnostic Explanations – a local, model-agnostic method for explaining model predictions.

MDA Multiple Discriminant Analysis

ML Machine Learning

MLE Maximum Likelihood Estimation

NN Neural Network

NP Non-deterministic Polynomial

RF Random Forest

SHAP Shapley Additive Explanations – a model explanation method based on Shapley values.

SMOTE Synthetic Minority Over-sampling Technique

SVM Support Vector Machine

- TN** True Negative – the number of customers who do not actually default and are correctly predicted by the model as non-defaulting.
- TP** True Positive – the number of customers who actually default and are correctly predicted by the model as defaulting.
- VE** Variable Elimination – an inference algorithm that eliminates variables.
- XAI** Explainable Artificial Intelligence

Chapter 1

Thesis Overview

This chapter provides an overview of the background and motivation of the thesis, including the rationale for choosing the topic, the problems that remain in applying machine learning models to credit risk management today, as well as the objectives, scope and approach. In addition, we introduce the overall structure and the specific research scenarios in order to orient the reader for the chapters that follow.

1.1 Problem Statement

The rapid development of Artificial Intelligence (AI) and Machine Learning has created a major turning point in the financial sector. The shift from traditional linear statistical models (such as Logistic Regression) to complex machine learning algorithms (such as Random Forest, XGBoost or Neural Networks) has substantially improved the accuracy of default prediction. However, the sharp trade-off between predictive performance (accuracy) and the interpretability of black-box models is becoming a serious obstacle.

Under the pressure of strict legal regulations, most notably Article 22 of the General Data Protection Regulation in Europe [20], financial institutions are required to guarantee the right to explanation for customers with respect to any automated decision that directly affects their financial interests. Merely returning a "Loan rejected" outcome without being able to trace back the underlying cause is no longer acceptable, either legally or operationally.

In addition, traditional machine learning models are essentially correlational models. They learn to recognise patterns from historical data but entirely lack an understanding of the underlying causal structure of that data. When the macroeconomic context changes, these surface-level correlational patterns are easily broken. To address this, many current systems employ post-hoc explainable AI (post-hoc XAI) methods such as LIME [54] or SHAP [43]. Although they can explain the contribution of individual features, these post-hoc tools are in fact merely an outer wrapper: they inadvertently overlook the network of mutual interactions within the original data and readily lead to a breakdown of statistical information.

Against this backdrop, the Causal Machine Learning approach, and in particular the Bayesian Network, emerges as an optimal solution. A Bayesian Network represents knowledge in the form of a directed graph in which the nodes and edges express causal relationships and probabilistic dependencies. This approach offers intrinsic

interpretability, directly resolving the trade-off between accuracy and transparency, and creating a solid, safe basis for credit risk appraisal decisions [42].

1.2 Research Objectives

The general objective of this thesis is to thoroughly investigate and apply the Causal Machine Learning framework in order to fundamentally address the lack of transparency in current closed credit risk assessment systems, while maintaining strong classification capability on complex financial data. From this general objective, the thesis aims at the following specific objectives:

1. *Refining a domain-specific data preprocessing procedure*: Building an automated data pipeline to overcome the class imbalance commonly found in credit data, while also denoising the feature space so as to be compatible with the probabilistic nature of Bayesian networks.
2. *Resolving the bottleneck in network structure learning*: Optimising the process of learning a Directed Acyclic Graph (DAG) by combining a local search algorithm with the Bayesian Information Criterion (BIC).
3. *Performing causal inference and model interpretation*: Fully exploiting the architecture of the Bayesian network to carry out forward inference, backward inference, and what-if scenario analysis in order to make the entire financial decision flow transparent.

1.3 Research Subject and Scope

The subject of this thesis is Causal Machine Learning models, specifically the graph structure of Bayesian networks, together with credit profile datasets that are characteristic of the financial sector.

The scope of the research focuses on modelling the structure of the interaction network in order to interpret credit risk. Rather than aiming merely to produce reports of predicted outcomes, the research places its emphasis on building a robust, automated and stable data pipeline for preprocessing information. This pipeline comprises techniques for addressing data imbalance, denoising the feature space, and discretising continuous variables so as to be fully compatible with the probabilistic statistical nature of Bayesian networks. The research is limited to structured, tabular credit data and does not extend to unstructured data types such as contract documents or real-time series data.

1.4 Main Contributions of the Thesis

The core contribution of this thesis does not stop at the successful application of a risk prediction algorithm; more importantly, it lies in realising and optimising an in-depth causal machine learning methodological framework for financial data. Specifically, the research fundamentally resolves the sharp trade-off between classification performance

and interpretability by establishing a comprehensive Bayesian network learning procedure. By demonstrating the superiority of the probabilistic graph structure in preserving and making transparent the root-level flows of influence, compared with traditional post-hoc XAI methods (such as LIME and SHAP), the thesis provides a practical solution capable of directly capturing causal logic. The system thereby minimises the breakdown of information and the learning of spurious surface-level correlations, a factor that is vital for any credit appraisal system required to comply with strict legal regulations on algorithmic transparency such as the General Data Protection Regulation (GDPR) [20].

In practical terms, the research results are transformed into a Business Intelligence tool that acts as a highly reliable risk analysis assistant, helping to personalise the diagnosis of default causes for each individual profile and supporting safe decision-making for credit officers. The thesis further contributes a reference methodology (the five-stage framework: S_Pro, L_Pro, K_Pro, BN_Pro and In_Pro) for resolving the bottleneck of enormous computational cost (an NP-hard problem) in learning network structures over imbalanced and complex data spaces. This demonstrates that combining causal graph thinking with the power of machine learning is an inevitable direction for handling densely interconnected business data networks, laying the groundwork for extending and scaling the model to large-scale enterprise-level risk management problems.

1.5 Overall Structure of the Thesis

In order to help the reader move from the general to the specific, the thesis is organised into five main chapters as follows:

Chapter 1: Thesis Overview. Presents the background, the rationale for choosing the topic, the objectives, subject and scope of the research, the main contributions, and the research scenarios of the system.

Chapter 2: Theoretical Background. Provides foundational knowledge on credit risk, causal machine learning, Bayesian network graph theory, and an assessment of current XAI methods.

Chapter 3: Proposed Credit Risk Prediction Model Based on Causal Machine Learning. Presents in detail the architecture of the system, the data preprocessing pipeline (SMOTE, Lasso, K-means), and the algorithms for causal network structure learning and inference.

Chapter 4: Experiments and Evaluation of Results. Describes the dataset, the experimental procedure, the analysis of model performance metrics, and an illustration of the interactive BI Dashboard interface.

Chapter 5: Conclusion. Summarises the results achieved, analyses the remaining limitations, and proposes directions for further research and development.

1.6 Research Scenarios of the Credit Risk Assessment System

In the credit appraisal process, analysts often face complex questions that go beyond “Will this customer default or not?” to “Why?”. These are not merely isolated operational queries, but demanding practical problems posed to financial AI models. However, because the scenarios arising in practice are extremely diverse, this research does not aim to build a system that knows everything. Instead, we adopt a more pragmatic approach: clearly defining and delimiting the key research scenarios that the system needs to focus on handling.

Each defined scenario directly reflects a core need in the credit risk management process, such as: explaining the reasons for a local application rejection, analysing the sensitivity of financial variables, or identifying the overall risk structure of a customer portfolio. By designing the system specifically to address these case studies rather than merely building a traditional classification model, the thesis ensures the highest transparency and reliability for the resulting decisions. Specifically, Table 1.1 below presents in detail the research scenarios that have been formulated, thereby clarifying the operational boundaries as well as the core interpretive capabilities of the system.

Bảng 1.1: List of the formulated scenarios.

Scenario	Business meaning / Input question	Output of the Bayesian network system
1 Forward inference	Assessing how the risk probability changes given a particular profile. <i>“If the customer changes the loan term or the loan amount, how does the probability of default increase or decrease?”</i>	The system propagates the evidence through the directed graph and recomputes the posterior probability distribution at the target node (Risk level).
2 Backward inference	Tracing the common characteristics of a group of outcomes. <i>“For the profiles that have been flagged as defaults, which variables account for the most common causes?”</i>	Reversing the direction of inference from the Outcome node back to the Feature nodes, allowing the profile of a high-risk customer to be identified.
3 What-if analysis	Supporting officers in testing new credit-granting scenarios in order to advise customers. <i>“If customer A settles their existing credit card (reducing outstanding debt), will they qualify for a new loan?”</i>	By setting a conditional value (an intervention) at a specific node, the model directly quantifies the percentage (%) change in the approval decision.

Chapter 2

Theoretical Background

This chapter establishes the theoretical framework underpinning the study and application of Causal Machine Learning to the problem of credit risk prediction. We begin by analysing the nature of credit data together with the shift from traditional statistical methods to complex machine learning models, thereby clarifying the core obstacles of closed systems and the necessity of integrating XAI methods into the risk management process. Next, in order to directly address the conflict between classification performance and transparency, the focus of the research turns to the structure of causal machine learning and the quantitative mathematical foundations of Bayesian networks. The principles of directed acyclic graphs, the Conditional Probability Table (CPT), and the mechanism of probabilistic inference clarified here will serve as the solid logical basis for the entire process of designing, training and interpreting the model in the chapters that follow.

2.1 Credit Risk Prediction

Credit risk prediction is a methodological system established to help financial institutions judge whether or not to grant a loan to a particular individual or organisation. This process typically operates through an in-depth analysis of identity information as well as the transaction and repayment history of the borrower, from which the potential probability of default is computed. In essence, it is the fundamental line of defence in the entire credit cycle, serving to prevent the risk of financial loss before it actually materialises.

Beyond its operational importance, the credit modelling process regularly faces a core mathematical challenge stemming from the very nature of financial data: the phenomenon of class imbalance. In practice, the sample size representing the group of defaulting customers always accounts for a very small proportion compared with the group of ordinary customers who fulfil their financial obligations on time. This structural disparity creates serious obstacles during the training phase of machine learning algorithms. Instead of objectively learning the features that distinguish risk, the system tends to be biased towards the majority class, thereby establishing distorted decision boundaries. Without specialised preprocessing and intervention techniques, the inevitable consequence is that the model will repeatedly produce overly optimistic assessments, inadvertently overlooking profiles that carry latent credit risk.

Fundamentally resolving data imbalance and optimising the predictive model brings

substantial value to enterprise-level risk management. An accurate credit assessment system not only supports the efficient allocation of capital but also helps banks strictly comply with capital adequacy regulations. Moreover, when combined with causal analysis methods capable of transparent interpretation, the prediction system moves beyond the limits of an ordinary binary classification tool. Instead, this foundation gives managers deep insight into the core drivers governing default behaviour, laying a solid basis for designing risk mitigation strategies and managing the investment portfolio in a proactive and comprehensive manner.

2.2 Credit Scoring Models

In the modern financial system, credit activity plays a central role in allocating capital among economic agents; however, the process of granting credit always entails a certain degree of uncertainty regarding the borrower's ability to repay. Credit risk is understood as the danger that a borrower will not fully meet their financial obligations as committed, resulting in losses for the lending institution. Building models for default prediction has therefore become an important task in financial risk management.

Credit scoring is regarded as a quantitative system for assessing the creditworthiness of an individual or an enterprise based on characteristics of credit history, demographic information, financial capacity and repayment behaviour. The core objective of credit scoring is to convert a complex set of input information into a probability or a score reflecting the likelihood of an adverse credit event, most commonly the failure to repay debt on time. Liu et al. (2024) [42] argue that credit risk prediction models are usually built on the historical data of borrowers in order to determine the probability of default of a new customer, thereby helping financial institutions make appropriate credit-granting decisions.

In essence, credit scoring is a binary or probabilistic classification problem in the field of machine learning, in which the target variable typically represents the state of “default” or “non-default”. Unlike ordinary classification problems, however, credit scoring has its own particular character, since the decisions the model produces directly affect an individual's access to finance and at the same time bear on the stability of the banking system. A predictive model with high accuracy but lacking interpretability may give rise to credit decisions that are difficult to verify; balancing predictive performance against transparency has therefore become an increasingly important requirement in modern credit scoring research [40].

Liu et al. (2024) [42] divide approaches to credit risk modelling into two main groups: statistical techniques and machine learning methods. This shift reflects the transition from linear models based on statistical assumptions to non-linear models capable of exploiting complex structures in large financial datasets.

2.2.1 Classical Statistics

The earliest studies on bankruptcy and credit risk prediction appeared in the early twentieth century. FitzPatrick (1932) [22] is regarded as one of the pioneering scholars in using financial ratios to distinguish bankrupt from non-bankrupt firms. This work laid the foundation for the idea that default behaviour can be explained and predicted through quantitative indicators rather than relying solely on the subjective judgement of credit officers.

For several decades, statistical methods played a foundational role in building credit assessment systems. What the models of this school have in common is that they are based on classical statistical probability theory, on the assumption that the relationship between the input variables and the state of default can be described through determinate mathematical functions.

An important turning point in this period was the Multiple Discriminant Analysis (MDA) model of Altman (1968) [3], which marked a significant shift in the field of credit prediction. Altman's Z-score model seeks an optimal linear combination of financial variables so as to maximise the distance between the bankrupt and the non-bankrupt groups. Mathematically, MDA assumes that data belonging to different groups can be separated by a linear hyperplane:

$$Z = w_1X_1 + w_2X_2 + \dots + w_nX_n, \quad (2.1)$$

where X_i are the input financial variables and w_i are the weights optimised so as to form the discriminant function. The resulting value of Z is compared against a determined threshold in order to classify the customer.

Although it achieved high effectiveness in the context of traditional financial data, MDA requires strict assumptions such as multivariate normal distribution, homogeneity of variance across groups, and a linear relationship between the input variables and the target variable. To overcome these limitations, Ohlson (1980) [49] proposed the Logistic Regression model, which quickly became one of the most widely used methods in the financial industry thanks to its ability to model the probability of a binary event.

Unlike MDA, Logistic Regression does not require the data to follow a normal distribution and does not directly predict the class label; instead, it estimates the probability through the sigmoid function:

$$P(Y = 1|X) = \frac{1}{1 + \exp(-(\beta_0 + \sum_{i=1}^n \beta_i x_i))}, \quad (2.2)$$

where:

- $P(Y = 1|X)$: The probability that the customer falls into the state of default, conditional on the input feature set X .
- e : The base of the natural logarithm.

- β_0 : The intercept of the model.
- β_i : The weight (regression coefficient) corresponding to the i -th feature, expressing the degree of linear influence of that variable.
- x_i : The value of the i -th feature in the borrower's credit profile.
- n : The total number of features (independent variables) considered.

The Logistic model quickly became the gold standard in academic research and in practical banking applications thanks to its simplicity, stability and high interpretability. In practice, Logistic Regression to this day remains the reference model in internal credit rating systems under the Basel II and Basel III standards [6].

Nevertheless, these statistical methods have inherent drawbacks, being constrained by strict assumptions about the data distribution and relying mainly on a mechanism of linear fitting. In the modern data environment, credit behaviour is often affected simultaneously by many non-linearly interacting factors such as transaction history, consumption behaviour, social network data, cash flow over time and demographic characteristics. Traditional statistical models are hardly capable of learning these complex relationships [40].

Consequently, although it retains an important role in systems requiring high interpretability, the statistical school increasingly reveals its limits in terms of data representation capacity and predictive performance.

2.2.2 Individual Machine Learning Models

The success of the Z-Score model created the academic foundation for the entire modern credit scoring industry. However, the development of Big Data, financial technology (FinTech), peer-to-peer lending (P2P Lending), digital banking and e-commerce has given rise to datasets that are increasingly complex, high-dimensional and non-linear in structure. Together with the rapid development of artificial intelligence, this has produced a fundamental shift in the field of credit prediction. Instead of constructing mathematical functions based on statistical assumptions, machine learning algorithms allow the system to learn latent rules from the data automatically. This has driven the transition from traditional statistical models to machine learning and artificial intelligence methods [30].

One of the first algorithms to be widely applied is the Support Vector Machine (SVM) developed by Vapnik [71]. The advantage of SVM lies in its ability to construct an optimal separating hyperplane in a high-dimensional feature space through the kernel trick, maximising the distance to the data points closest to the boundary. Shen et al. (2022) [61] developed a Semi-Supervised SVM model for credit prediction, exploiting both labelled and unlabelled data in order to improve predictive effectiveness.

The basic optimisation objective of SVM is:

$$\min \frac{1}{2} \|w\|^2$$

subject to: $y_i(w^T x_i + b) \geq 1$ where w is the normal vector of the hyperplane and b is the bias. Optimising the margin gives SVM a strong capacity to generalise to unobserved data.

In parallel, models based on decision trees have also proved highly effective. Random Forest, proposed by Breiman (2001) [9], operates on the principles of bagging and random feature selection. Ye et al. (2018) [77] applied Random Forest to Lending Club data and recorded predictive performance superior to that of traditional Logistic Regression.

In the field of Deep Learning, studies have begun to exploit the ability of neural networks to learn feature representations automatically. Fonseca et al. (2020) [24] built a Two-Stage Fuzzy Neural Network model to handle uncertainty in credit data in Brazil. Zhou et al. (2020) [83] developed a CNN model for the credit scoring problem and achieved performance considerably higher than that of traditional models.

Although individual machine learning models substantially improve predictive accuracy, they still have a number of limitations. Each algorithm typically exploits only certain aspects of the data effectively. For example, SVM works well on small datasets but encounters difficulties when scaled up; neural networks have a strong capacity to learn non-linear relationships but are prone to overfitting; decision trees are easy to interpret but lack stability. This has led to the emergence of the hybrid and ensemble learning school.

2.2.3 Ensemble Models

Although individual machine learning algorithms have brought considerable progress, each method still has certain limitations in terms of generalisation capability or susceptibility to overfitting on noisy data. To overcome these barriers, researchers have integrated multiple artificial intelligence algorithms in order to build ensemble models [42]. This combination is typically carried out through a strategy of using one model to optimise the parameters of another method, or of aggregating the predictions of several different classifiers to create a new algorithm that is more powerful and more stable.

In general, ensemble models are usually represented as a weighted sum of the base models:

$$F(x) = \sum_{k=1}^K w_k f_k(x) \quad (2.3)$$

where:

- $F(x)$: The final predicted output (the probability or the default label) of the ensemble system.

- K : The total number of base classifiers used.
- w_k : The contribution weight of the k -th base model, usually optimised according to the reliability or the performance of each component model on the dataset.
- $f_k(x)$: The independent prediction of the k -th base model for the input feature set x .

In practical research on credit risk prediction, the superiority of ensemble learning has been demonstrated through a wide variety of hybrid architectures. With regard to integrating deep learning algorithms with classical machine learning, Huang and Yen (2019) [33] successfully combined Deep Belief Networks with SVM, producing predictions markedly more accurate than those obtained from individual classifiers alone such as SVM or the Deep Boltzmann Machine. Following a different approach aimed at optimising deep learning architectures, regularisation techniques can be improved by simultaneously applying L_1 constraints to a single L_2 regularisation scheme. This refinement considerably improves the structure of the neural network and delivers superior effectiveness when handling the problem of corporate credit risk prediction in high-dimensional data spaces [76]. In addition, merging ensemble machine learning methods with probabilistic theoretical foundations has also opened up a number of breakthrough directions. A representative example is the combination of the Random Forest algorithm with evidence theory through the Dempster–Shafer (D–S) rule of combination, which has delivered predictive performance considerably higher than that of the original algorithms as well as of traditional risk assessment models [85].

In particular, ensemble learning has attracted great attention thanks to its good generalisation and excellent classification performance in the field of credit risk assessment, most notably the families of tree-based ensembles such as Random Forest and Gradient Boosting [53], [82]. However, although risk assessment performance is improved, the increased complexity of the ensemble structure aggravates the problem of opacity and lack of transparency. Sacrificing intrinsic interpretability in exchange for accuracy poses a major challenge in the practice of financial risk management, where explaining why a customer has been refused a loan becomes extremely difficult. This is precisely the driving force behind the emergence of Explainable Artificial Intelligence (XAI) and Causal Machine Learning [42].

2.3 Interpretability in XAI

2.3.1 The Emergence of XAI Methods

Along with the rapid development of financial technology and the growth of data-driven decision-making, the interpretability of credit risk assessment models has become a core and urgent research topic in the modern financial field. In recent years, the emergence of Explainable Artificial Intelligence methods, aimed at improving the transparency, comprehensibility and trustworthiness of intelligent systems, has reshaped the approach to risk management. The emergence of XAI methods has been strongly driven by the

following three core factors:

1. *The conflict between performance and transparency:* The development of machine learning has produced complex ensemble models and deep neural networks with outstanding accuracy. However, these architectures operate as black-box models, making it impossible for humans to directly observe or understand the processing logic inside them. The consequence is that one must confront a trade-off between accuracy and interpretability. This trade-off has prompted the need to develop new techniques in order to narrow the gap between predictive capability and the ability to understand the model.
2. *Pressure from legal regulations:* As algorithmic applications become ever more pervasive, international legal safeguards have been established to protect consumer rights. A prime example is the General Data Protection Regulation (GDPR) issued by the European Union, which clearly stipulates the customer's right to explanation with respect to fully automated decisions made by AI. This obliges lending institutions to make the decision logic of personal credit algorithms transparent.
3. *Practical requirements in risk management and bias prevention:* A model with high accuracy but lacking interpretability harbours substantial systemic risk, since the soundness of its predictions cannot be verified. A lack of transparency prevents managers from detecting confounding variables or discriminatory data (such as gender, ethnicity or income). XAI has therefore emerged as an indispensable tool for monitoring sensitive data, ensuring fairness and building trust among financial institutions, borrowers and regulatory authorities.

Overall, XAI methods are not merely technical post-hoc tools; they have become a strategic component of credit risk prediction systems, fully serving both the goal of profit optimisation and that of compliance with technology ethics [42].

2.3.2 Classification of XAI Methods and Techniques

In order to systematise the approaches to making models transparent, the research community generally classifies XAI techniques according to the point at which the explanation is generated relative to the life cycle of the original model. In essence, these methods can be divided into two main streams: intrinsic interpretability and post-hoc interpretability.

Intrinsically Interpretable Models

Intrinsic XAI refers to the group of algorithms designed from the outset with a simple and transparent architecture, allowing users to directly understand the learning mechanism and the internal decision logic without the need for any auxiliary tool. This transparency is achieved mainly by limiting the complexity of the algorithm, whereby the relationships between the input variables and the output are represented clearly through mathematical coefficients or a system of branching rules.

Representative examples of this approach include linear regression, decision trees and, in particular, Bayesian networks. The essence of a Bayesian network lies in its use of a directed acyclic graph and Conditional Probability Tables to visually express the causal flows of influence within the data, thereby helping to isolate the root causes of risk [42]. Although this approach provides an absolutely accurate explanation of what the algorithm actually computes, traditional intrinsic models are often limited in their learning capacity. Specifically, when having to process a dataset D that is non-linear and complex, their predictive performance is usually considerably lower than that of modern machine learning models.

Post-hoc Interpretation Techniques

In order to address the problem of optimising accuracy over n observed samples when applying complex black-box models, a second stream of approaches known as post-hoc interpretability has been adopted. Post-hoc XAI techniques must be used to explain the results after the model has completed training and produced its predictions. The salient characteristic of these techniques is their independence from the internal structure of the original model (model-agnostic). They treat the machine learning algorithm as a latent function and attempt to approximate or analyse its behaviour from the outside by perturbing the input data and observing the resulting changes in the output.

From a mathematical standpoint, many prominent post-hoc XAI algorithms (typically LIME or SHAP) approach explanation through the additive feature attribution method. This local approximation method is expressed in general form by the formula:

$$g(z') = \phi_0 + \sum_{j=1}^M \phi_j z'_j \quad (2.4)$$

where:

- $g(z')$: The linear *explanation model*, used to approximate the local behaviour of the original black-box model around the data point under consideration.
- M : The total number of *features* introduced into the explanation space.
- ϕ_0 : The *base value* of the prediction, usually the expected value of the model when no feature exerts any influence.
- ϕ_j : The weight (degree of contribution) of the j -th feature to the difference between the current prediction and the base value.
- z'_j : A binary state variable taking the value 1 if the j -th feature is present and 0 if it is withheld.

Although post-hoc XAI partly removes the barriers posed by complex algorithms, it still has a major limitation in that it provides only approximate results rather than

reflecting the mechanism of the original model with complete fidelity. Moreover, these methods mainly exploit surface-level statistical correlations instead of identifying the genuine causal mechanism leading to a credit default event [42].

In the context of the credit risk problem, the over-reliance on closed models combined with post-hoc XAI still fails to fundamentally resolve the trade-off between accuracy and interpretability. The limitations of post-hoc XAI in terms of causal reasoning are precisely what motivate the move towards a more powerful methodological framework: incorporating non-linear predictive capability into structures with high intrinsic interpretability through causal machine learning techniques. In the study by Liu, Zhang and Xiong (2024) [42], the causal Bayesian network is proposed as a complete causal machine learning framework for the credit risk prediction problem, comprising five successive stages: S_Pro, L_Pro, K_Pro, BN_Pro and In_Pro.

2.4 Overview of Causal Machine Learning

2.4.1 Definition and Basic Properties

Throughout many decades of development in data science and machine learning, most predictive models have been built on the basis of detecting correlational patterns between input and output variables. From Logistic regression, decision trees and Random Forest to modern deep learning architectures, the central objective of these algorithms is to optimise predictive capability by exploiting the statistical patterns present in historical data [40], [71], [9], [26]. This approach has brought major achievements in many fields, particularly in credit risk prediction, where machine learning models are able to detect complex behavioural patterns that traditional statistical methods struggle to identify [40].

However, the success of traditional machine learning has simultaneously revealed a fundamental limitation: models may predict accurately yet cannot explain the true cause of that prediction. In other words, classical machine learning algorithms mainly learn positive or negative associations between variables, but have no capacity to distinguish what is a cause from what is merely an effect or a co-occurring variable [50]. A model may detect that customers with low incomes are frequently accompanied by a high probability of default, but it cannot answer whether increasing a customer's income would genuinely reduce that probability. This is precisely the boundary between correlational inference and causal inference.

The theoretical foundation for this shift was laid by Judea Pearl, who is regarded as the father of modern causal theory. In his classic work *Causality: Models, Reasoning and Inference*, Pearl (2009) [50] argues that data science cannot stop at the question of “what occurs together” but must move on to answer the question of “what causes what”. This perspective opened up an entirely new line of research known as causal machine learning, in which the objective is not only to predict but also to discover, quantify and verify the cause–effect relationships existing within the data. This epoch-making shift helps AI systems escape their dependence on the assumption of an unchanging

data distribution (the i.i.d. assumption), which is frequently violated in economic practice owing to temporal drift or macroeconomic shocks. It thereby opens the way to building predictive models that combine near-optimal accuracy with robust intrinsic interpretability [63], [42], [58].

The emergence of Causal Machine Learning has also arisen from the practical needs of high-risk fields such as finance, healthcare, insurance and public administration. In these fields, a wrong decision not only causes economic damage but also leads to serious legal and ethical consequences. Modern regulations such as the European Union’s GDPR require artificial intelligence systems to be able to explain the basis on which automated decisions are formed [20]. This makes purely correlation-based black-box models increasingly challenging to deploy in practice.

2.4.2 From Correlation-Based Machine Learning to Causation-Based Machine Learning

In traditional statistics, correlation reflects the degree to which two variables change together. If two variables frequently appear together, the model treats this as a data pattern with predictive value. However, the existence of a correlation does not mean that a cause–effect relationship exists. This is the well-known principle that “correlation does not imply causation”, widely acknowledged in modern statistics [50], [65].

For example, in the financial domain, historical data may show that customers holding many credit cards tend to have a higher default rate (as presented in Table 4.1 and Figure 4.1). This does not mean, however, that opening additional credit cards is the cause of default. The real cause may stem from the level of latent financial pressure or from poor cash-flow management. Relying on correlation alone, a model may arrive at misleading conclusions and lead to inaccurate credit decisions.

The core foundation separating causal machine learning from classical statistical and machine learning algorithms lies in how the relationship between correlation and causation is defined and handled. Traditional machine learning models optimise an objective function based on the ordinary conditional probability $P(Y|X)$, that is, they search for statistically co-occurring patterns between the input features X and the target variable Y [8, 10].

The limitation of this approach is its particular sensitivity to confounders and selection bias, which makes it easy for the algorithm to learn and exploit spurious correlations in producing its predictions [26, 30]. In the field of credit scoring, a correlation-based model may assess an individual as low-risk merely because they possess attributes commonly associated with the group of safe borrowers; yet this correlation may be entirely broken when the boundary conditions of the economy change, causing the credit score to lose its actual predictive value [66], [30], [42].

To overcome this limitation, Pearl (2009) argues that data science needs to move through three levels of inference [50]:

- *The first level (Association)* is the level of observation, answering the question: “What is happening?”. This is the level of most current Machine Learning models.
- *The second level (Intervention)* answers the question: “What will happen if we actively change a variable?”. This is precisely the foundation of what-if analysis and intervention policies.
- *The third level (Counterfactual)* answers the question: “What would have happened if we had acted differently in the past?”. This is the highest level of inference in causal artificial intelligence.

While ordinary machine learning models are locked at the first level, where the question stops at “If we observe the event X , what will happen to Y ?”, Causal ML allows the mathematical system to rise to the second and third levels by using do-calculus to answer questions such as “If we actively intervene to change X , how will the outcome Y vary?” [50], [52]. By shifting from computing the observational probability $P(Y|X)$ to the interventional probability $P(Y|do(X))$, causal machine learning isolates the true structure of influence of each variable, helping to clearly distinguish variables bearing a direct causal relationship to default behaviour from those that are merely superficially correlated [42], [10]. This enables financial institutions not merely to look at an opaque predictive score but to genuinely understand the nature of what creates the risk in each customer profile [6], [42].

2.4.3 The Structure of Causal Machine Learning

Causal machine learning is a combination of three major scientific fields: causal statistics, machine learning and probabilistic graphical models [50], [37].

The process of building a causal machine learning system usually begins with identifying the variables under study and constructing a causal graph. This graph represents the hypotheses about the causal relationships among the variables. Each node stands for a variable and each directed edge expresses the causal influence from the cause variable to the effect variable [50], [65]. After the DAG has been constructed, the next step is to identify confounders, mediators and colliders. This is a particularly important stage, since mishandling these types of variables can introduce serious bias into causal inference [65]. Next, the system carries out Structure Learning and Parameter Learning. In modern research, the DAG structure can be discovered automatically through algorithms such as the PC Algorithm, Hill Climbing Search, Tabu Search or Max-Min Hill Climbing (MMHC) [37], [34], [65]. The model then estimates the conditional probability of each node in the network. For BNs, this process typically uses Maximum Likelihood Estimation (MLE) or Bayesian Estimation [17], [34]. The final stage is causal inference. This is the step that converts historical data into knowledge capable of supporting decision-making. Common techniques include Pearl’s do-calculus [50], Bayesian probabilistic inference [17], Sensitivity Analysis [35], What-if Analysis [42], [35] and Counterfactual Analysis [10].

2.5 Bayesian Networks

Post-hoc interpretability through popular tools such as LIME [54] or SHAP [43] is in fact merely a method of locally approximating the behaviour of a complex model, rather than accurately reflecting the operating mechanism of the real world. These methods typically introduce small perturbations at the input in order to measure the change in the output of the closed model, which creates a risk of misleading interpretation if the model itself has learned faulty correlations or has been seriously affected by multicollinearity among the economic variables [40], [23]. Conversely, the interpretive mechanism of causal machine learning, particularly when combined with the Bayesian network structure, is a form of intrinsic, self-contained interpretation that is entirely transparent by virtue of its structure [37], [17]. The cause-effect relationships are made explicit through a directed acyclic graph, in which each node represents an economic variable and each directed arrow expresses a direct causal influence [34], [42].

The Bayesian network, also known as the belief network, represents a revolutionary advance in extending and applying Bayesian statistical methods to multidimensional data spaces. From a mathematical standpoint, it is one of the most compelling and internally coherent theoretical models for representing and inferring uncertain knowledge [50]. The core architecture of a Bayesian network is built on a directed acyclic graph in which each node represents a random variable, and the directed edges serve to depict directly the conditional dependencies among the variables [37], [34].

What gives Bayesian networks their outstanding power compared with traditional statistical methods is the ability to construct a joint probability distribution for the entire multivariate system through the chain rule of probability. This mechanism provides an extremely rigorous mathematical framework for quantifying the uncertainty of both the input data and the model structure, while also conferring an outstanding natural advantage in interpretability [17], [65]. Unlike traditional closed algorithms, which are confined to the first rung of the causal ladder (assessing only surface-level statistical correlation), the Bayesian network model has a structure focused on modelling the mechanism by which uncertainty in the data arises. By profoundly fusing graph theory with probability space, Bayesian networks provide logical, transparent and thoroughly causal interpretations of every interaction and intersection among economic or social variables [58], [52].

Precisely because of this capacity to quantify uncertainty together with its excellent intrinsic transparency, the Bayesian network has risen to become a leading foundation and is widely applied in the field of credit risk assessment and prediction. Representative of this trend, the study by Masmoudi et al. (2019) [45] pioneered the structuring of a Bayesian network incorporating latent variables, opening up the possibility of exploiting risk characteristics that cannot be directly observed in order to improve credit appraisal performance comprehensively. From another angle, approaching the problem from the perspective of data quality, Leong (2016) [39] successfully addressed core intractable problems in real-world credit scoring systems, including data censorship, class distribution imbalance handled by the SMOTE algorithm [11], and the challenges

of real-time implementation, through a specialised Bayesian network model.

Alongside the explosion of data science and machine learning technology, the Bayesian network ecosystem has become ever more deeply embedded in the foundations of modern finance. For instance, Wu and Han (2021) [75] proposed a breakthrough inference model designed specifically for credit investigation systems based on fuzzy Bayesian networks, making a major contribution to handling ambiguous risk boundaries. To respond to the continuously fluctuating nature of the market, Lin et al. (2020) [41] developed an optimal Bayesian method for efficiently estimating multivariate risk measures within a dynamic framework, thereby considerably improving the accuracy of the operating system. Going further in exploiting the temporal dimension, Shin (2021) [62] introduced an event-based Bayesian structure that allows variables to be modelled over different time intervals in order to assess the elasticity of financial standards and an institution's capacity to withstand risk.

Not stopping at the boundary of credit risk, the comprehensiveness of the Bayesian network architecture has also demonstrated outstanding capability when extended to problems of portfolio risk management, project management and large-scale fraud detection. The study by Bai et al. (2020) [4] made excellent use of fuzzy Bayesian networks to assess the risk of the resource system within a project portfolio, allowing key risk factors with cross-interdependency to be identified and systematic risk mitigation strategies to be proposed. Similarly, Akhavan et al. (2021) [2] built a specialised Bayesian network modelling framework to capture the risks of start-up and knowledge-based projects, providing a powerful logical tool for analysing multidimensional risk scenarios and quantifying their direct impact on overall success. It can be affirmed that Bayesian networks have established a firm foundation across the entire financial ecosystem, covering credit risk assessment, investment portfolio optimisation, market prediction and the detection of sophisticated financial fraud [42]. Notably, the resonance between modern computational capability and causal theory has laid the groundwork for Bayesian networks to penetrate deeply into other key scientific fields. This methodological framework has featured strongly in transportation systems for predicting the risk of flight delays (Truong, 2021) [69], in medical diagnosis through variational Bayesian networks supporting biomedical image segmentation (Chen, Zhao, & Liu, 2022) [13], as well as in maritime safety risk assessment with fuzzy Bayesian networks (Chen, Zhang, et al., 2022) [12] and in socio-technical risk control processes at industrial megaprojects (Zangeneh & McCabe, 2022) [80].

In order to realise causal interpretability explicitly and to quantify precisely the influence of each input feature, sensitivity analysis is integrated into the Bayesian network as a core post-hoc inference tool. This method is used specifically to probe and measure the sensitivity of a target node (such as the probability of default) to variation in the satellite nodes (such as income or debt ratio) within the network system. The mechanism of sensitivity analysis operates mainly on the control variable method; accordingly, the analyst actively intervenes to pin or alter the state of a cause node in order to observe directly the fluctuation in the posterior probability distribution of the effect nodes [42], [35]. This technique quantifies precisely how adjustments to the

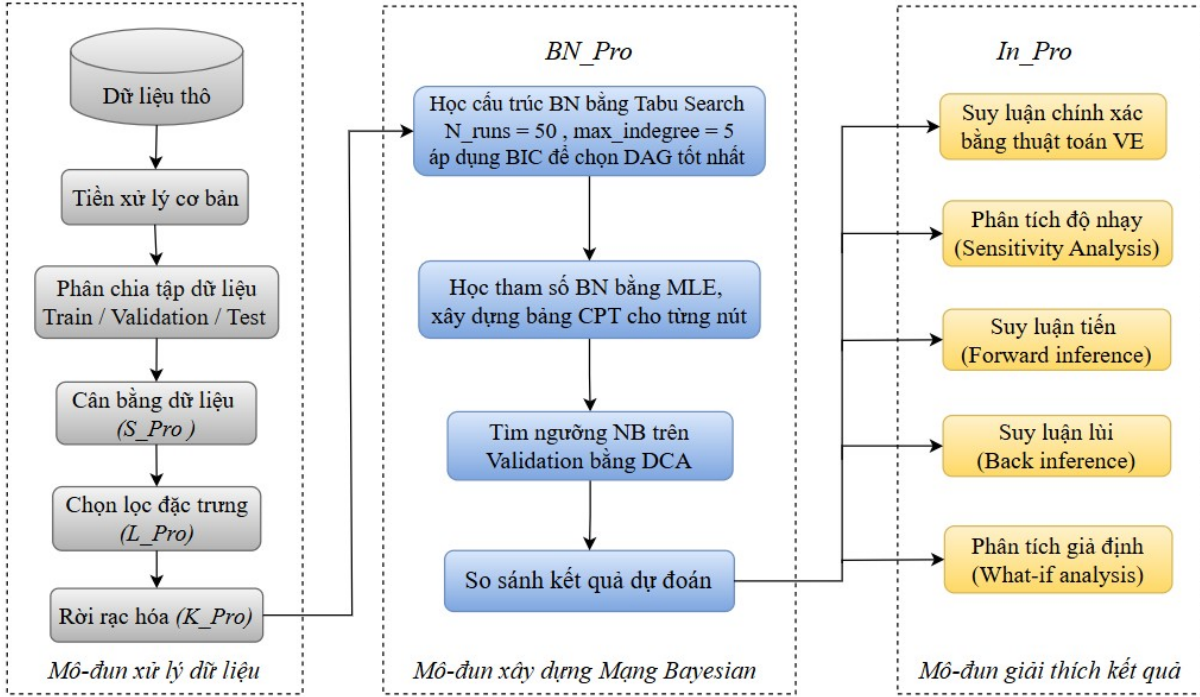
input data of the Bayesian network will amplify or cancel out the magnitude of risk at the output. The larger the change in probability (the delta), the more sensitive the node under examination, demonstrating mathematically that the intrinsic causal link between the observed node and the varied node is extremely tight and dominant (Kamal & Kutay, 2021) [35]. Successfully locating nodes of high sensitivity not only opens up the algorithmic black box but also provides a solid basis for the decision-making process, ensuring that the system operates safely and stably and laying the groundwork for planning purposeful risk interventions [10].

In Bayesian network research practice, the control variable method continues to assert its position as the backbone. Typically, the study by Chen, Zhang et al. (2022) [12] constructed an evidence-based fuzzy Bayesian network to model the probability of maritime accidents, while using sensitivity analysis to isolate in full the core factors governing the occurrence of accidents. Continuing along this line of approach, Kaptan (2022) [36] flexibly employed Bayesian networks combined with sensitivity analysis to explore the risks in the process of loading vehicles onto RO-RO ships, demonstrating the versatility of this tool. Not stopping at the traditional control variable method, the constructive basis of sensitivity analysis has become increasingly diversified with the application of beta regression methods to measure the sensitivity of distribution parameters (Rohmer & Gehl, 2020) [55], or the fusion of principal component analysis (PCA) to handle dynamic data spaces (Zhao et al., 2023) [81]. Overall, the results obtained from Bayesian network sensitivity analysis go far beyond the limits of statistical figures; they have become the ultimate logical tool for explaining mechanisms, making structures transparent and vouching for every prediction of a causal machine learning model in the face of the most demanding requirements of science and of financial law [20], [42], [58], [52].

Chapter 3

Proposed Credit Risk Prediction Model Based on Causal Machine Learning

This chapter presents in detail the methodology and the overall architecture of the credit risk prediction model based on Causal Machine Learning. We begin by establishing a specialised data processing pipeline that integrates techniques for class balancing (S_Pro), feature selection (L_Pro) and discretisation (K_Pro) in order to construct an optimal sample space fully compatible with the probabilistic nature of the model. Next, the focus of the research turns to the core procedure that constitutes the Bayesian network (BN_Pro), including the resolution of the optimisation problem in graph structure learning and the methods for estimating the parameter system. Finally, in order to affirm the system's capacity to make the financial decision flow transparent, we present the techniques for interpreting the results (In_Pro), from sensitivity analysis and forward-backward inference to hypothetical intervention scenarios, while also establishing comprehensive criteria for evaluating predictive performance. The proposed model framework is described in Figure [3.1](#)



Hình 3.1: Framework of the proposed model.

3.1 Data Processing Procedure

3.1.1 Data Balancing with SMOTE (S_Pro)

In the credit risk prediction problem, the training dataset often exhibits severe class imbalance, with the number of default profiles (the minority class) considerably smaller than that of ordinary profiles (the majority class) (Table 4.1, Figure 4.1). Without intervention, machine learning algorithms tend to be biased towards the majority class, resulting in a distorted decision boundary and a severe reduction in the ability to identify genuine risk. To remedy this, the present study applies the Synthetic Minority Over-sampling Technique (SMOTE) in order to rebalance the class distribution before the data enters the Bayesian network structure learning procedure. SMOTE operates by synthesising new minority samples rather than directly duplicating existing observations, thereby avoiding local overfitting [11]. Specifically, for each minority sample x_i , the algorithm identifies the k nearest neighbours in the feature space according to Euclidean distance, then randomly interpolates along the line segment joining x_i to a randomly chosen neighbour x_{nn} in order to create a new synthetic sample x_{new} according to the formula:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{nn} - x_i), \quad \lambda \sim \mathcal{U}(0, 1) \quad (3.1)$$

where λ is a random value sampled uniformly from the interval $[0, 1]$. This process is repeated until the number of samples in the minority class is balanced with that of the majority class. The algorithm is presented as follows:

Algorithm 1 Pseudocode of the SMOTE algorithm**Input:** Minority sample set X_{min} , number of neighbours k , synthesis ratio N .**Output:** Synthetic sample set X_{syn} .

```

1:  $X_{syn} \leftarrow \emptyset$ 
2: for each  $x_i \in X_{min}$  do
3:   Compute the  $k$  nearest neighbours of  $x_i$  in  $X_{min}$  according to Euclidean distance
4:   for  $n = 1$  to  $N$  do
5:     Randomly select  $x_{nn}$  from the set of  $k$  neighbours
6:     Sample  $\lambda \sim \mathcal{U}(0, 1)$ 
7:      $x_{new} \leftarrow x_i + \lambda \cdot (x_{nn} - x_i)$ 
8:      $X_{syn} \leftarrow X_{syn} \cup \{x_{new}\}$ 
9:   end for
10: end for
11: return  $X_{syn}$ 

```

The key point of the interpolation mechanism is that the synthetic samples created lie on the line segment joining two real samples in the feature space, and therefore always fall within a neighbourhood of genuine data density. This ensures that new samples are not generated in regions of space far removed from the original distribution, a common drawback of synthesis methods based on random noise. As a result, the decision boundary between the two classes is expanded and smoothed more naturally, helping the model to learn the discriminative characteristics of the minority class rather than merely memorising repeated samples.

3.1.2 Feature Selection Using Lasso Regression (L_Pro)

After the data balancing stage, the input feature space often contains many variables with low contribution or high mutual correlation, causing information redundancy and increasing the computational cost of the Bayesian network structure learning process — which is already an NP-hard problem. The feature selection step therefore plays an important role in reducing the dimensionality of the variable space while still preserving the discriminative power of the model. This study applies Logistic regression with L1 regularisation (commonly referred to as Lasso regression) as the principal feature selection mechanism. Mathematically, Lasso adds to the objective function a penalty term proportional to the sum of the absolute values of the coefficients, forcing the coefficients of certain variables exactly to 0 when the improvement in model fit from including a variable is not sufficient to offset its penalty in increasing the number of probability parameters to be estimated, thereby performing automatic feature selection within the parameter estimation process itself [68]. The corresponding optimisation problem is written as:

$$\hat{\beta} = \arg \min_{\beta} \left[-\ell(\beta; \mathbf{X}, \mathbf{y}) + \frac{1}{C} \sum_{j=1}^p |\beta_j| \right], \quad (3.2)$$

where,

- $\hat{\beta}$: The optimal set of coefficients (weights) to be estimated for the model.
- $\ell(\beta; \mathbf{X}, \mathbf{y})$: The log-likelihood function of the logistic model with respect to the feature matrix $\mathbf{X}$ and the vector of actual labels $\mathbf{y}$.

$$\ell(\beta; \mathbf{X}, \mathbf{y}) = \sum_{i=1}^n \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \quad p_i = \frac{1}{1 + e^{-\beta^\top \mathbf{x}_i}} \quad (3.3)$$

- β_j : The weight coefficient corresponding to the j -th feature in the data.
- p : The total number of input features.
- C : The inverse of the penalty strength ($C > 0$), used to control the degree of regularisation of the model:
 - *When C is small*: The strong penalty forces many coefficients β_j decisively to 0; specifically, a variable is eliminated when the improvement it brings to the log-likelihood function is smaller than the penalty that L_1 regularisation imposes on its coefficient, that is, when the marginal contribution of the variable is not sufficient to justify its retention in the model.
 - *When C is large*: The penalty is weak, and the objective function tends towards the form of ordinary logistic regression without constraints.
- $\sum_{j=1}^p |\beta_j|$: The L1 regularisation term (L1 norm), measuring the sum of the absolute values of the coefficients.

3.1.3 Data Discretisation Using K-means (K_Pro)

Bayesian networks require the input variables to be in discrete form so that conditional probability tables can be constructed through frequency statistics. When the data contains continuous variables, computing conditional probabilities directly over a continuous value domain is computationally infeasible and also leads to severe data sparsity in the cells of the CPT [37][27]. The discretisation step is therefore a prerequisite for ensuring the validity of the network parameter learning process.

This study uses the K-means algorithm as its principal discretisation mechanism [42]. To understand the reason for this choice, we consider the differences between three common discretisation strategies. The equal-width method divides the entire value domain into k intervals of equal width, regardless of the actual density of the data distribution. As a consequence, the bins in the tail regions of financial data distributions, which are typically strongly skewed, will contain very few observations, leading to

unreliable estimates of conditional probability in the CPT. The equal-frequency method overcomes the density problem by ensuring that each bin contains approximately the same number of observations, but it may sever the natural clusters in the data when bin boundaries are placed arbitrarily according to quantiles. K-means, by contrast, divides the value domain on the basis of the actual geometric density structure, seeking bin boundaries such that the observations within the same bin are as similar to one another as possible. This is particularly meaningful for Bayesian networks, since more homogeneous bins produce CPT cells that more accurately reflect the true data-generating mechanism.

With k bins determined in advance, K-means seeks a set of cluster centroids $\{\mu_1, \mu_2, \dots, \mu_k\}$ that minimises the total within-cluster sum of squared distances [44]:

$$\min_{\{S_i\}} \sum_{i=1}^k \sum_{x \in S_i} (x - \mu_i)^2, \quad (3.4)$$

where:

- k : The total number of clusters (the number of bins) defined in advance.
- S_i : The set of data points (observations) assigned to the i -th cluster.
- x : A particular data point with a continuous value belonging to cluster S_i .
- μ_i : The centroid of the i -th cluster, usually the mean value of all points x lying within that cluster.
- $(x - \mu_i)^2$: The squared distance from the data point x to its corresponding cluster centroid μ_i .

In the centroid update step, the centroid of each cluster is recomputed as the arithmetic mean of all the points belonging to that cluster. This iterative process is guaranteed to converge because the objective function is non-negative and decreases monotonically after each iteration. The assignment step does not increase the total sum of squared distances, and the centroid update step likewise does not increase the value of the objective function, since the arithmetic mean is the minimiser of the sum of squared distances within a cluster. Each observation is then assigned the bin label corresponding to the index of the nearest cluster, converting the continuous value into a discrete ordinal variable $\{0, 1, \dots, k - 1\}$ suited to the input requirements of the Bayesian network. The pseudocode of the algorithm is presented as follows:

Algorithm 2 Pseudocode of the K-means algorithm for discretisation

Input: Dataset X , number of bins k .

Output: Bin label $b(x) \in \{0, 1, \dots, k-1\}$ for each observation.

```

1: Initialise  $k$  cluster centroids  $\{\mu_1, \mu_2, \dots, \mu_k\}$  randomly from  $X$ 
2: repeat
3:   Assignment step:  $S_i \leftarrow \{x \in X : |x - \mu_i| \leq |x - \mu_j|, \forall j \neq i\}$ 
4:   Centroid update step:  $\mu_i \leftarrow \frac{1}{|S_i|} \sum_{x \in S_i} x$ 
5: until the cluster centroids no longer change
6: Assign the bin label  $b(x) \leftarrow \arg \min_i |x - \mu_i|$  for each  $x \in X$ 
7: return  $\{b(x)\}$ 

```

3.2 Building the Bayesian Network (BN_Pro)

Unlike black-box machine learning models such as Random Forest, XGBoost or Deep Neural Networks, which focus solely on optimising a predictive loss function, the Bayesian network aims to model the structure of probabilistic dependence among variables by establishing a qualitative structure through the concept of a directed acyclic graph, thereby allowing the mechanism by which credit risk is formed to be clearly described and probabilistic inference to be performed across the entire system [50], [37], [17]. A DAG is defined mathematically as a graph pair $G = (V, E)$, in which V is the set of vertices representing the random variables in the credit model, and E is the set of directed edges expressing the relationships of direct dependence between pairs of variables. The acyclicity property plays a foundational role, ensuring that if one starts from any node and moves along the direction of the arrows, there never exists a closed path in the system returning to the initial node. In the credit risk prediction problem, this property completely eliminates causal loops that are illogical in economic terms, while also establishing a clear structural ordering among the influencing factors, for example from demographic characteristics and financial history to repayment behaviour and finally to the state of default (Liu et al., 2024) [42]. Applying Bayesian networks to credit risk prediction requires an extremely rigorous model construction procedure that transforms raw financial datasets into a system of causal knowledge capable of mathematical inference. According to Liu et al. (2024) [42], the procedure for building the BN_Pro model for the credit default prediction problem comprises three core stages: structure learning, estimation of the parameter space, and deployment of the probabilistic inference algorithm. These are three architectural layers tightly linked to one another. The structure learning stage aims to discover the dependency relationships among the variables in the data; the parameter learning stage quantifies the degree of that dependence through conditional probabilities; and the inference stage uses all the knowledge that has been learned to compute the probability of the target event occurring, such as the likelihood that a customer defaults (Default = 1). The organic

combination of these three stages fundamentally resolves the barrier of computational complexity that always exists in multidimensional data spaces, while maximising the explanatory capacity of the model in the face of complex credit-granting decisions. The learning process of a Bayesian network is not merely the optimisation of a predictive function as in Logistic Regression or Gradient Boosting, but the construction of a complete probabilistic model for the entire data system. This forms the foundation for interpretability, sensitivity analysis, counterfactual analysis and causal inference in modern financial research [50], [42], [52].

3.2.1 Network Structure Learning

The process begins with determining the network structure, which the academic community has established as an NP-hard problem owing to the exponential explosion of the space of Directed Acyclic Graphs as the number of feature variables increases [37], [34]. To overcome this computational barrier, instead of using exhaustive search methods over the entire space, which are infeasible in terms of resources, the structure learning process gives priority to heuristic search algorithms for approximating the globally optimal solution. Within the framework of this study, the Tabu Search algorithm proposed by Glover (1986) [37] is deployed as a breakthrough local search tool, combined with the Bayesian information criterion in order to shape a causal network that both fits the data closely and remains structurally parsimonious [42].

The essence of the Tabu Search algorithm is that it fundamentally overcomes the fatal weakness of ordinary hill climbing algorithms, namely becoming trapped at local optima. The algorithm operates by continuously traversing neighbouring graphs through spatial transformations such as adding an edge, deleting an edge or reversing the direction of an arrow. The core difference is the mechanism of maintaining a Tabu list that acts as a short-term memory; when a network structure or a transformation operation has just been evaluated, it is placed on this list and forbidden from being repeated for a certain number of iterations [65]. This enforcement mechanism prevents the algorithm from returning to previous moves, compelling the system to cross the valleys of the descending function in order to escape the local optimum trap and explore new regions of the structure space, thereby approaching the globally optimal structure [42].

To evaluate the quality of each neighbouring structure during the traversal, the BIC score function is integrated as the ultimate objective function [37]. The BIC score acts as a balancing mechanism, resolving the strict trade-off between the goodness of fit of the model (measured by the residual sum of squares or the Likelihood function) and the complexity of the model (measured by the number of free parameters) [42]. By imposing a logarithmic penalty on the network size and the number of samples, the BIC criterion effectively prevents overfitting and eliminates edges that cause an unnecessary explosion of parameters.

$$BIC = \log P(D|\hat{\theta}, G) - \frac{\log N}{2} \cdot k, \quad (3.5)$$

where:

- D : the training data.
- G : the DAG structure.
- $\hat{\theta}$: the optimal parameter set of the network.
- $\log P(D|\hat{\theta}, G)$: the Log-Likelihood, assessing the ability of the DAG to explain the data
- N : the number of samples.
- k : the number of parameters to be estimated.

with $\frac{\log N}{2} \cdot k$ being the penalty coefficient for model complexity.

This entire discovery process is placed within a meta-loop of 50 independent network initialisations and training runs in order to eliminate the risk of bias due to the initial starting point, ensuring that the final causal structure reflects the mechanism generating credit risk as faithfully and robustly as possible [42].

$$G^* = \arg \max_{r=1, \dots, 50} BIC_r, \quad (3.6)$$

where:

- G^* : the optimal DAG structure selected.
- $r = 1, \dots, 50$: the index of the models/DAG structures tested.
- BIC_r : the BIC score of the r -th model.

The construction and adjustment process is described in Algorithm 3.

Algorithm 3 Pseudocode of the BIC-based Tabu Search algorithm

Initialisation: A discrete Bayesian network in DAG form. Generate a random initial DAG as the current solution. Set the current best solution = the current solution = the DAG. Initialise the Tabu list and the BIC score.

Output: Return the Bayesian network structure with the highest BIC score.

```

1: while (the stopping criterion has not been met) do
2:   Generate feasible neighbouring network structures.
3:   for (each feasible neighbouring solution  $N$ ) do
4:     if  $N$  is not on the Tabu list then
5:       Compute the evaluation index  $Q(N)$  of the neighbouring solution
        $N$  using the BIC score.
6:       if  $Q(N) > Q(\text{the current optimal solution})$  then
7:         Update the current solution =  $N$ .
8:       end if
9:     end if
10:  end for
11:  Select the neighbouring solution with the largest  $Q(N)$  as the new current
  solution.
12:  Add the current solution to the Tabu list.
13:  Decrease the Tabu memory of the solutions already in the Tabu list by
  one unit.
14: end while
15: Repeat the Tabu Search algorithm several times with different initial DAGs.
16: Return the DAG with the highest BIC score.

```

3.2.2 Parameter Estimation

After the structure learning process has been completed and the directed acyclic graph determined, the Bayesian network has so far only represented the relationships of probabilistic dependence among the variables in the form of a graph structure. The DAG by itself, however, is not sufficient for performing probabilistic inference or predicting credit risk. For the Bayesian network to become a complete probabilistic model, as soon as the DAG skeleton has been firmly established, the system enters the parameter learning stage in order to quantify the causal space [17]. The central task is to determine the Conditional Probability Tables for each network node, thereby delineating the degree of statistical dependence among the financial variables [34]. This process operates on the basis of Maximum Likelihood Estimation (MLE), a canonical mathematical foundation that allows the observed frequencies from the historical dataset to be transformed directly into predictive probability distributions [42].

According to the Bayesian network theory presented by Pearl [50] and by Koller and Friedman [37], each edge in the DAG reflects only the existence of a relationship of

probabilistic dependence between two variables, while the strength and quantitative nature of that relationship is encoded through the probability parameters in the CPT. Hence, if structure learning answers the question “which variable influences which”, parameter learning answers the question “how great the influence is” [50], [37].

Consider a Bayesian network comprising a set of n random variables: $\mathbf{X} = (X_1, X_2, \dots, X_n)$ with a known DAG structure. According to the probability decomposition rule of Bayesian networks [50], [37], [17]:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i)), \quad (3.7)$$

where:

- X_i is the i -th variable in the network;
- $Pa(X_i)$ denotes the set of direct parent nodes of X_i ;
- $P(X_i | Pa(X_i))$ is the probability of the variable X_i given the states of its parent variables in the DAG.

The expression above shows that the joint probability distribution of the entire system is decomposed into a product of local probability distributions.

In the credit risk prediction problem, this means that a customer’s probability of default is not modelled directly by a single global function, but is derived from the combination of many conditional probabilities among the financial variables, credit behaviour, borrowing history and the target variable Default [42], [45], [39].

Conditional Probability Table

Suppose the variable X_i has s_i different states. Its parent set $Pa(X_i)$ has q_i value configurations. The parameter set to be learned is then:

$$\theta = \{\theta_{ijk} | i = 1, \dots, n; j = 1, \dots, q; k = 1, \dots, s_i\}$$

with:

$$\theta_{ijk} = P(X_i = k | Pa(X_i) = j), \quad (3.8)$$

where:

- i : the variable under consideration.
- j : the j -th configuration of the parent set.
- k : the k -th state of the variable.

According to Jensen and Nielsen [34], each row in the CPT must satisfy the

normalisation constraint:

$$\sum_{k=1}^{s_i} \theta_{ijk} = 1$$

for all:

$$j = 1, \dots, q_i$$

This ensures that the total probability of all possible states of a variable always equals 1.

The Likelihood Function

Analytically, the algorithm seeks the optimal parameter set such that the likelihood function attains its maximum value. In order to handle the enormous computational load, the algorithm converts the likelihood function into logarithmic form (the log-likelihood), turning complex products of simultaneous probabilities into a series of linear additions [37], [42]. By scanning the entire dataset, counting the number of samples satisfying the joint condition between the current node and its parent set, and applying the method of Lagrange multipliers to resolve the mathematical constraints, the system fills in the CPT with precisely the optimal probability percentages [42]. The resulting parameter grid not only describes the structure of the training data but also becomes the core quantitative knowledge base for every subsequent inference query [17], [42].

Suppose the training dataset:

$$D = d_1, d_2, \dots, d_m$$

consists of m independent observations.

According to Koller and Friedman [37], the likelihood function is defined as:

$$L(\theta|D) = P(D|\theta) = \prod_{l=1}^m P(d_l|\theta), \quad (3.9)$$

where:

- $L(\theta|D)$: the likelihood function, representing the probability of observing the data D given the parameters θ .
- D : the training dataset.
- θ : the parameter set of the Bayesian network.
- $P(d_l|\theta)$: the probability of the l -th data sample given the parameter set θ .
- m : the total number of data samples in the training set.

The objective of MLE is to find the parameter set:

$$\theta^* = \arg \max_{\theta} L(\theta|D)$$

that is, the parameter set that makes the observed data most probable.

Because the likelihood function is a product of a great many small probabilities, it is prone to arithmetic underflow. Studies on Bayesian networks therefore all switch to the log-likelihood [37], [17], [42].

Let m_{ijk} denote the number of samples satisfying: $X_i = k$ and $Pa(X_i) = j$

Then:

$$\begin{aligned} \ell(\theta|D) &= \log L(\theta|D) \\ &= \log \prod_{l=1}^m P(d_l|\theta) \\ &= \sum_{i=1}^n \sum_{j=1}^{q_i} \sum_{k=1}^{s_i} m_{ijk} \log(\theta_{ijk}), \end{aligned} \tag{3.10}$$

where:

- $\ell(\theta|D)$: the log-likelihood function of the Bayesian network with parameter set θ and data D .
- n : the number of variables in the Bayesian network.
- q_i : the number of states of the parent set of the variable X_i .
- s_i : the number of states of the variable X_i .
- m_{ijk} : the number of times the variable X_i is observed in state k while the parent set is in state j .
- θ_{ijk} : the conditional probability of the variable X_i being in state k given the j -th parent state.

In order to maximise the log-likelihood function $\ell(\theta|D)$ in formula 3.10, the probability normalisation constraint must be satisfied for each node X_i and each combination of parent states j :

$$\sum_{k=1}^{s_i} \theta_{ijk} = 1, \quad \forall i \in \{1, \dots, n\}, \quad \forall j \in \{1, \dots, q_i\} \tag{3.11}$$

This is an optimisation problem with equality constraints, and the method of Lagrange multipliers is therefore applied. The Lagrangian function is constructed as

follows:

$$\mathcal{L}(\theta, \lambda) = \sum_{i=1}^n \sum_{j=1}^{q_i} \sum_{k=1}^{s_i} m_{ijk} \log(\theta_{ijk}) + \sum_{i=1}^n \sum_{j=1}^{q_i} \lambda_{ij} \left(1 - \sum_{k=1}^{s_i} \theta_{ijk} \right) \quad (3.12)$$

where λ_{ij} is the Lagrange multiplier corresponding to the constraint at node X_i and the j -th combination of parent states.

Taking the partial derivative of $\mathcal{L}$ with respect to θ_{ijk} and setting it equal to 0:

$$\frac{\partial \mathcal{L}}{\partial \theta_{ijk}} = \frac{m_{ijk}}{\theta_{ijk}} - \lambda_{ij} = 0 \implies \theta_{ijk} = \frac{m_{ijk}}{\lambda_{ij}} \quad (3.13)$$

Substituting into the normalisation constraint to determine λ_{ij} :

$$\sum_{k=1}^{s_i} \theta_{ijk} = \sum_{k=1}^{s_i} \frac{m_{ijk}}{\lambda_{ij}} = \frac{1}{\lambda_{ij}} \sum_{k=1}^{s_i} m_{ijk} = 1 \implies \lambda_{ij} = \sum_{k=1}^{s_i} m_{ijk} = m_{ij.} \quad (3.14)$$

where $m_{ij.} = \sum_{k=1}^{s_i} m_{ijk}$ is the total number of observations whose parent set is in state j , regardless of the state of X_i .

Substituting back, the explicit solution of the MLE problem is:

$$\boxed{\hat{\theta}_{ijk} = \frac{m_{ijk}}{m_{ij.}}} \quad (3.15)$$

This solution has a clear intuitive meaning: the optimal conditional probability of X_i being in state k when the parent set is in state j is precisely the relative frequency observed in the training data. This is the mathematical basis on which the MLE class in the pgmpy library fills in the values of the CPT: instead of iterative optimisation, it need only count the frequency of occurrence of each combination of states ($X_i = k, Pa(X_i) = j$) in the training set and compute the ratio. This is the central formula of the entire parameter learning process in Bayesian networks [37], [42].

3.2.3 Initialising the Inference Mechanism and Markov Blanket Analysis

Immediately after the complete Bayesian network has been built, an exact inference engine based on the variable elimination algorithm is initialised in conjunction with the model, serving as the sole interface through which all probabilistic queries in the In_Pro phase will pass. Before exploiting the inference mechanism for prediction and

explanation purposes, a preliminary sanity check is carried out: the unconditional prior probability $P(\text{target} = 1)$ is queried directly from the Variable Elimination engine and compared against the empirical prior probability `prior_default` computed directly from the frequencies in the training data. Agreement between these two values (with a discrepancy smaller than a conventionally agreed tolerance threshold) is a necessary (though not sufficient) condition for asserting that the graph structure and the learned Conditional Probability Distributions (CPD) correctly encode the latent dependency relationships in the original data, since if inference over the network cannot exactly reproduce the marginal distribution of the target variable that was used to learn those parameters, there is a high likelihood of an error in constructing the graph or in estimating the CPT. In parallel with this validation step, an important structural analysis is performed, namely determining the Markov Blanket set of the target variable, denoted $MB(\text{target})$, which is formally defined as the union of three subsets: the set of direct parents (Parents), the set of direct children (Children), and the set of parents of the children (Parents of Children, also known as the set of “spouses”):

$$MB(\text{target}) = \text{Parents}(\text{target}) \cup \text{Children}(\text{target}) \cup \text{Parents}(\text{Children}(\text{target}))$$

According to Pearl’s definition [50] within the framework of Bayesian networks, the core theoretical significance of the Markov Blanket is the property of conditional independence separation: once the states of all the variables in $MB(\text{target})$ are known, the target variable becomes conditionally independent of every remaining variable in the network that does not belong to the Markov Blanket

$$\text{target} \perp (\text{all other variables}) \mid MB(\text{target})$$

The direct practical consequence of this property is that the Markov Blanket is precisely the minimal and sufficient feature set required to predict the target variable accurately within the framework of the learned model: adding any variable outside $MB(\text{target})$ to the evidence set of an inference query will not contribute any additional marginal information for estimating $P(\text{target} \mid \text{evidence})$, because the entire influence of those variables (if any) has already been transmitted and fully “absorbed” through the intermediate variables belonging to the Markov Blanket. This is the theoretical basis for the study’s use of the Markov Blanket as a tool for structural explanation, running in parallel with the specific probabilistic inference analyses in the In_Pro phase.

3.3 Interpretation of Results (In_Pro)

Successfully constructing a Bayesian network does not stop at prediction; the graph structure must also be fully exploited in order to explain the mechanism by which risk is formed. The Interpretation Process (In_Pro) block serves as a bridge between the computational power of machine learning and the human need to understand root causes. This stage operates mainly on two core foundations: the mechanism of multidimensional

probabilistic inference and the technique of sensitivity analysis combined with mutual information.

3.3.1 Exact Inference Using the Variable Elimination Algorithm

Once the Bayesian network has been completed in terms of both its qualitative geometric structure and its quantitative parameter system, the next core problem is to exploit the model in order to produce scientific judgements when real-world information appears. This process is called probabilistic inference, in which the system updates the probability distribution of one or a group of target variables given the specific known states of another group of observed variables, which serve as empirical evidence (Liu et al., 2024) [42]. Within the scope of rigorous mathematical methods, exact inference is the approach that computes posterior probability values that are absolute and analytically precise, rather than using stochastic approximation techniques that carry uncertainty. The results of exact inference ensure that every risk decision made rests on a purely mathematical logical foundation, unaffected by simulation error.

In the context of credit risk prediction research, the ultimate goal is not to determine the probability of each individual variable but to infer the probability of a customer defaulting given an observed set of specific credit information. This is equivalent to computing the posterior probability:

$$P(\text{Default} = 1 \mid X_1 = x_1, X_2 = x_2, \dots, X_m = x_m) \quad (3.16)$$

in which the set of observed variables serves as the evidence, while the variable Default serves as the query variable [42].

According to Pearl (2009) [50], the inference process in a BN is precisely the process of belief updating when new information appears. Mathematically, this is carried out through Bayes' theorem combined with the probability decomposition structure of the network. This form of decomposition is the foundation that allows a BN to handle complex inference problems by exploiting conditional independence. This is precisely the important advantage that makes BNs suitable for the credit risk assessment problem, in which factors such as income, credit history, financial behaviour and loan status are mutually dependent [50], [37], [17].

Although the probabilistic representation becomes more efficient, performing exact inference on a Bayesian network remains a problem of very high computational complexity. Cooper (1990) [14] proved that the problem of exact probabilistic inference on a BN belongs to the NP-hard class. This means that the computational cost may grow exponentially as the number of variables or the density of connections in the network increases. For this reason, the choice of an appropriate inference algorithm plays a decisive role in the practical applicability of Bayesian networks in large-scale credit risk prediction systems.

In the study by Liu et al. [42], as well as in the model proposed in this thesis, the Variable Elimination (VE) algorithm is chosen as the exact inference mechanism for computing a customer's probability of default from the learned Bayesian network structure.

The Variable Elimination Algorithm

VE is one of the classic exact inference algorithms for Bayesian networks, developed on the ideas of non-sequential dynamic programming and the technique of decomposing the inference problem into local operations on probabilistic factors [17], [34], [18].

The core idea of the algorithm is to avoid computing the entire joint probability distribution of the network directly. Instead, VE takes advantage of the graph structure and the relationships of conditional independence to carry out probability additions and multiplications in an optimal order so as to reduce the number of operations that must be performed [17].

Suppose we need to compute:

$$P(Y|E = e)$$

where Y is the query variable and E is the set of evidence variables. By the definition of posterior probability:

$$P(Y | E = e) = \frac{P(Y, E = e)}{P(E = e)} = \frac{\sum_z P(Y, E = e, Z = z)}{\sum_{y,z} P(Y = y, E = e, Z = z)}, \quad (3.17)$$

where:

- $P(Y | E = e)$: The posterior probability of the target state variable Y (for example, the likelihood of default) after the evidence set $E = e$ has been observed.
- Y, y : The target variable to be queried and its specific state.
- E, e : The set of evidence variables in the Bayesian network and the specific values actually recorded (for example, the customer has a low income or a history of late payment).
- Z, z : The set of intermediate variables belonging neither to the target group Y nor to the evidence group E , and their state space.
- $\sum_z$: The marginalisation summation over the entire state space of Z , in order to eliminate the influence of these intermediate variables from the query result.
- $\sum_{y,z}$: The summation over the entire state space of both Y and Z in order to compute the marginal probability $P(E = e)$ in the denominator, which acts as a normalising constant ensuring that the total probability equals 1.
- $P(E = e)$ is the probability of the evidence occurring

- $P(Y, E = e)$ is the joint probability of the query variable and the evidence.

If the summation is performed directly over the entire state space of Z ($O(k^n)$), where k is the average number of states per variable and n is the number of hidden variables, the computational cost will grow exponentially. VE resolves this problem by changing the order of the multiplication and summation operations:

$$\sum_Z \prod_i f_i$$

into a sequence of local operations:

$$\sum_{Z_k} (f_1 \cdot f_2 \cdots f_m)$$

such that only one intermediate variable Z_k is removed from the system at a time.

It is precisely this interchange of the order of multiplication and summation that allows VE to reduce inference complexity considerably compared with the brute-force method [17], [18], [48]. Algorithm 4 below is the pseudocode of the VE algorithm.

Algorithm 4 Pseudocode of the variable elimination inference algorithm

Input: Bayesian network BN .

Output: Return the posterior probability $P(Y|E = e)$.

- 1: Initialise the VE inference engine with BN .
 - 2: Set the query variable Y and the evidence $E = e$.
 - 3: Extract the factors F from BN .
 - 4: **while** hidden variables remain **do**
 - 5: Select a variable X from the elimination order ρ .
 - 6: Multiply the factors containing X .
 - 7: Eliminate X by summation.
 - 8: **end while**
 - 9: Combine the remaining factors to obtain $h(Y)$.
 - 10: Return $P(Y|E = e) = h(Y) / \sum_Y h(Y)$.
-

A Bayesian network represents the joint distribution as a product of local CPDs: $P(X_1, \dots, X_n)$. From the computational standpoint of the VE algorithm, each of these CPDs is regarded as a factor, that is, a function defined over a subset of variables, mapping each combination of states of those variables to a non-negative probability value. A factor $f(A, B, C)$ need not be a normalised probability distribution; it need

only be a table of values capable of taking part in multiplication and aggregation operations.

1. Factor multiplication

When two factors share one or more variables, for instance $f_1(A, B)$ and $f_1(B, C)$, factor multiplication produces a new factor over the union of the participating variables:

$$f(A, B, C) = f_1(A, B) \times f_2(B, C) \quad (3.18)$$

where, for each specific combination of states of (A, B, C) , the value of the resulting factor equals the product of the corresponding values of the two component factors at the shared state combination of the variable B . This operation is the basic computational building block that allows the local joint distribution to be reconstructed gradually from the initial discrete CPDs, without explicitly constructing the entire joint distribution table over all variables.

2. Variable elimination

Instead of computing the summation directly over the entire state space of the hidden variables, the VE algorithm marginalises the hidden variables one at a time, immediately after multiplying the factors related to that variable, according to the distributive law of addition over multiplication:

$$\sum_{\text{hidden}} \prod_i f_i \Rightarrow \sum_{\{Z_1\}} \left(\sum_{\{Z_2\}} (\dots (f_1 \times f_2 \times \dots \times f_m) \dots) \right) \quad (3.19)$$

Specifically, for each variable X_k to be eliminated in the predetermined elimination order $\rho = (X_1, X_2, \dots, X_k)$, the algorithm groups together all the existing factors that still contain X_k , multiplies them into an intermediate factor, then sums out X_k from this intermediate factor to obtain a new factor no longer containing X_k , and replaces the old factors with this new factor in the set of factors being processed. The process is repeated until all hidden variables have been eliminated, leaving factors containing only the query variable Y , which are multiplied together to obtain $h(Y)$ and finally normalised into a valid posterior distribution:

$$P(Y|E = e) = \frac{h(Y)}{\sum_Y h(Y)} \quad (3.20)$$

The complexity advantage of VE over brute-force enumeration lies in the fact that the aggregation order is arranged so that each aggregation operation acts only on small local factors rather than on the global joint distribution table. However, the computational complexity of VE depends directly on the treewidth of the graph corresponding to a particular elimination order, and finding the elimination order that minimises treewidth is itself an NP-hard problem. In practice, pgmpy applies heuristics

for selecting the elimination order based on the smallest node degree (the min-degree heuristic) in order to approximate an efficient elimination order, and with the credit network structure used in this research, the computational cost of VE remains within feasible limits for the repeated queries required by sensitivity and what-if analysis.

3.3.2 Probabilistic Inference in Bayesian Networks

Forward Inference

The core essence of the interpretive power of Bayesian networks lies in their capacity for probabilistic inference, which operates on the firm foundation of Bayes' theorem. Unlike closed algorithms that establish only a one-directional mapping from input to output, the Bayesian network uses the organic relationship between the joint probability distribution function and the conditional probability distribution functions to carry out multidirectional inference. In terms of graph representation, each node in the network stands for a random variable, while the directed edges connecting the nodes depict directly the relationships of conditional dependence among these variables. Accompanying that geometric structure, each node possesses a conditional probability distribution table, which describes quantitatively the shift in the probability distribution at that node when the values of its dependent variables (the parent variables) change. This decomposition allows the system to represent the joint probability of the entire variable space $X = \{X_1, X_2, \dots, X_n\}$ through a product of local probabilities according to the formula:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i)) \quad (3.21)$$

On the basis of that architecture, the forward inference mechanism is defined as the process of computing the posterior probability distribution of a target variable given evidence provided by one or more feature variables $P(Y|E = e)$.

Considered in the experimental context of the credit-granting problem, by learning from the historical dataset of individuals who have previously taken out loans, the model can establish the structure and the intrinsic conditional probability distributions among the feature nodes, for example computing the probability that an individual has a borrowing need and a risk of default under the constraining conditions of their current income level and their past credit rating score [42]. As soon as the system receives an entirely new application for a personal credit limit, the forward inference process is immediately activated. The empirical values of that individual's income and credit score are assigned directly into the network as evidence variables. Through the Bayesian inference engine, the system propagates probabilities along the directed edges of the graph, thereby automatically computing and outputting the posterior probability distribution expressing that individual's borrowing need as well as their potential default rate [48]. This process is not a mere approximation of a function but a circulation and

updating of belief entirely grounded in the laws of statistical probability.

Backward Inference

In parallel with forward inference, the Bayesian network also provides a superior mathematical basis for backward inference, otherwise known as diagnostic reasoning. In this direction of inference, the algorithm computes the posterior probability distribution of one or more feature variables (causes) once the specific state of one or more target variables (effects) has been observed [50]. For instance, when the system records a profile that has genuinely fallen into the state of default, backward inference will diagnose and reconstruct the probability of occurrence of the income levels or credit behaviours that led to that outcome.

Suppose:

- Y : the target variable, which in this study is Default.
- X : the feature variable to be inferred (for example, Income, CreditScore, etc.).

Given that the customer has defaulted ($Y = 1$), the posterior probability of the feature variable is determined by:

$$P(X|Y = 1) = \frac{P(Y = 1|X) P(X)}{P(Y = 1)}, \quad (3.22)$$

where:

- $P(X|Y = 1)$: the posterior probability of the variable X given that the customer has defaulted.
- $P(Y = 1|X)$: the probability that the customer defaults given the value of X .
- $P(X)$: the prior probability of the variable X .
- $P(Y = 1)$: the prior probability of the target variable, acting as a normalising constant.

The ability to combine forward inference (to predict the future) and backward inference (to diagnose causes) seamlessly is precisely the theoretical reason why Bayesian networks have become the gold standard in artificial intelligence systems specialising in probabilistic inference and financial decision-making [17], [42]. By continuously adding and updating new evidence in the observation space, the graph continuously refines the probability distribution curves, thereby sharpening the precision and accuracy of credit judgements in subsequent cycles.

3.3.3 Sensitivity Analysis Method

In order to measure quantitatively and to isolate the mechanism of influence within the network structure, sensitivity analysis is deployed as a key posterior evaluation tool. Within the Bayesian network ecosystem, sensitivity analysis is usually decomposed into two specialised lines of reasoning: structural sensitivity analysis and parameter sensitivity analysis [35], [55]. On the structural analysis side, the degree of influence on the target variables is assessed by directly adjusting the network architecture: researchers actively intervene in the geometric skeleton of the DAG through operations such as grafting on, or eliminating, nodes and links (edges). After each structural transformation, the algorithm recomputes the posterior probability distribution and compares it with the original network in order to measure the discrepancy; the larger the discrepancy, the more it demonstrates the acute sensitivity of the target to that structural change. Parameter sensitivity analysis, by contrast, focuses on preserving the graph architecture while actively intervening in the intrinsic parameters of the network, specifically the values inside the conditional probability distribution table.

Within the scope of this study, the focus of the analysis is placed entirely on parameter sensitivity analysis, in order to exploit to the fullest the local interpretability of the model with respect to credit scenarios. By applying the control variable method thoroughly, the researcher can actively change the value of a single input variable (for example, the borrower's income level) while freezing all the remaining variables. By repeating this trial many times and recording the continuous change in the probability of default, the model quantifies precisely the magnitude of that feature's influence on overall risk. The variable that produces the largest shift in probability is precisely the variable with the highest sensitivity.

Suppose the input variable X changes from the value x_1 to x_2 ; the degree of change in the posterior probability is determined by:

$$\Delta P = P(Y = 1|X = x_2) - P(Y = 1|X = x_1), \quad (3.23)$$

where:

- ΔP is the change in the probability of default;
- X is the variable being analysed;
- x_1, x_2 are two states of the variable.

The larger $|\Delta P|$ is, the more sensitive that variable is for the predictive model.

Successfully locating the nodes of maximum sensitivity not only provides management with a solid basis for operating the system safely and stably, but also guides purposeful risk intervention measures. At the same time, the results of this process can be used to optimise the model directly by readjusting the weights of the variables or by

deciding to collect additional new variables with higher explanatory potential [41], [7].

To provide a powerful quantitative mathematical basis for determining sensitivity and the degree of association, the study applies the theory of mutual information. In a Bayesian network, mutual information quantitatively expresses the degree of correlation and information sharing between two random variables. Measuring the mutual information between network nodes helps to indicate whether the conditional probability distributions between them are mutually constrained [48], [16].

The mathematical formula expressing the mutual information between the feature variable X and the target variable Y is defined as the integral (or the sum, for discrete variables) of the joint distribution function multiplied by the logarithm of the ratio between the joint distribution and the product of the marginal distributions:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right) \quad (3.24)$$

If the mutual information between two nodes equals $I(X; Y) = 0$, this is absolute mathematical proof that the two nodes are entirely conditionally independent in the context of the neighbouring nodes already determined. By analysing the mutual information spectrum of the target node, analysts can distil and rank precisely the degree of association between the input feature nodes and the danger of default, thereby eliminating noisy features and establishing an optimal inference space that is extremely convincing for the predictive results.

3.4 Evaluation of Predictive Performance

3.4.1 Receiver Operating Characteristic Curve

In the binary classification problem, the Receiver Operating Characteristic curve (ROC), or ROC curve, is a graphical representation illustrating the performance of a binary classification system as its discrimination threshold is varied continuously over the entire range of possible values. The ROC curve is created by plotting the ratio of true positives to the total number of actual positive samples (TPR) against the ratio of false positives to the total number of actual negative samples (FPR) at different threshold settings [60]. In the context of credit risk prediction, positive is usually taken by convention to be the case of a customer defaulting, while negative is the case of a customer repaying on time (non-default).

The classification model takes as input the customer's financial features and produces a risk score s , expressing the degree to which a default event is likely to occur. Based

on a classification threshold $c \in (0, 1)$, customers are classified into two groups:

$$\hat{Y} = \begin{cases} 1, & \text{if } s \geq c \\ 0, & \text{if } s < c \end{cases}, \quad (3.25)$$

where:

- $\hat{Y} = 1$: the customer is predicted by the model to be at risk of default (Default).
- $\hat{Y} = 0$: the customer is predicted by the model not to default (Non-default).
- c : the classification threshold chosen to convert the continuous risk score into a binary classification decision.

After comparing the model's predictions with the customers' actual states, we obtain a Confusion Matrix comprising four cases:

- TP (True Positive): the number of customers who actually default and are correctly predicted by the model as defaulting.
- TN (True Negative): the number of customers who do not actually default and are correctly predicted by the model as non-defaulting.
- FP (False Positive): the number of customers who do not actually default but are incorrectly predicted by the model as defaulting.
- FN (False Negative): the number of customers who actually default but are incorrectly predicted by the model as non-defaulting.

From the quantities above, the two important components constituting the ROC curve are determined to be sensitivity and the false positive rate (FPR).

Sensitivity, or the rate of correctly detecting defaulting customers, is denoted $Se(c)$ or $TPR(c)$:

$$TPR(c) = Se(c) = \frac{TP(c)}{TP(c) + FN(c)} \quad (3.26)$$

This quantity reflects the model's ability to identify customers with high credit risk accurately. The larger the value of $TPR(c)$, the fewer potentially defaulting customers the model overlooks.

Conversely, specificity expresses the ability to identify the group of non-defaulting customers accurately:

$$Sp(c) = \frac{TN(c)}{TN(c) + FP(c)} \quad (3.27)$$

From specificity, the rate at which good customers are wrongly predicted to be risky customers is determined by:

$$FPR(c) = 1 - Sp(c) \quad (3.28)$$

or:

$$FPR(c) = \frac{FP(c)}{FP(c) + TN(c)}, \quad (3.29)$$

where:

- $FPR(c)$ expresses the proportion of customers who do not default but are misclassified by the model as defaulting at the threshold c .
- $Sp(c)$ reflects the model's ability to correctly rule out non-risky cases.

Suppose the risk score output by the model has the corresponding probability density functions:

$$f(s|Y = 1) \quad \text{and} \quad f(s|Y = 0),$$

where:

- $f(s|Y = 1)$: the probability density function of the risk score for the group of customers who actually default.
- $f(s|Y = 0)$: the probability density function of the risk score for the group of customers who do not actually default.
- s : the risk score predicted by the model.

With a classification threshold c , a customer is predicted to default if: $s \geq c$. The probability that the model correctly detects a defaulting customer is then expressed as:

$$TPR(c) = Se(c) = P(s \geq c|Y = 1) \quad (3.30)$$

or:

$$TPR(c) = \int_c^{\infty} f(s|Y = 1)ds \quad (3.31)$$

Similarly, the probability that the model misclassifies a non-defaulting customer as a defaulting one is determined as:

$$FPR(c) = P(s \geq c|Y = 0) \quad (3.32)$$

or:

$$FPR(c) = \int_c^\infty f(s|Y = 0)ds \quad (3.33)$$

Thus, for each different threshold value c , the model produces a pair of values:

$$(FPR(c), TPR(c))$$

When c is allowed to vary continuously over the entire range of the score s , the set of these points forms a curve in the two-dimensional space $[0, 1] \times [0, 1]$. The resulting curve is precisely the ROC curve.

Unlike overall accuracy, the ROC curve does not depend on the prevalence of the default class in the dataset — a key property, since real-world credit data is usually severely imbalanced [11], with defaulting customers typically accounting for only a small fraction of the whole. At the same time, the ROC curve allows researchers to evaluate classification performance across the entire range of decision thresholds, thereby helping credit institutions to adjust their credit-granting threshold flexibly according to their specific risk appetite, for instance prioritising the minimisation of missed defaulting customers (increasing Se) or prioritising the minimisation of wrongful rejection of good customers (increasing Sp).

3.4.2 Area Under the ROC Curve

While the ROC curve provides a comprehensive visual representation of classification performance across all thresholds, it does not directly yield a single aggregate figure for quantitative comparison between models. To address this limitation, the Area Under the ROC Curve (AUC) is introduced as a single scalar measure reflecting the entire discriminative capacity of a classifier across all possible thresholds. The area under a receiver operating characteristic curve, abbreviated AUC, is a single scalar numerical value measuring the overall performance of a binary classifier [31].

The landmark theoretical contribution to the statistical interpretation of AUC belongs to Hanley and McNeil (1982) [31], who proved an important result: AUC is equivalent in value to the Mann–Whitney U test (also known as the Wilcoxon rank-sum statistic). Given a set of positive samples P and negative samples N , AUC equals the proportion of all (positive, negative) pairs in which the classifier assigns a higher score to the positive sample. This interpretation gives AUC a highly intuitive probabilistic meaning of great practical value: AUC is precisely the probability that the test result of a subject chosen at random from the diseased group (or the defaulting group) exceeds that of a subject chosen at random from the non-diseased group (or the non-defaulting group) [5]. Applied to the credit risk problem, this means that AUC reflects the probability

that the model assigns a higher risk score to a randomly chosen defaulting customer than to a randomly chosen non-defaulting customer, making it a criterion for evaluating the ranking ability of the model independently of the choice of any particular threshold.

AUC is defined as the integral of the TPR function with respect to FPR over the entire domain $[0, 1]$:

$$AUC = \int_0^1 TPR(FPR^{-1}(x))dx = P(S_1 > S_0), \quad (3.34)$$

where:

- S_1 : the risk score assigned by the model to a customer chosen at random from the defaulting group;
- S_0 : the risk score assigned by the model to a customer chosen at random from the non-defaulting group, independent of S_1 ;
- $P(S_1 > S_0)$: the probability that a defaulting customer receives a higher score than a non-defaulting customer.

In practice, with finite data, AUC is usually estimated through the formula equivalent to the Mann–Whitney U statistic that Hanley and McNeil showed to be:

$$\widehat{AUC} = \frac{U_1}{n_1 \cdot n_0} = \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \psi(s_i, s_j) \quad (3.35)$$

with:

$$\psi(s_i, s_j) = \begin{cases} 1, & \text{if } s_i > s_j \\ 0.5, & \text{if } s_i = s_j \\ 0, & \text{if } s_i < s_j, \end{cases} \quad (3.36)$$

where:

- n_1 : the number of defaulting customers (positive samples) in the test dataset;
- n_0 : the number of non-defaulting customers (negative samples) in the test dataset;
- s_i : the risk score of the i -th defaulting customer ($i = 1, \dots, n_1$);
- s_j : the risk score of the j -th non-defaulting customer ($j = 1, \dots, n_0$);
- U_1 : the Mann–Whitney U statistic computed over the positive–negative sample pairs;
- $\psi(s_i, s_j)$: the indicator function comparing each pair of scores, taking the value 1 if the pair is correctly ordered (a concordant pair), 0.5 in the case of a tie, and 0 if incorrectly ordered (a discordant pair).

When the data is represented as discrete points on the empirical ROC curve $(FPR_k, TPR_k)_{k=0}^m$, AUC can also be computed approximately by the trapezoidal rule:

$$\widehat{AUC} = \sum_{k=1}^m \frac{(TPR_k + TPR_{k-1})}{2} \times (FPR_k - FPR_{k-1}) \quad (3.37)$$

Here m is the number of discrete threshold points used to construct the empirical ROC curve. The AUC value lies within the interval $[0.5, 1.0]$, where the minimum value represents the performance of a random classifier and the maximum value corresponds to a perfect classifier [31]. Regarding estimation accuracy, Hanley and McNeil (1982) also proposed a closed-form analytical formula for computing the variance of AUC based on the AUC value, the number of positive samples and the number of negative samples, although bootstrap resampling is usually preferred in modern experimental work because it provides better coverage for extreme AUC values [31].

In research, AUC serves as the most common benchmark criterion for comparing performance between different models, from classical statistical models such as linear discriminant analysis [3] and logistic regression [49] to modern machine learning models such as random forests [9], support vector machines [71], [61] and ensemble methods [8], [26]. Because AUC is independent of the choice of any particular threshold, it allows an objective assessment of the model's intrinsic discriminative capacity before the researcher must make a decision about the optimal classification threshold for practical deployment.

3.4.3 Criteria for Finding the Optimal Classification Threshold

Although AUC provides a comprehensive measure of the model's discriminative capacity across the entire range of thresholds, in operational practice credit institutions are obliged to select a specific decision threshold c in order to classify customers as granted credit or rejected/placed under risk monitoring. The choice of this threshold cannot be arbitrary but must rest on optimisation criteria with a clear theoretical basis, constructed directly from the geometry and statistics of the ROC curve; in this section we present the mathematical definitions of the methods for determining the optimal threshold that are most commonly used in research on diagnostic accuracy [70]. Below is a detailed analysis of the three criteria used in this study.

a) The Youden Index Maximisation Criterion

The Youden index (Youden's J statistic, or Youden's index) was first introduced by the statistician William J. Youden [79] in 1950, within the framework of studies aimed at improving the accuracy of medical diagnostic tests. Youden's J statistic is a single number summarising the performance of a binary diagnostic test. It combines sensitivity and specificity into a single index, making it easy to compare tests and to find the optimal decision threshold on a continuous score.

Geometrically, the Youden index is precisely the greatest vertical distance between the ROC curve and the reference diagonal (the chance line), that is, the greatest vertical distance between the ROC curve and the chance diagonal, and it is at the same time mathematically equivalent to the Kolmogorov–Smirnov distance statistic¹ measuring the degree of separation between the two score distributions of the defaulting and non-defaulting groups [57].

An important theoretical advantage of the Youden index, which has led to its wide application in credit rating research, is its independence from the prevalence of the positive class in the population. Because it does not depend on the prevalence of the disease, Youden’s J index can be transferred between populations with different base rates, unlike predictive values [79]. This property is particularly well suited to the credit risk prediction problem, where the actual default rate often varies considerably across customer segments, credit products, or stages of the economic cycle.

The Youden index at the threshold c is defined as:

$$J(c) = Se(c) + Sp(c) - 1 \quad (3.38)$$

and the optimal threshold according to this criterion is determined by:

$$c_J = \arg \max_{c \in \mathbb{R}} J(c), \quad (3.39)$$

where:

- $J(c)$: the Youden index at the classification threshold c ;
- $Se(c)$: the sensitivity at the threshold c , that is, the proportion of defaulting customers correctly identified by the model;
- $Sp(c)$: the specificity at the threshold c , that is, the proportion of non-defaulting customers correctly identified by the model;
- c_J : the optimal classification threshold according to the Youden index maximisation criterion;
- $\arg \max_c$: the operator finding the value of c that makes the function $J(c)$ attain its maximum over the entire real domain of the threshold.

The index $J(c)$ takes values in the interval $[-1, 1]$, where $J = 0$ corresponds to a model with no discriminative ability whatsoever (the threshold lies on the reference diagonal of the ROC), while $J = 1$ corresponds to a perfect classification model with neither false positives nor false negatives. A J value of zero means

¹The Kolmogorov–Smirnov distance statistic is the value measuring the greatest distance between two cumulative distribution functions

that the test provides no useful information, while a value of one indicates perfect classification with no false positives or false negatives [79]. Youden's J index is one of the oldest and most widely used optimal-threshold frameworks, determining the threshold by maximising the sum of sensitivity and specificity across the different threshold points [70].

However, an important theoretical limitation of the Youden criterion was pointed out by Smits (2010) [64]: maximising $J(c)$ implicitly assumes a misclassification cost ratio equal to 1 minus the prevalence of the positive class, an assumption that does not always accurately reflect the decision-maker's real priorities.

b) The Closest-to-(0,1) Criterion

Also known as the Euclidean distance criterion, formally introduced by Perkins and Schisterman in 2006 [51], it is based on the geometric intuition that the most ideal operating point on the ROC curve is the point corresponding to a perfect classifier, that is, the point with coordinates $(FPR, TPR) = (0, 1)$. Perkins and Schisterman introduced the closest-to-(0, 1) criterion, which selects the optimal threshold by minimising the distance of the points on the ROC curve to the perfect classification point $(0, 1)$, where both sensitivity and specificity attain their maximum values. In essence this is a direct geometric rule: the optimal threshold is determined at the empirical operating point on the ROC curve with the shortest Euclidean distance to the ideal point $(0, 1)$ [29].

The Euclidean distance from the operating point at the threshold c to the ideal point $(0, 1)$ is computed as:

$$D(c) = \sqrt{(1 - Se(c))^2 + (1 - Sp(c))^2} \quad (3.40)$$

and the optimal threshold according to this criterion is determined by:

$$c_D = \arg \min_{c \in \mathbb{R}} D(c) \quad (3.41)$$

This criterion is equivalent to simultaneously minimising the sum of the squares of $FNR(1 - Se)$ and $FPR(1 - Sp)$; an estimator based on the smallest sum of squares of 1 minus sensitivity and 1 minus specificity is preferred when sensitivity and specificity are regarded as equally important. Compared with the Youden criterion, the closest-to-(0,1) criterion tends to react more sensitively to asymmetric deviations between the two types of classification error, since it applies a quadratic penalty to the deviation along each axis separately, instead of considering only a linear sum as the Youden index does.

c) The Symmetry Point Criterion

The Symmetry Point criterion (the Equal Sensitivity-Specificity Criterion, or the intersection point of the sensitivity and specificity curves) is based on the principle of determining the optimal classification threshold at the position where sensitivity and specificity attain balanced values. When the ROC curve has an area under the curve smaller than one, a trade-off always exists between the ability to correctly detect positive subjects (sensitivity) and the ability to correctly rule out negative subjects (specificity). The optimal threshold can therefore be determined by finding the value of c such that the discrepancy between sensitivity and specificity is smallest.

This approach has a theoretical basis closely related to research on threshold selection when sensitivity and specificity are valued equally, a topic systematically examined in the study by Froud and Abel (2014) [28] comparing methods for determining the minimally important change threshold in clinical outcome measurement Using ROC curves to select the threshold of minimally important change when sensitivity and specificity are valued equally. The core principle of this group of methods is that, when there is no basis for assigning different cost weights to the two types of classification error (FP and FN), the balance point where the two indices Se and Sp are equal is regarded as the fairest and most neutral operating point.

The optimal symmetry threshold is determined as the solution c of the equation:

$$Se(c) = Sp(c)$$

or equivalently, is determined by minimising the absolute deviation between these two quantities:

$$c_S = \arg \min_{c \in \mathbb{R}} |Se(c) - Sp(c)| \quad (3.42)$$

The threshold c_S corresponds to the intersection between the empirical ROC curve and the auxiliary line $TPR + FPR = 1$ in ROC space. This criterion has an important practical advantage in the prediction problem. It produces a balanced operating point at which the error rates between the two customer groups (those wrongly rejected and those wrongly granted credit) are equivalent as a percentage within each group, thereby helping credit institutions to maintain a consistent and transparent credit-granting policy that is not excessively biased towards either of the two types of risk (credit risk and the risk of losing customers/business opportunities). It should be noted, however, that the symmetry point criterion, like the Youden criterion, still implicitly assumes cost parity between the two types of classification error, and therefore needs to be applied with consideration for the specific risk management strategy of each financial institution, particularly given that credit data usually exhibits a high degree of class imbalance [11].

3.4.4 The Decision Curve Analysis Method

Decision Curve Analysis (DCA) was first formally introduced by Vickers and Elkin [72] in 2006 in the journal *Medical Decision Making*, with the aim of addressing an important gap remaining in traditional methods of evaluating predictive models. While accuracy measures such as sensitivity, specificity and the area under the receiver operating characteristic curve are essential for evaluating predictive models, they cannot capture the practical consequences of deploying such models in the real world. Specifically, measures such as AUC or the Youden index, although they reflect the intrinsic discriminative capacity of the model across the entire range of thresholds, entirely fail to quantify the practical value of a specific decision (for instance the decision to grant or refuse a loan) to the decision-maker, since they treat all classification thresholds as equally important, whereas in reality the consequences of a false positive and a false negative usually carry very different meanings and costs.

The theoretical foundation of DCA stems from formalising the relationship between the decision probability threshold c and the exchange rate between the benefit of a TP and the harm of a FP . In the context of credit risk prediction, the concept of the threshold c has a practical interpretation as the minimum probability of default at which a credit institution decides to refuse credit (or to require additional risk mitigation measures such as raising the interest rate or requiring collateral). A low threshold c reflects a prudent credit policy, in which the institution regards the harm of wrongly granting credit to a customer who will default (a false negative in the original logic, or correspondingly “approve a bad loan”) as far more serious than the harm of wrongly rejecting a good customer; conversely, a high threshold c reflects a credit expansion policy that prioritises maximising the number of loans disbursed and accepts a higher risk of default.

The original formula for Net Benefit proposed by Vickers and Elkin (2006) [72], computed on a test dataset of N observations, is expressed as follows; in the original paper, the net benefit is computed as:

$$NB(c) = \frac{TP(c)}{N} - \frac{FP(c)}{N} \left(\frac{c}{1-c} \right), \quad (3.43)$$

where:

- $NB(c)$: the net benefit of the model at the decision probability threshold c ;
- $TP(c)$: the number of customers who genuinely default and are correctly predicted by the model at the decision threshold c ;
- $FP(c)$: the number of customers who genuinely do not default but are wrongly predicted by the model as going to default at the decision threshold c ;
- N : the total number of observations (customers) in the test dataset;
- $\frac{c}{1-c}$: the odds ratio at the threshold c , acting as a weighting factor that converts

the number of false positives into the same unit of harm as the number of true positives.

The formula above is often expressed in an equivalent form in terms of sensitivity, specificity and prevalence, which is more convenient for linking it to the ROC framework presented in the previous sections [73]:

$$NB(c) = Se(c) \times \pi - (1 - Sp(c)) \times (1 - \pi) \times \frac{c}{1 - c} \quad (3.44)$$

The formula above shows that DCA is not a method entirely separate from the ROC framework; rather, it uses the very structural components of the ROC curve (Se , Sp) as its input material, while adding two elements that ROC/AUC entirely ignore: the actual prevalence of the event (π) and the relative cost weighting between the two types of error (through the factor $c/(1 - c)$) — moving beyond AUC: decision curve analysis to quantify the net benefit of risk prediction models [56].

The Decision Curve and the Reference Lines

After computing $NB(c)$ at a range of different values of c (usually spread evenly over a practically meaningful probability interval, for example from 0.01 to 0.5), plotting $NB(c)$ against c forms the decision curve; plotting the net benefit against the threshold probability produces the decision curve [72]. For the decision curve to have practical interpretive meaning, it is always presented together with two reference strategies serving as comparison standards:

1. *Treat All*: assumes that every observation is classified as positive (in the credit context, equivalent to rejecting/monitoring every customer), regardless of the model score. The net benefit of this strategy is computed as:

$$NB_{All}(c) = \pi - (1 - \pi) \times \frac{c}{1 - c} \quad (3.45)$$

2. *Treat None*: assumes that every observation is classified as negative (no customer is rejected on the basis of the model), in which case $TP(c) = FP(c) = 0$ and the net benefit is therefore always zero at every threshold; the treat-none strategy assumes that no observation receives the intervention, and by definition its net benefit is always zero across all thresholds:

$$NB_{None}(c) = 0 \quad \forall c \in (0, 1), \quad (3.46)$$

where:

- $NB_{All}(c)$: the net benefit of the treat-all strategy at the threshold c ;

- $NB_{\text{None}}(c)$: the net benefit of the treat-none strategy at the threshold c (always equal to 0);
- π : the prevalence of the positive class.

A predictive model is considered to have practical value at a specific threshold c if and only if its decision curve lies above both of these reference lines at that threshold; the model provides genuine benefit only within the range of thresholds where its curve lies above both reference strategies. The intersection point of these two reference lines, which occurs precisely at $c = \pi$, carries important guiding significance for determining the range of practically meaningful thresholds to examine in a given study.

Although DCA was developed and popularised mainly in the field of medical statistics, where it has become a routinely recommended tool to supplement traditional measures of discrimination and calibration when evaluating clinical predictive models — decision curve analysis addresses this limitation by incorporating the concept of net benefit, allowing a more comprehensive assessment of a model’s clinical usefulness [72] — the underlying mathematical structure of the method is entirely transferable to the credit risk prediction problem, since both problems are essentially binary classification problems governed by a decision threshold with asymmetric costs between the two types of error.

Research on default prediction has long been well aware of the core problem that DCA seeks to resolve: the cost of wrongly rejecting a good customer, leading to lost business opportunity and lost interest revenue, and the cost of wrongly approving a customer who will default, leading to loss of principal, are entirely asymmetric and depend strongly on each institution’s own risk management strategy — an evaluation that takes into account the financial consequences of false positives (rejecting good borrowers) and false negatives (approving risky borrowers). The DCA theoretical framework provides a more systematic and quantitatively rigorous approach to formalising this trade-off, going beyond reliance on the qualitative intuition of risk experts or on heuristic rules for making subjective decisions when assessing a user’s credit risk.

The decision curve allows the researcher or the risk management department of a credit institution to answer an important practical question: “at what level of default probability threshold does using the predictive model to make credit-granting decisions yield a higher net benefit than the two extreme strategies of approving all loan applications or rejecting all loan applications?” This is of far greater practical significance than merely reporting a single AUC figure or Youden index, because it allows the entire range of thresholds at which the model genuinely creates added value to be illustrated visually, while also exposing cases in which a model with a high AUC but poor calibration — that is, where the predicted probabilities do not accurately reflect the actual frequency of default — may still be harmful (yielding a negative net benefit) over certain ranges of thresholds. This constitutes an important premise, warning researchers of the latent risk in evaluating complex machine learning models, where the problem of probability calibration is often overlooked when the researcher focuses solely on optimising AUC or classification accuracy.

Chapter 4

Experiments and Evaluation

This chapter presents the entire experimental process of the credit risk prediction framework based on Causal Machine Learning, implemented and evaluated on three real-world credit datasets with different characteristics and scales: the Australian Credit Dataset, the German Credit Dataset, and Lending Club 2007–2014. In addition to comparing the classification performance between the Bayesian network and baseline machine learning algorithms, this chapter also presents in detail the results of each stage in the data processing pipeline, the structure of the learned causal graph, as well as model interpretation analyses through forward inference, backward inference, and what-if scenario analysis.

4.1 Experimental Data

To comprehensively evaluate the effectiveness of the credit risk prediction framework based on Causal Machine Learning, this study conducts experiments on three real-world credit datasets with different scales, origins, and structural characteristics, namely the Australian Credit Dataset, the German Credit Dataset, and Lending Club 2007–2014. Each dataset represents a distinct credit appraisal context, thereby allowing an assessment of the model’s generalisation capability across multiple types of financial data.

4.1.1 Description of the Datasets

The Australian Credit Dataset was collected from the UCI Machine Learning Repository, reflecting personal credit profile information in Australia, with data that has been fully anonymised. The dataset comprises 690 observations and 14 input features after removing the postal code column (ZipCode), as it carries no risk-discriminating value. Among the remaining 14 features, 5 variables are continuous in nature, namely Age, Debt, YearsEmployed, CreditScore and Income; the remaining 8 are nominal categorical variables, namely Gender, BankCustomer, EducationLevel, Ethnicity, CreditHistory, Employed, DriversLicense and Citizen. The distribution of the target label is relatively balanced, with 383 default samples (55.5%) and 307 non-default samples (44.5%), and there are no missing values. The dataset is split at a ratio of 60/40 for the training set and the test set.

The German Credit Dataset was likewise obtained from the UCI Machine Learn-

ing Repository, representing the credit risk classification problem in the context of German banking. The dataset comprises 1,000 observations with 20 input features. Of these, 3 continuous variables are month_duration, credit_amount and age; the remaining 17 are categorical variables, including status_account, credit_history, purpose, status_savings, years_employment, status_and_sex, secondary_obligor, collateral, other_installment_plans, housing, job, telephone, is_foreign_worker and several other auxiliary variables. The target label was re-encoded from string format (“good” $\rightarrow$ 0, “bad” $\rightarrow$ 1), with a distribution of 700 good samples (70%) and 300 bad samples (30%), reflecting a moderate degree of class imbalance. The dataset contains no missing values and is split at a ratio of 60/40.

Lending Club 2007-2014 is a large-scale peer-to-peer lending dataset collected from the Lending Club platform, comprising 466,345 loan records with 151 original information columns. Owing to the scale and particular complexity of this dataset, the pre-processing stage required considerably more in-depth cleaning steps than the other two datasets. First, the target variable was binarised from the loan_status column: the statuses “Fully Paid”, “Current” and “Does not meet the credit policy. Status: Fully Paid” were assigned label 0 (non-default), while the statuses “Charged Off”, “Late”, “In Grace Period” and “Does not meet the credit policy. Status: Charged Off” were assigned label 1 (default). Next, columns carrying no appraisal information (id, url, free-text descriptions) and columns causing future data leakage (total_pymnt, recoveries, last_pymnt_amnt, and other variables reflecting outcomes after loan disbursement) were removed entirely. Columns with a missing-value rate exceeding 50% were likewise excluded. On the remaining set of numerical variables, variance inflation factor (VIF) analysis and correlation filtering were applied to address multicollinearity, reducing the number of continuous variables from 36 to an optimal level before feeding them into Lasso. Combined with 10 categorical variables, namely term, grade, emp_length, verification_status, home_ownership, purpose, addr_state, initial_list_status, hardship_flag and debt_settlement_flag, the final feature space was determined. Due to computational resource constraints when learning the Bayesian network structure on such an extremely large dataset, the study performed controlled subsampling with 21,000 non-default samples and 9,000 default samples, forming a working set of 30,000 observations with an imbalance ratio of approximately 7:3, closely approximating the natural distribution of the original data. This set was then split at a ratio of 60/40 for training and testing.

Summary information on the three datasets is presented in Table 4.1.

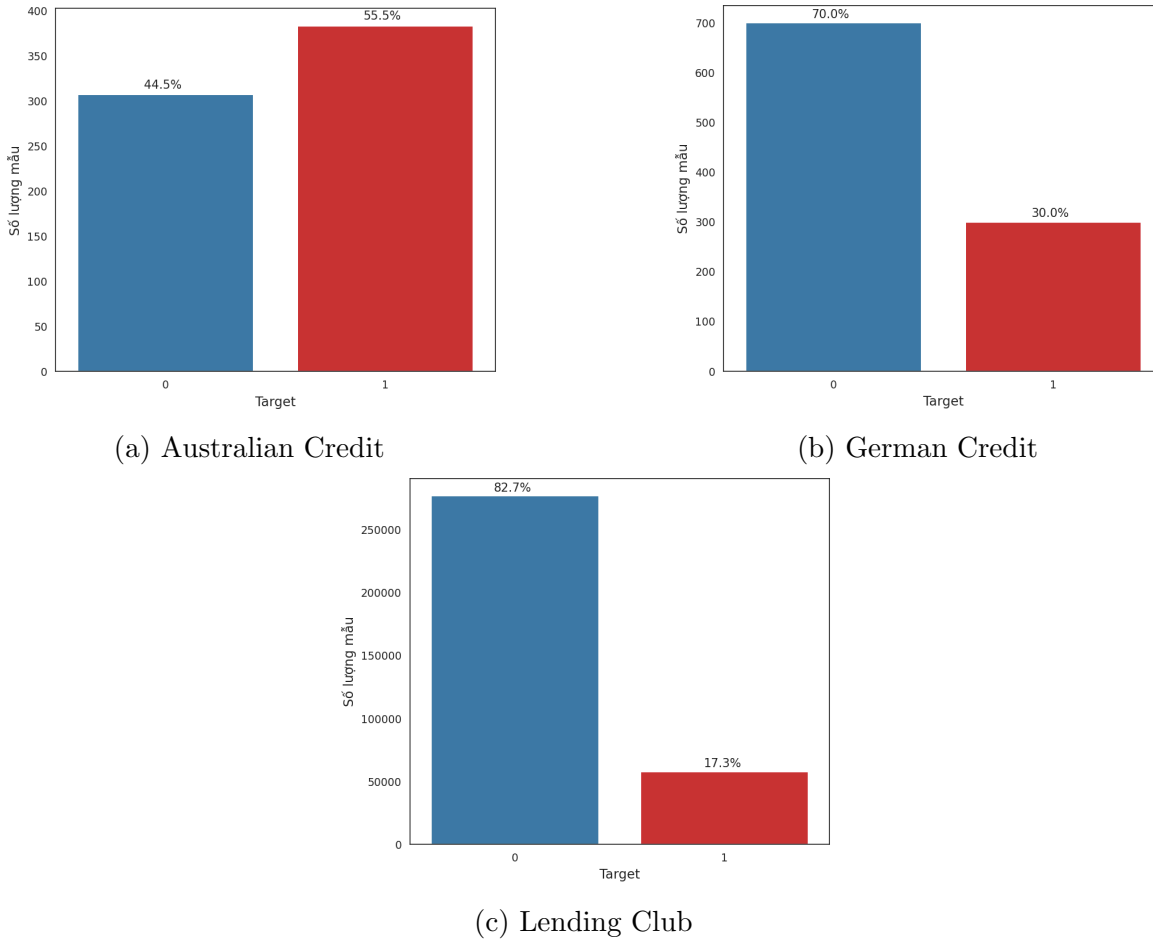
Bảng 4.1: Summary information on the experimental datasets

Dataset	No. of samples	No. of features	Non-default	Default
Australian	690	14	307	383
German	1,000	20	700	300
Lending Club	30,000	46	21,000	9,000

4.1.2 Exploratory Data Analysis (EDA)

Before proceeding to model construction, exploratory data analysis (EDA) was carried out on all three datasets in order to clarify the distributional characteristics, the degree of class imbalance, and the correlation structure among the features. The results of this analysis also provide the empirical basis for the design decisions made in the data pre-processing pipeline at subsequent steps.

Degree of Class Imbalance



Hình 4.1: Distribution of the target label across the three experimental datasets

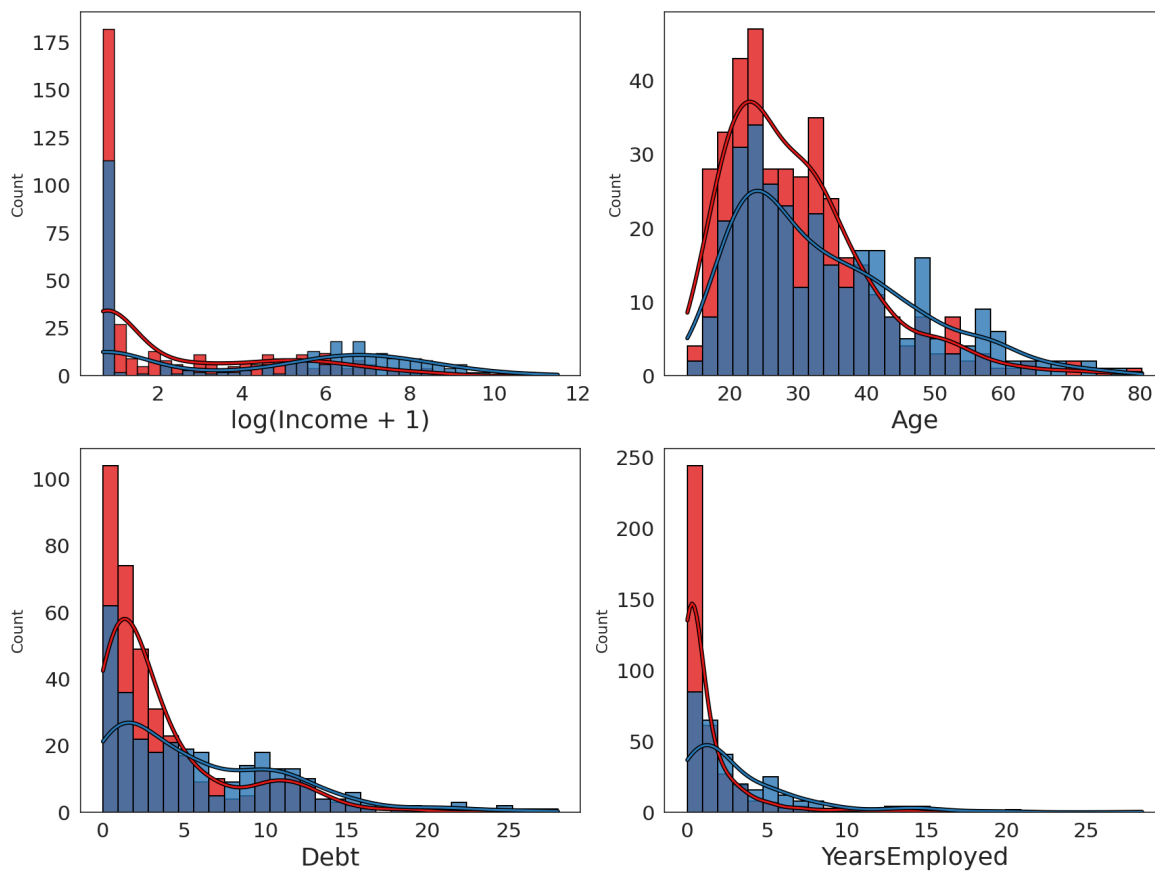
Figure 4.1 shows the distribution of the target label for the three datasets. Australian Credit exhibits the best degree of balance, with a ratio of 44.5% non-default to 55.5% default, which is close to natural balance. German Credit shows a moderate imbalance at a ratio of 70/30, whereas the original Lending Club dataset exhibits the most severe imbalance, with a ratio as high as 82.7% non-default versus 17.3% default. This discrepancy confirms the necessity of the S_Pro step in the processing pipeline.

Distribution of Continuous Variables

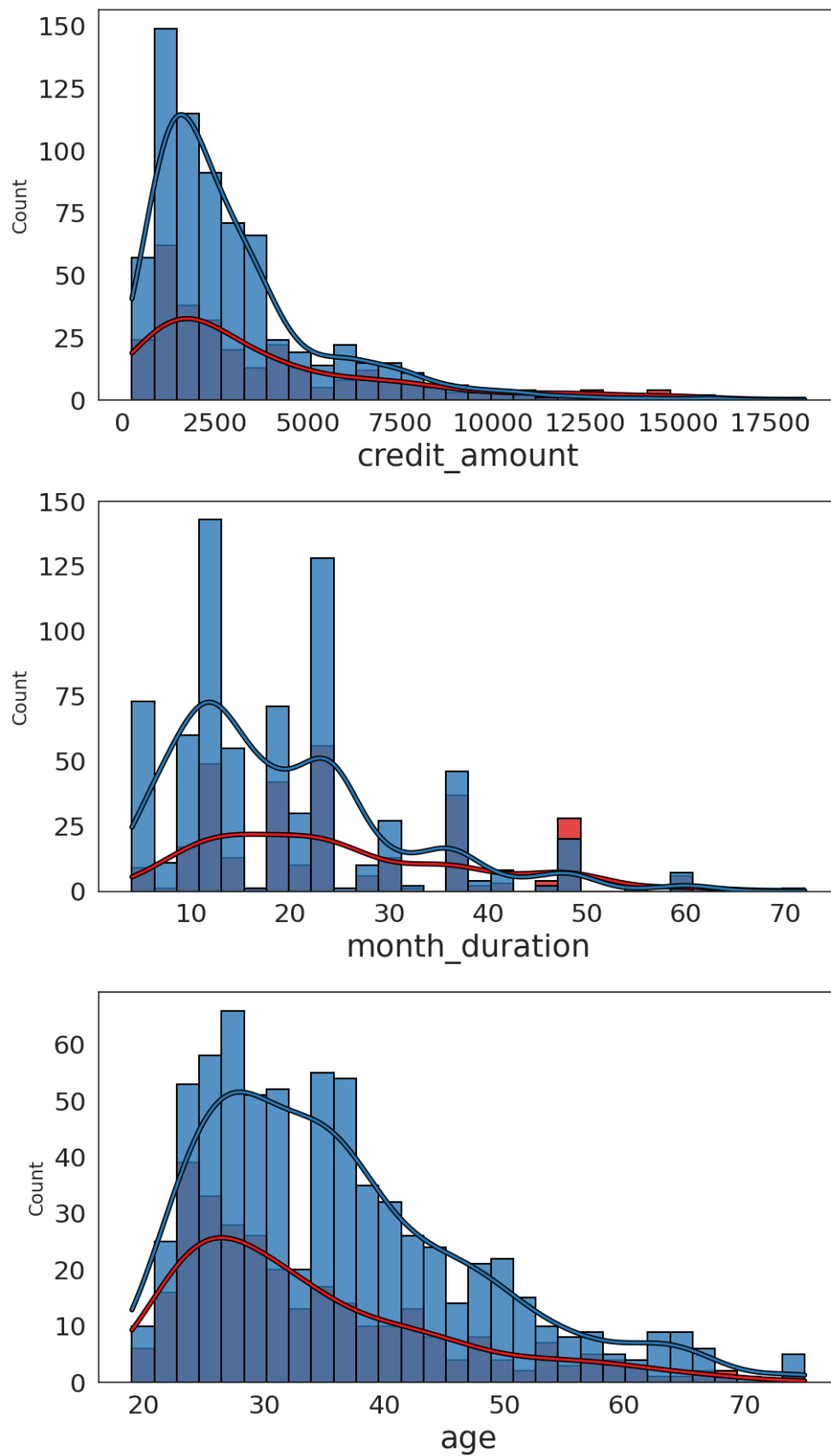
On the Australian dataset (Figure 4.2), `Income` exhibits an extremely strong right-skewed distribution. After the transformation $\log(\text{Income} + 1)$, the default group is overwhelmingly concentrated in the very low value region, whereas the non-default group is more evenly spread and clearly dominant at medium and high income levels. `Age` is concentrated mainly in the 18–35 age range, with the default group tending to be younger, while the non-default group tends to be older, as reflected in the distribution’s tail extending towards ages 45–60. `Debt` and `YearsEmployed` are both strongly right-skewed, with the two label groups overlapping considerably, indicating limited univariate discriminative power. On the German dataset (Figure 4.3), `credit_amount` is right-skewed and concentrated below 5,000 DM, with the default group spread over higher value ranges. `month_duration` exhibits a clearly multimodal distribution at the 12, 24 and 36-month marks, reflecting the reality of commonly fixed loan terms, with the default group tending towards longer terms. On the Lending Club dataset (Figure 4.4), `int_rate` is the most discriminative variable among the continuous variables — the KDE curve of the default group is shifted markedly to the right relative to the non-default group, reflecting the credit logic whereby higher-risk borrowers are charged higher interest rates. `fico_range_low` shows the opposite trend: the credit scores of the non-default group are consistently higher.

Distribution of Categorical Variables

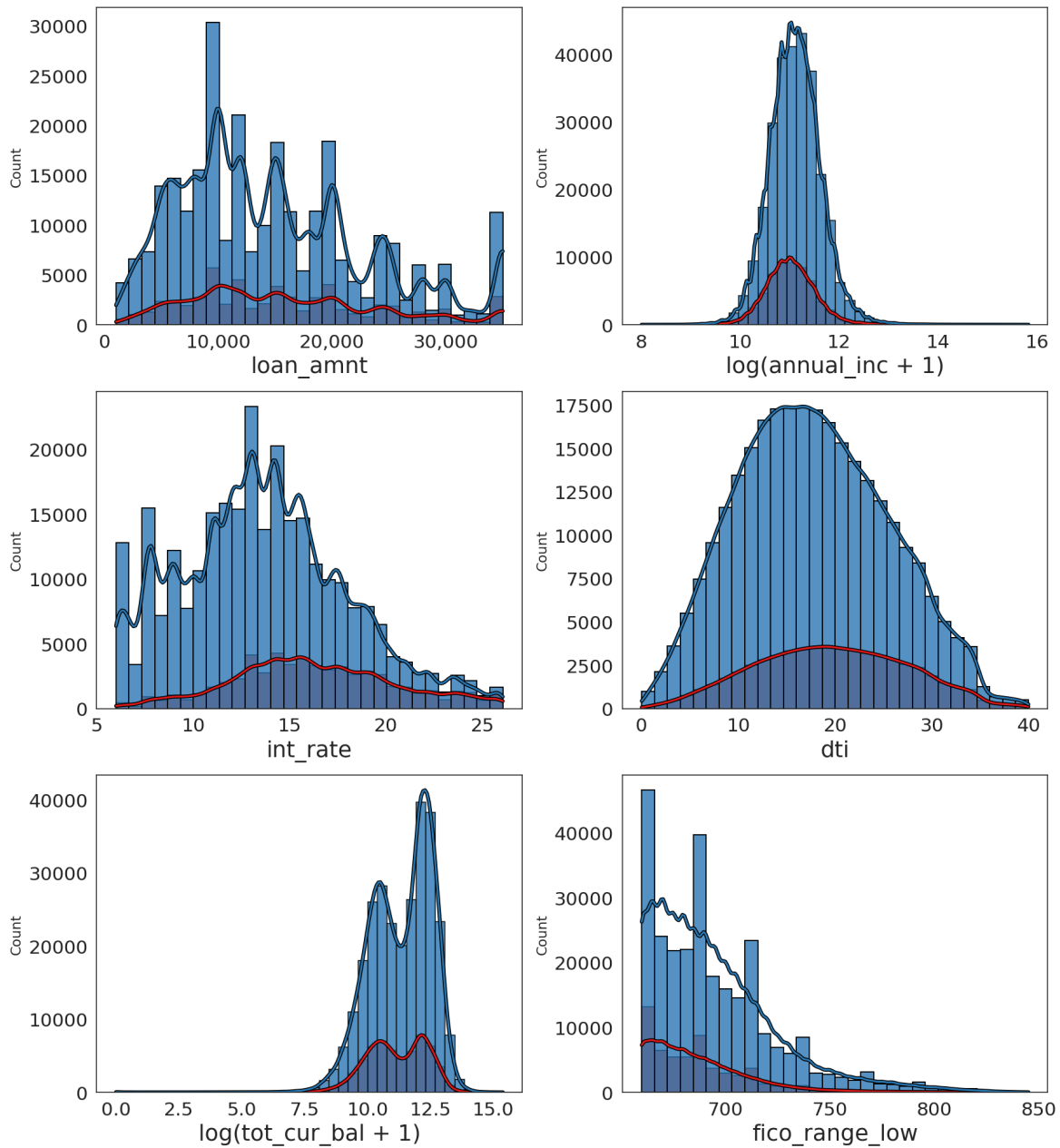
The distribution charts show that several categorical variables have very clear risk-discriminating power. On Australian (Figure 4.5), `CreditHistory` is the most discriminative variable: the group `CreditHistory` = 1 consists almost entirely of non-default samples, whereas the group `CreditHistory` = 0 has a dominant proportion of defaults. `Employed` exhibits a similar discriminative pattern. On German (Figure 4.6), `status_account` is a strong risk signal — the group with no checking account (“no checking account”) has an overwhelming proportion of non-defaults, whereas the group with a negative balance (“< 0 DM”) concentrates a higher proportion of default samples. `status_savings` confirms a similar trend: the group with savings below 100 DM accounts for the majority and has the highest default rate. On Lending Club (Figure 4.7), `grade` is the categorical variable with the strongest discriminative power — the default rate increases monotonically from grade A to G, showing that Lending Club’s internal rating system correlates directly with actual risk. The 60-month loan term also has a markedly higher default rate than the 36-month term.



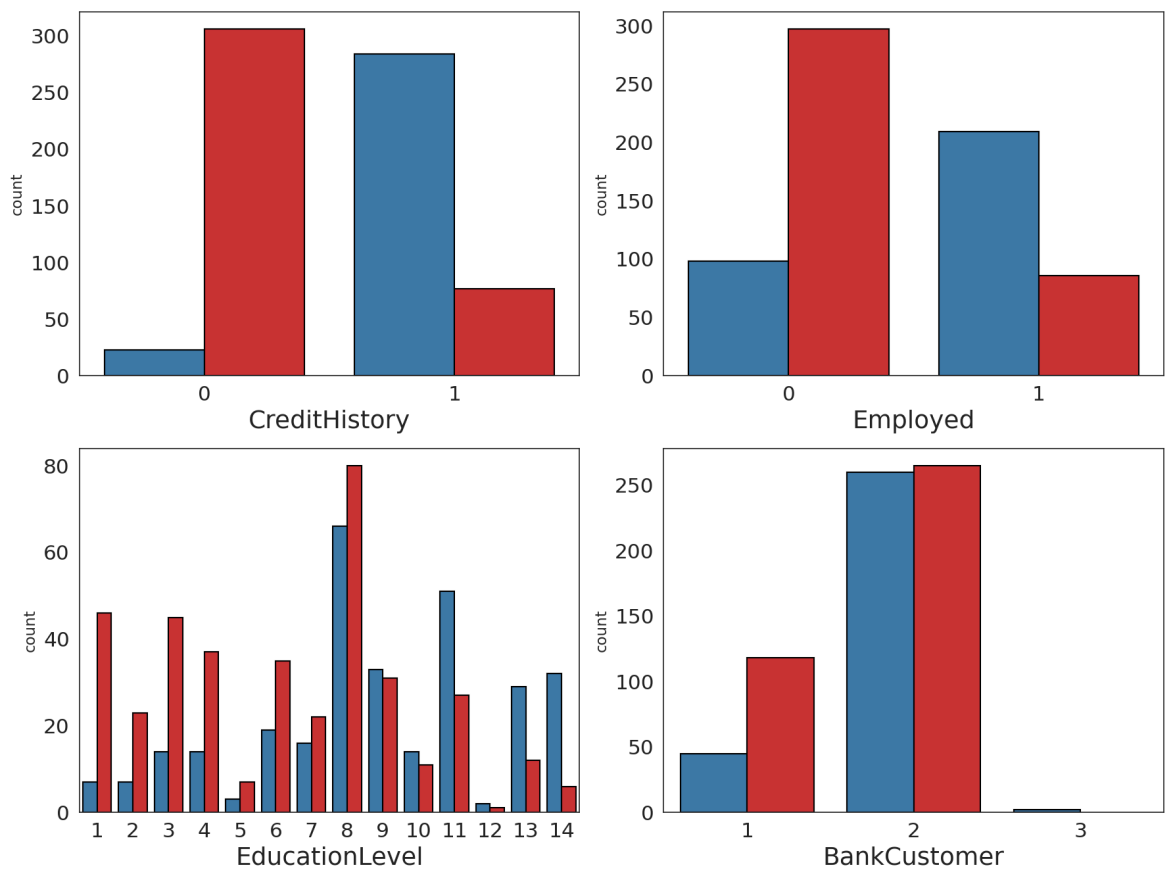
Hình 4.2: Distribution of continuous variables by the **target** label — Australian Credit.
(Where blue denotes **target** = 0 and red denotes **target** = 1)



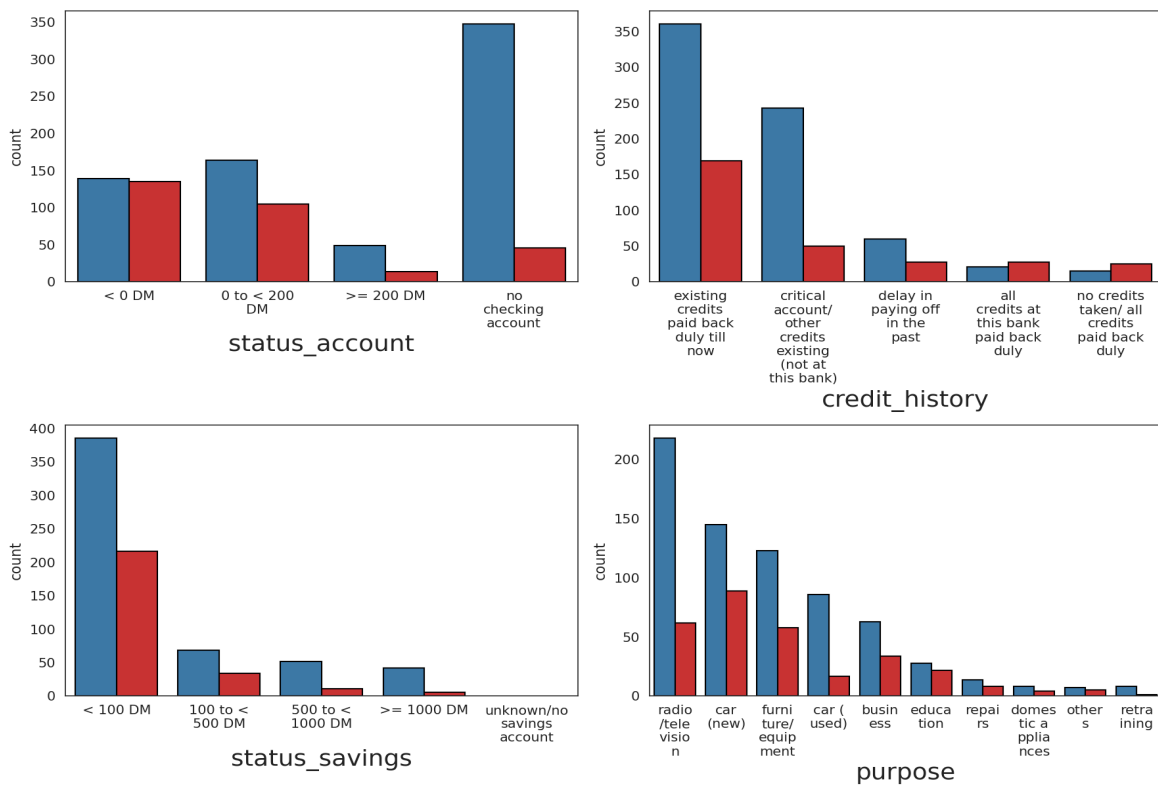
Hình 4.3: Distribution of continuous variables by the `target` label — German Credit.
(Where blue denotes `target = 0` and red denotes `target = 1`)



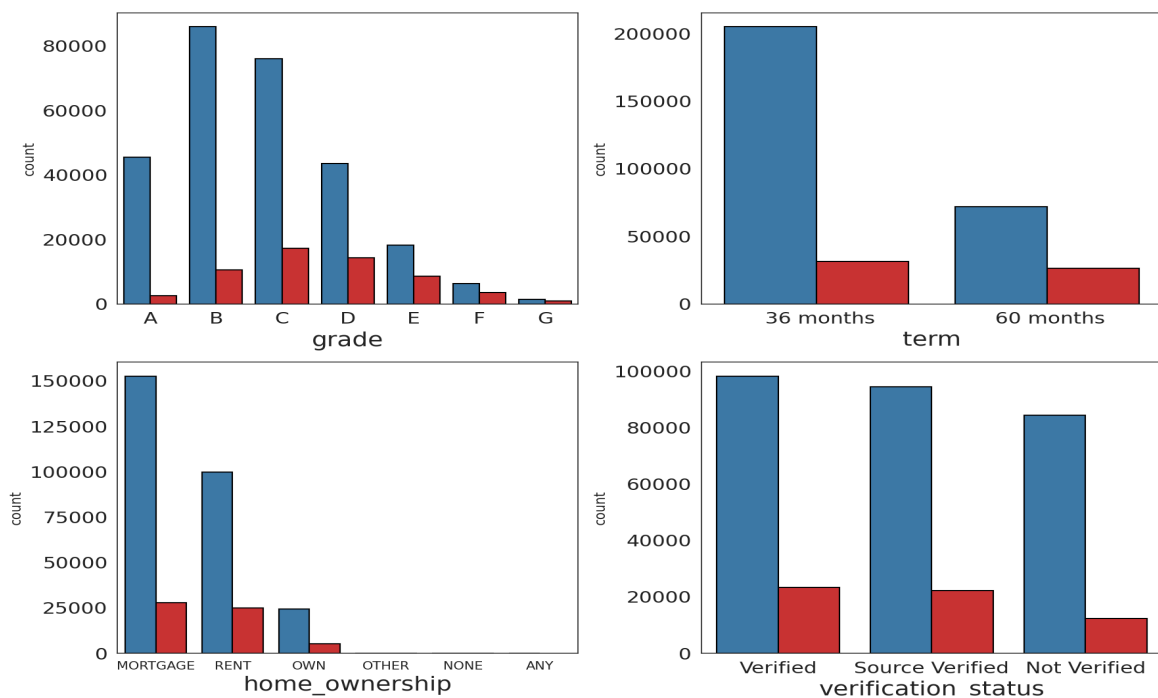
Hình 4.4: Distribution of continuous variables by the **target** label — Lending Club.
(Where blue denotes **target** = 0 and red denotes **target** = 1)



Hình 4.5: Distribution of categorical variables by the **target** label — Australian Credit.
(Where blue denotes **target** = 0 and red denotes **target** = 1)



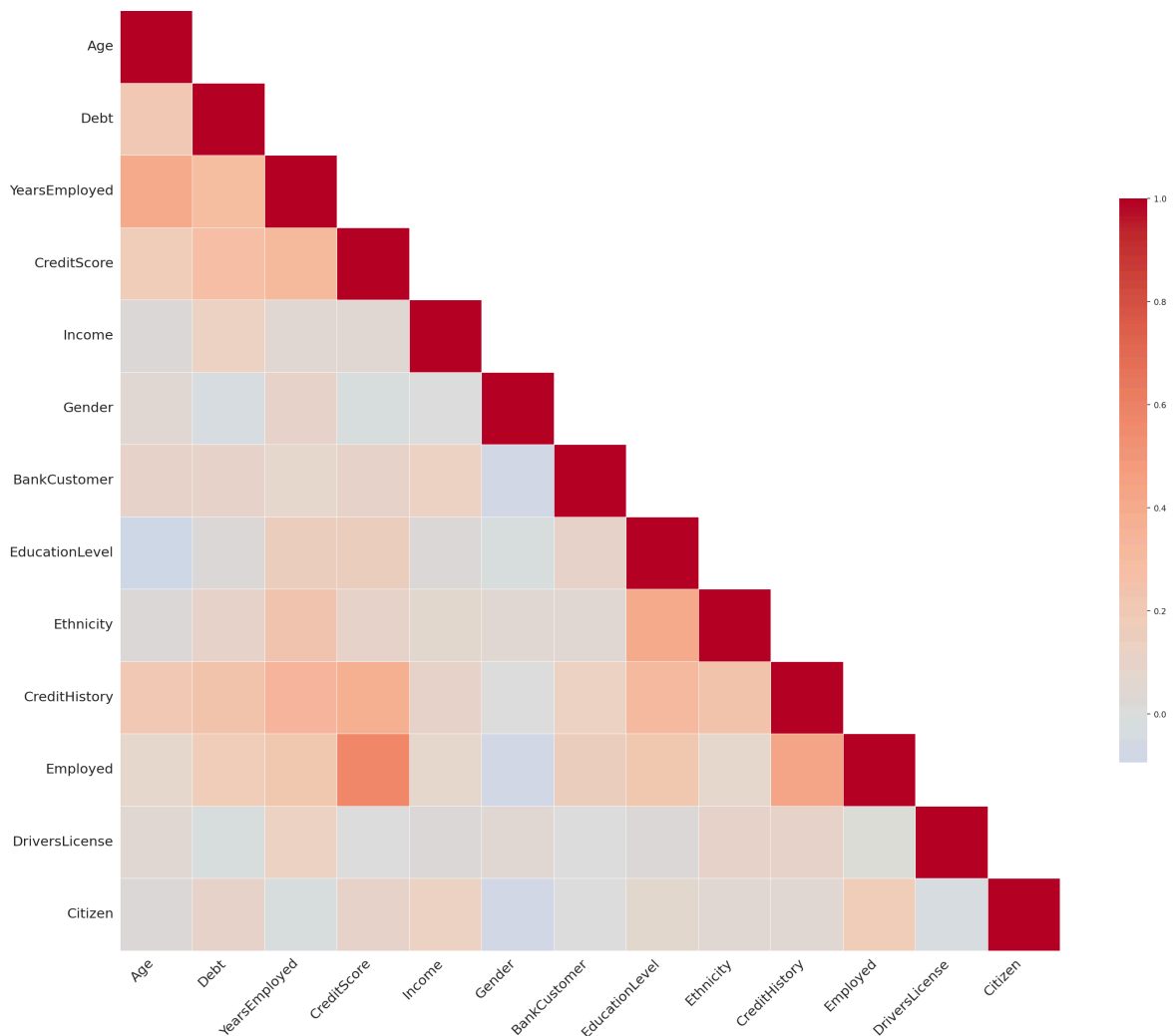
Hình 4.6: Distribution of categorical variables by the **target** label — German Credit.
(Where blue denotes **target** = 0 and red denotes **target** = 1)



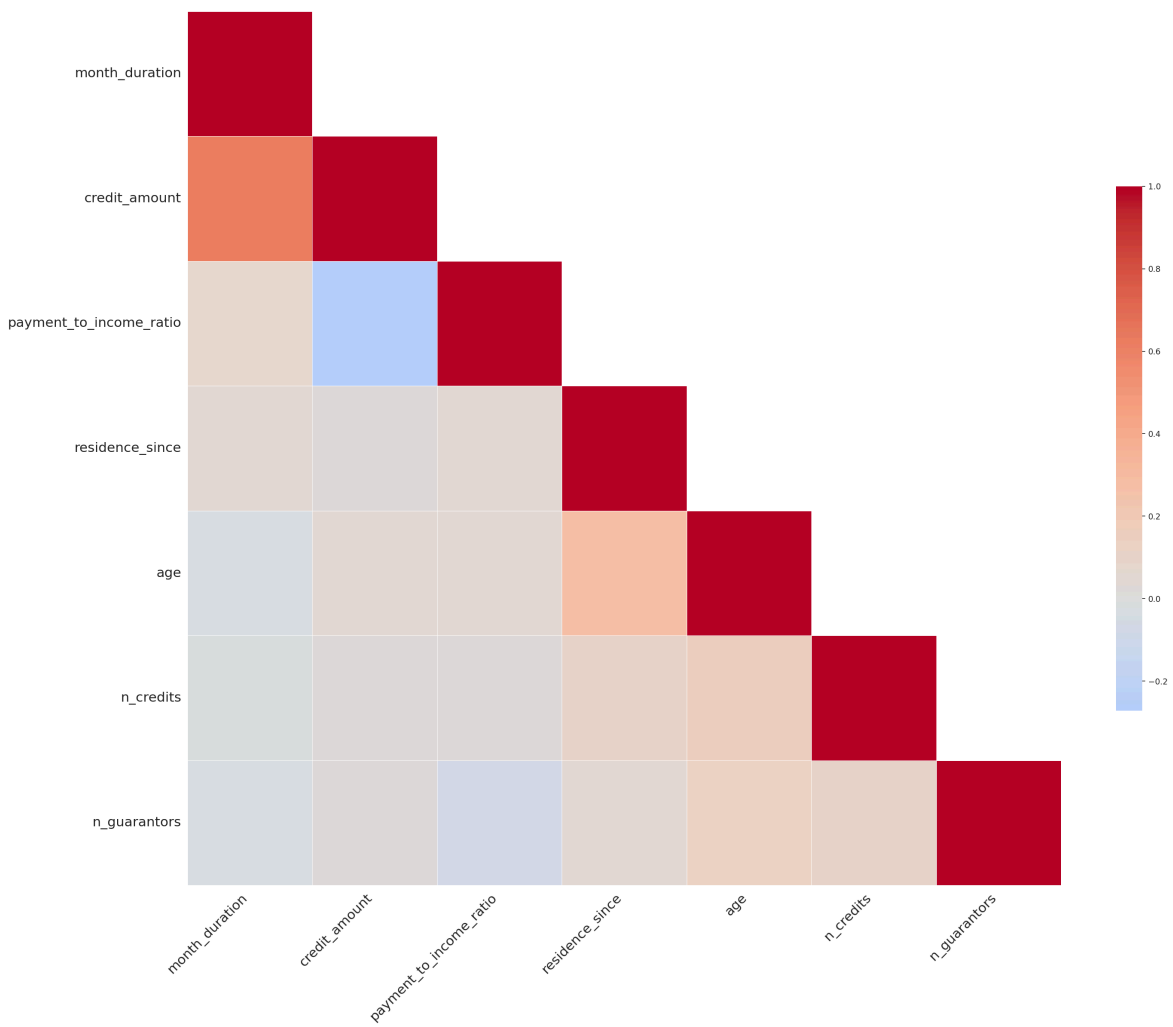
Hình 4.7: Distribution of categorical variables by the **target** label — Lending Club.
(Where blue denotes **target** = 0 and red denotes **target** = 1)

Correlation Matrix

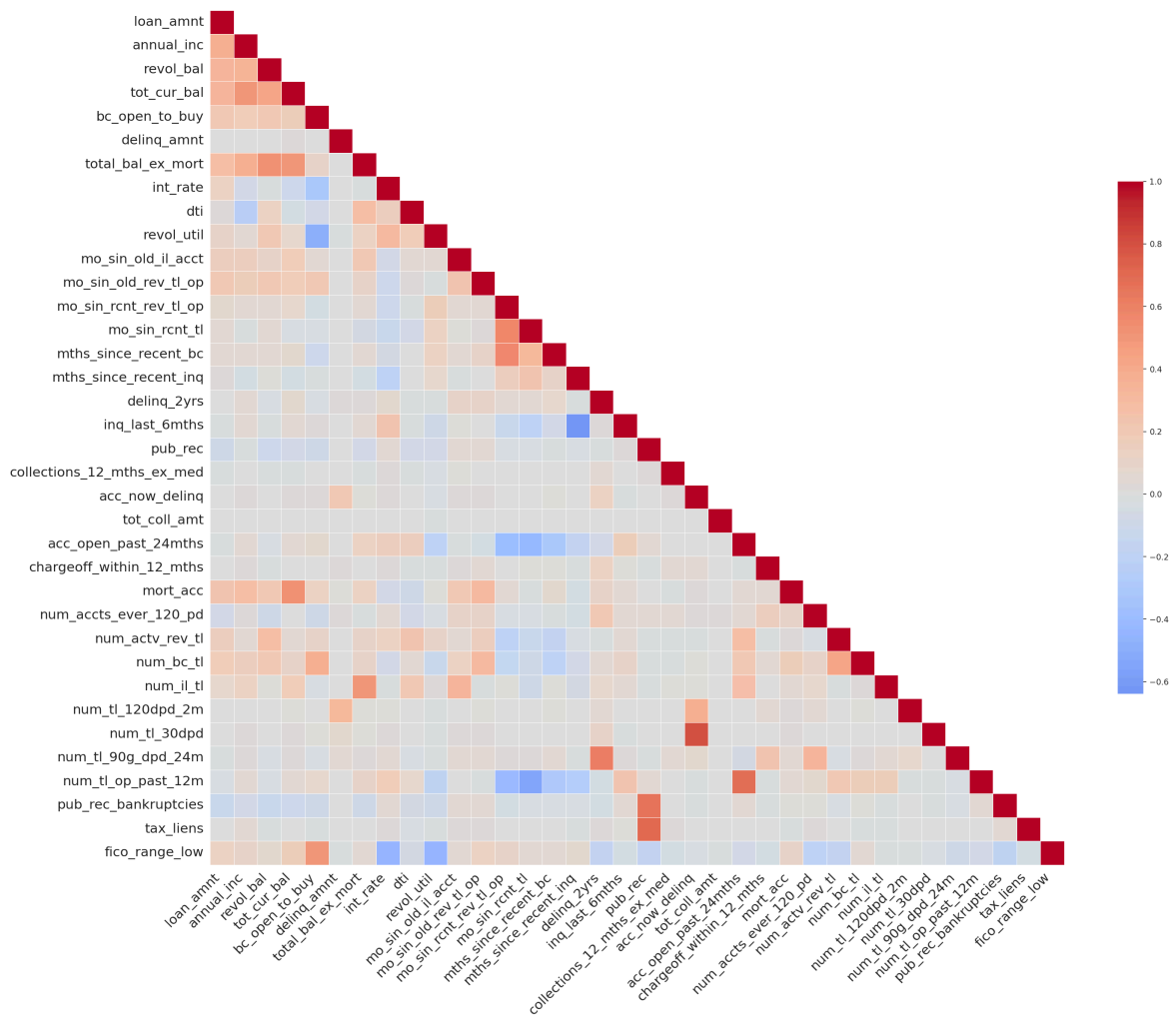
On Australian (Figure 4.8), most pairs of variables have very low correlation, with the exception of the **YearsEmployed-Employed** pair, which shows a moderate positive correlation, reflecting the natural relationship between employment status and tenure. On German (Figure 4.9), the pair **month_duration** and **credit_amount** shows a notable positive correlation — larger loans tend to be accompanied by longer repayment terms, a relationship that is reasonable from a credit business standpoint. On Lending Club (Figure 4.10), the variable space is much wider, with 36 numerical variables, and several internal correlation clusters emerge among variables belonging to the same group, such as the group of account-count variables (**num_bc_tl**, **num_il_tl**, **num_actv_rev_tl**) and the group of monthly time variables. Notably, the pair **fico_range_low** and **int_rate** shows a clear negative correlation, correctly reflecting the risk-pricing mechanism: a higher credit score corresponds to a lower interest rate. The presence of these correlation clusters provides the motivation for the subsequent L_Pro step.



Hình 4.8: Correlation matrix among continuous variables on Australian Credit



Hình 4.9: Correlation matrix among continuous variables on German Credit



Hình 4.10: Correlation matrix among continuous variables on Lending Club

4.2 Data Processing

4.2.1 Encoding of Categorical Features

Before feeding the data into the data-processing pipelines, categorical variables need to be converted into numerical form so that the algorithms can process them. However, not every categorical variable can be encoded in the same way. This study clearly distinguishes two groups of features according to their statistical nature: ordinal variables, which have an inherent hierarchical relationship among their values, and nominal variables, which have no natural ordering among categories. Applying the wrong encoding method to each type could introduce spurious ordinal relationships into the model, leading to bias in learning the structure of the Bayesian network.

For the Australian dataset, all categorical variables in the original dataset were already encoded as integers, so no additional encoding step was required.

For the German dataset, five ordinal variables were manually encoded according to a hierarchical mapping with business meaning. Specifically, `status_account` was mapped according to the level of account balance from negative to positive (0: below 0 DM $\rightarrow$ 3: no checking account); `credit_history` was ordered by increasing level of credit reliability, from problematic accounts to a perfect repayment history; `status_savings` reflects increasing savings scale; `years_employment` represents length of employment from unemployed to over 7 years; `job` reflects occupational level from unskilled labour to senior management. The remaining eight nominal variables, namely `purpose`, `status_and_sex`, `secondary_obligor`, `collateral`, `other_installment_plans`, `housing`, `telephone` and `is_foreign_worker`, have no natural hierarchical relationship among their values, and were therefore encoded automatically using `OrdinalEncoder`, serving purely as integer identifiers. This approach is consistent with the requirements of the Bayesian network, since the model learns the probabilistic structure from the observed frequencies, without assuming a linear relationship among the encoded values.

For the Lending Club dataset, four ordinal variables were manually mapped according to clear business logic. `grade` was mapped from A (0) to G (6) in order of increasing risk, correctly reflecting Lending Club’s internal credit rating system. `emp_length` represents length of employment from unspecified (0) to over 10 years (11). `verification_status` reflects the increasing level of income verification, from unverified to fully verified. `term` encodes the loan term, with 36 months as 0 and 60 months as 1. Six nominal variables, namely `home_ownership`, `purpose`, `addr_state`, `initial_list_status`, `debt_settlement_flag` and `hardship_flag`, were encoded automatically using `OrdinalEncoder`. In addition, the temporal variable `earliest_cr_line`, stored as a month-year string, was converted into the continuous numerical variable `mths_since_earliest_cr` — the number of months from the date the first line of credit was opened to the reference point of January 2015 — thereby preserving credit-history information in a quantitatively comparable form.

4.2.2 Splitting the Training and Test Sets

To ensure objectivity in model evaluation, the entire dataset was split into separate subsets before applying any pre-processing step related to machine learning. All three datasets used a 60/40 split ratio, in which 60% was used as the training set and the remaining 40% was held out as the test set. The splitting was carried out before the SMOTE, Lasso and K-means steps in order to avoid data leakage — that is, information from the test set inadvertently influencing the model training process, leading to artificially optimistic performance evaluation.

Once the entire pre-processing pipeline is complete, the initial test set (40%) is further split into two equal parts. The first part serves as the validation set, used to fine-tune the classification threshold, while the second part is retained as the true test set for reporting the final results. Thus, the overall split ratio is 60% training, 20% validation and 20% test.

Bảng 4.2: Distribution of the target variable after splitting the dataset

Dataset	Set	Non-default	Default	Total
Australian	Train	197	217	414
	Val	58	80	138
	Test	52	86	138
German	Train	418	182	600
	Val	145	55	200
	Test	137	63	200
Lending Club	Train	12,579	5,421	18,000
	Val	4,190	1,810	6,000
	Test	4,231	1,769	6,000

4.2.3 Data Balancing (S_Pro)

As analysed in 4.1, all three experimental datasets exhibit class imbalance to varying degrees. If left untreated, this imbalance would cause the Bayesian network structure-learning process to be biased towards the majority class, leading to biased estimates of the conditional probability tables and a reduced ability to identify genuine risk. Accordingly, the S_Pro step is applied to the training set immediately after the split, prior to any subsequent pre-processing operation, so as to ensure that no information from the test set leaks into the sample-generation process.

An important technical point is that all three datasets contain a mixture of continuous and categorical variables. The standard SMOTE algorithm [11], which was originally designed for purely continuous feature spaces, uses a linear interpolation mechanism given by Equation 3.1; when applied directly to categorical variables, this would produce intermediate values that are semantically meaningless, such as an account status with a decimal value of 2.4 or a loan-purpose code of 1.73. To overcome this limitation, the study uses the SMOTE-NC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features) variant. For each newly synthesised sample, the continuous variables are still interpolated according to Equation 3.1, while the value of each categorical variable is determined by a majority-voting mechanism over the set of k nearest neighbours, i.e. the categorical value occurring most frequently in the neighbourhood of the original sample is selected, thereby ensuring the semantic validity of the generated observations. The pseudocode of the algorithm is presented below:

Algorithm 5 Pseudocode of the SMOTE-NC algorithm

Input: Minority sample set X_{min} , set of categorical variable indices $\mathcal{C}$, number of neighbours k , oversampling ratio N .

Output: Synthetic sample set X_{syn} .

```

1:  $X_{syn} \leftarrow \emptyset$ 
2: for each  $x_i \in X_{min}$  do
3:   Compute the distance from  $x_i$  to the other samples in  $X_{min}$ , where con-
     tinuous variables use Euclidean distance and categorical variables use a value
     of 0 (same label) or  $\sigma_c$  (different label, where  $\sigma_c$  is the standard deviation of
     all continuous variables)
4:   Determine the  $k$  nearest neighbours  $\mathcal{N}(x_i)$ 
5:   for  $n = 1$  to  $N$  do
6:     Randomly select  $x_{nn}$  from  $\mathcal{N}(x_i)$ 
7:     Sample  $\lambda \sim \mathcal{U}(0, 1)$ 
8:     for each feature  $j$  do
9:       if  $j \notin \mathcal{C}$  then ▷ Continuous variable
10:         $x_{new}^{(j)} \leftarrow x_i^{(j)} + \lambda \cdot (x_{nn}^{(j)} - x_i^{(j)})$ 
11:       else ▷ Categorical variable
12:         $x_{new}^{(j)} \leftarrow \text{mode}(\{x^{(j)} : x \in \mathcal{N}(x_i)\})$ 
13:       end if
14:     end for
15:      $X_{syn} \leftarrow X_{syn} \cup \{x_{new}\}$ 
16:   end for
17: end for
18: return  $X_{syn}$ 

```

On the Australian dataset, 8 categorical variables were explicitly assigned to the set $\mathcal{C}$, namely Gender, BankCustomer, EducationLevel, Ethnicity, CreditHistory, Employed, DriversLicense and Citizen; the remaining 5 variables (Age, Debt, YearsEmployed, CreditScore, Income) were processed using the continuous interpolation mechanism. After application, the training set increased from 414 to 434 observations, with the class distribution reaching complete balance at 217 samples for each label. On the German dataset, 13 categorical variables were included in $\mathcal{C}$, namely status_account, credit_history, status_savings, years_employment, job, purpose, status_and_sex, secondary_obligor, collateral, other_installment_plans, housing, telephone and is_foreign_worker; the remaining 7 continuous variables were interpolated directly. The training set after S_Pro increased from 600 to 836 observations, reaching a distribution of 418 samples per class. On the Lending Club dataset, 10 categorical variables were assigned to $\mathcal{C}$, namely term, grade, emp_length, verification_status, home_ownership, purpose, addr_state,

`initial_list_status`, `hardship_flag` and `debt_settlement_flag`; the remaining 36 continuous variables were interpolated. After `S_Pro`, the training set increased from 18,000 to 25,158 observations, with a balanced distribution of 12,579 samples for each class.

It should be emphasised that the entire sample-generation process was performed on the training set only. The test set retained its natural distribution throughout the process, so as to ensure that the performance metrics reported later faithfully reflect the model’s generalisation capability under real-world conditions. The results after applying SMOTE-NC are summarised in Table 4.3.

Bảng 4.3: Results of data balancing using SMOTE-NC on the training set

Dataset	Non-default	Default	Total
Australian	217	217	434
German	418	418	836
Lending Club	12,579	12,579	25,158

4.2.4 Feature Selection (`L_Pro`)

The input feature space after `S_Pro` remains relatively large, particularly for Lending Club, which has as many as 46 variables, harbouring potential information redundancy and multicollinearity that are detrimental to the process of learning the Bayesian network structure. The `L_Pro` step is therefore carried out in order to reduce the variable space to a minimum while still preserving the ability to discriminate credit risk.

Since the target variable of the problem is binary (default/non-default) and most of the features after the `OneHotEncoder` encoding step are dummy in nature, the study uses Logistic regression with an L_1 penalty (Logistic-Lasso) instead of the classical linear-regression Lasso, in order to directly match the probabilistic structure of the classification problem. The parameter to be optimised in this model is the regularisation constant C , the inverse of the penalty strength λ , which determines the degree of sparsity of the retained feature set.

Three parallel criteria are used to determine the optimal C^* . The first criterion is the mean AUC over 5-fold cross-validation (AUC_{CV}), which directly measures the model’s discriminative ability at each value of C on a grid of 40 points spaced evenly on the $\log_{10}$ scale from 10^{-3} to 10^1 . The second criterion is AUC_{BIC} , which further penalises AUC_{CV} according to the number of features selected:

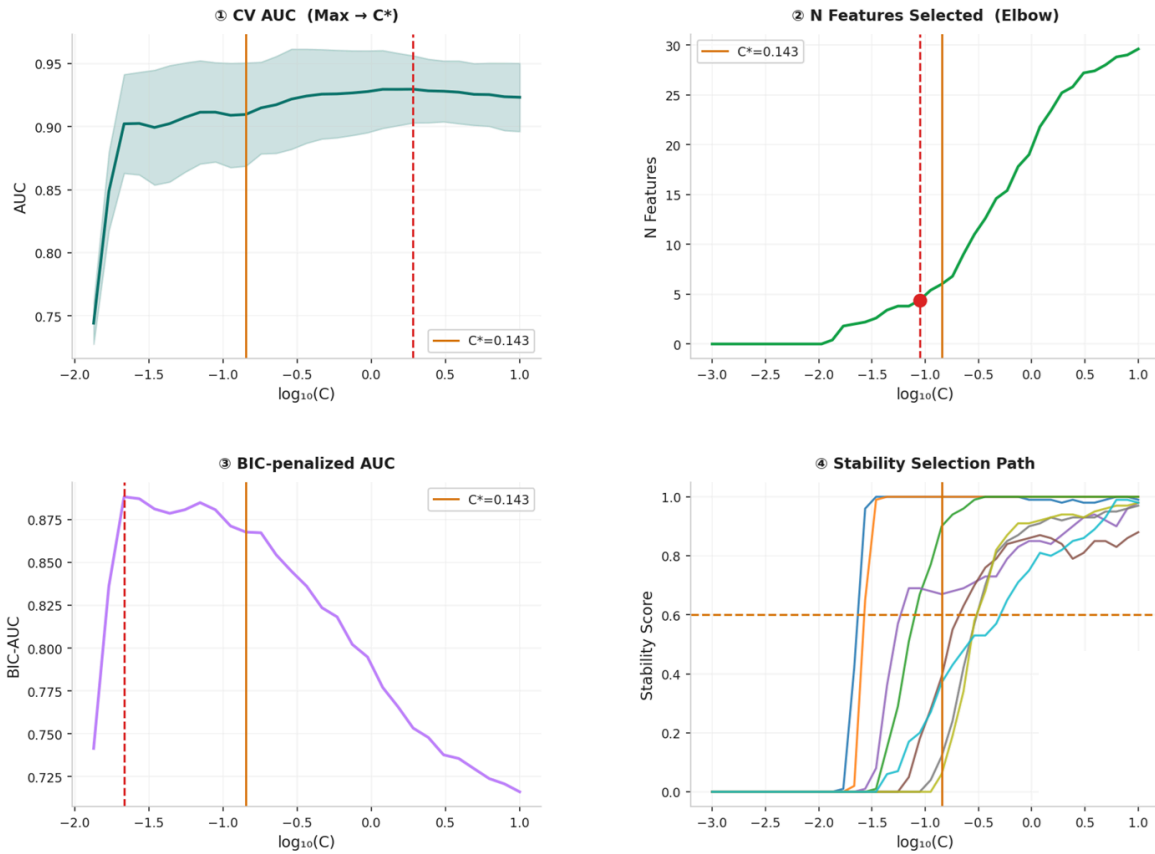
$$AUC_{BIC}(C) = AUC_{CV}(C) - \frac{|\mathcal{S}(C)|}{2n} \ln(n) \quad (4.1)$$

where $|\mathcal{S}(C)|$ is the number of features with a non-zero coefficient at the level C and n is the size of the training set; the penalty term plays a role similar to the BIC criterion, encouraging the sparsest possible model that still maintains an acceptable AUC. The

third criterion is the elbow point on the curve of the number of features against C , identified using the Kneedle algorithm — detecting the position at which a further increase in C no longer yields a meaningful increase in the number of features selected. The final value C^* is the geometric mean of the three candidates:

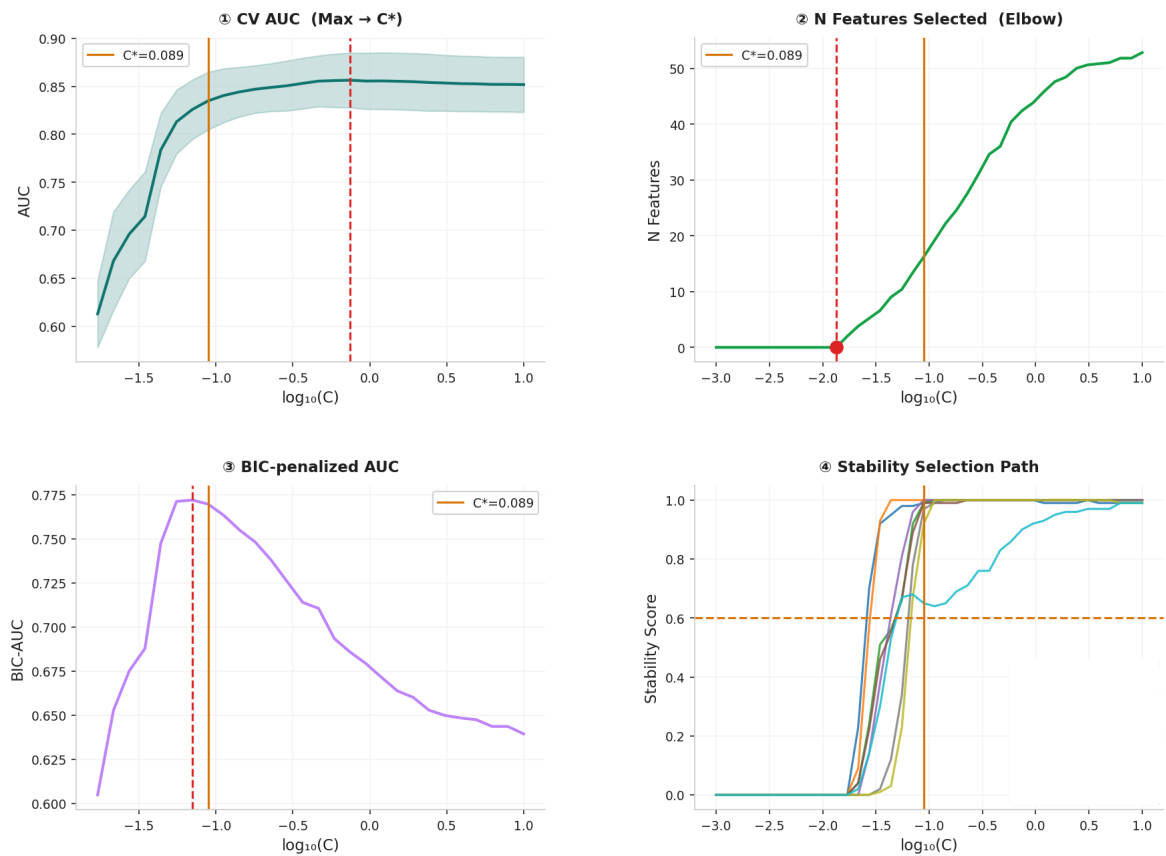
$$C^* = \exp \left(\frac{\ln C_{\text{AUC}} + \ln C_{\text{BIC}} + \ln C_{\text{elbow}}}{3} \right) \quad (4.2)$$

In addition to the three criteria above, Stability Selection with 100 bootstrap iterations and a stability threshold of 0.6 [46] is carried out in parallel to provide additional diagnostic information on the reliability of each feature across different data subsets. All four of these components are visualised on the same grid of C values, as shown in Figures 4.11, 4.12 and 4.13, allowing the convergence point of the criteria to be observed directly before being aggregated into C^* .



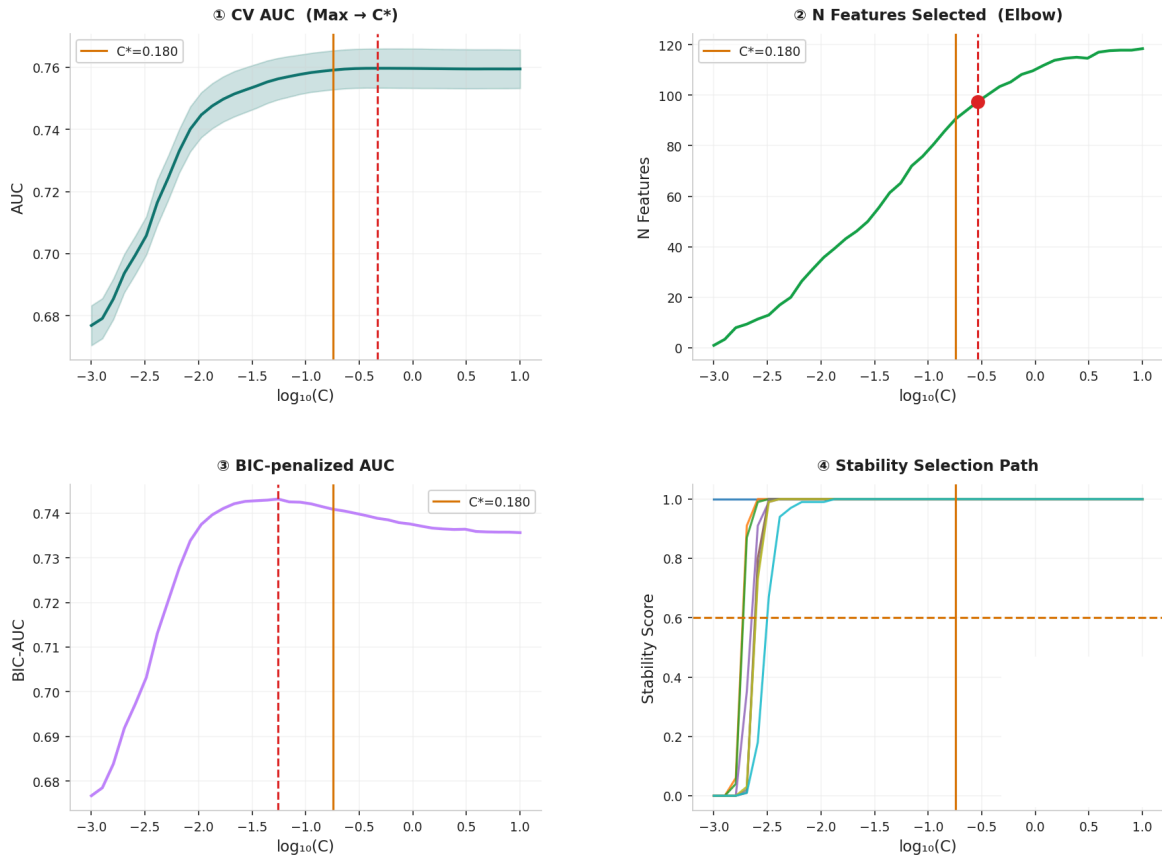
Hình 4.11: Optimisation process for the regularisation parameter C over the criteria grid on Australian Credit

(Where blue = `CreditScore`, orange = `CreditHistory_1`, green = `Income`, purple = `YearsEmployed`, brown = `Citizen_2`, grey = `Age`, olive = `Employed_1`, cyan = `Debt`; the dashed horizontal line is the `Threshold = 0.6`)



Hình 4.12: Optimisation process for the regularisation parameter C over the criteria grid on German Credit

(Where blue = month_duration, orange = age, green = secondary_obligor_2.0, purple = housing_1.0, brown = other_installment_plans_1.0, grey = status_account_3, olive = purpose_7.0, cyan = credit_amount; the dashed horizontal line is the Threshold = 0.6)



Hình 4.13: Optimisation process for the regularisation parameter C over the criteria grid on Lending Club

(Where blue = `int_rate`, orange = `acc_open_past_24mths`, green = `dti`, purple = `fico_range_low`, brown = `pub_rec_bankruptcies`, grey = `purpose_2.0`, olive = `delinq_2yrs`, cyan = `mths_since_recent_bc`; the dashed horizontal line is the Threshold = 0.6)

The three candidate values of C and the aggregated C^* for the three datasets are presented in Table 4.4.

Bảng 4.4: Value of the parameter C according to each criterion and the aggregated C^*

Dataset	C_{AUC}	C_{BIC}	C_{elbow}	C^*
Australian	1.9145	0.0215	0.0889	0.1425
German	0.7444	0.0702	0.0134	0.0889
Lending Club	0.4642	0.0554	0.2894	0.1805

It can be seen that C_{AUC} is always considerably larger than C_{BIC} across all three

datasets, reflecting the fact that pure AUC maximisation tends to retain a less sparse model than when a complexity penalty term is included. In the case of Lending Club specifically, C_{elbow} is even larger than C_{AUC} , showing that the feature-count curve for this dataset decreases slowly and the elbow point appears later than for the other two datasets, consistent with the particularly large input space of Lending Club. Taking the geometric mean helps prevent C^* from being entirely dominated by the most extreme of the three candidate criteria.

After determining C^* , a secondary filter based on Information Value (IV) is applied to remove variables whose actual discriminative power is too weak, even though they were retained by Lasso:

$$\text{IV} = \sum_i (\text{Dist}_{\text{Event},i} - \text{Dist}_{\text{Non-event},i}) \times \ln \left(\frac{\text{Dist}_{\text{Event},i}}{\text{Dist}_{\text{Non-event},i}} \right) \quad (4.3)$$

where i is the bin index of the feature variable after discretisation. The entire three-stage procedure — determining C^* , extracting the Lasso features, and secondary filtering using IV — is summarised in Algorithm 6.

On the Australian dataset, Lasso at $C^* = 0.1425$ selected 9 original features. After the IV filter with a threshold of 0.04, the variable **Gender** was excluded, as its IV reached only 0.0001, leaving a final feature set of 8 variables. On the German dataset, Lasso at $C^* = 0.0889$ retained 15 features; all of them exceeded the IV threshold of 0.05, led by **secondary_obligor** (IV = 0.6419), **other_installment_plans** (IV = 0.6278) and **purpose** (IV = 0.6040), so no variable was excluded. On the Lending Club dataset, Lasso at $C^* = 0.1805$ selected 45 out of 46 variables; the IV filter with a threshold of 0.03 further excluded 18 variables, leaving 27 features, among which **num_tl_op_past_12m** (IV = 1.1613), **inq_last_6mths** (IV = 0.9930) and **num_actv_rev_tl** (IV = 0.7156) are the three most discriminative variables, reflecting that recent credit behaviour and the level of account activity are the dominant risk signals in this dataset.

The results after feature selection are summarised in Table 4.5.

Bảng 4.5: Feature selection results after Lasso

Dataset	C^*	Lasso selected	IV threshold	After IV filter
Australian	0.1425	9	0.04	8
German	0.0889	15	0.05	15
Lending Club	0.1805	45	0.03	27

Algorithm 6 Pseudocode of the L_Pro feature selection procedure

Input: Training set $(X_{\text{train}}, y_{\text{train}})$ after S_Pro; parameter grid $\mathcal{C} = \{C_1, C_2, \dots, C_{40}\}$ spaced evenly on the $\log_{10}$ scale from 10^{-3} to 10^1 ; number of folds $K = 5$; number of bootstraps $B = 100$; stability threshold $\tau = 0.6$; IV threshold δ .

Output: Final feature set $\mathcal{F}^*$ to be passed into K_Pro.

```

// Stage 1: Determining the optimal  $C^*$ 
1: for each  $C \in \mathcal{C}$  do
2:   Train Logistic-Lasso with an  $L_1$  penalty and parameter  $C$  via  $K$ -fold
   cross-validation
3:    $\text{AUC}_{\text{CV}}(C) \leftarrow$  mean AUC over the  $K$  folds
4:    $|\mathcal{S}(C)| \leftarrow$  number of features with  $|\hat{\beta}_j| > 0$ , averaged over the  $K$  folds
5:    $\text{AUC}_{\text{BIC}}(C) \leftarrow \text{AUC}_{\text{CV}}(C) - \frac{|\mathcal{S}(C)|}{2n} \ln(n)$ 
6:   for  $b = 1$  to  $B$  do ▷ Stability Selection
7:     Randomly subsample 50% from  $(X_{\text{train}}, y_{\text{train}})$ 
8:     Train Lasso at  $C$ ; record the selected feature set  $\mathcal{S}^{(b)}(C)$ 
9:   end for
10:   $\Pi_j(C) \leftarrow \frac{1}{B} \sum_{b=1}^B \mathbf{1}[j \in \mathcal{S}^{(b)}(C)]$  for each feature  $j$  ▷ Selection frequency
11: end for
12:  $C_{\text{AUC}} \leftarrow \arg \max_C \text{AUC}_{\text{CV}}(C)$ 
13:  $C_{\text{BIC}} \leftarrow \arg \max_C \text{AUC}_{\text{BIC}}(C)$ 
14:  $C_{\text{elbow}} \leftarrow$  elbow point on the curve  $|\mathcal{S}(C)|$  using the Kneedle algorithm
15:  $C^* \leftarrow \exp\left(\frac{\ln C_{\text{AUC}} + \ln C_{\text{BIC}} + \ln C_{\text{elbow}}}{3}\right)$ , rounded to the nearest value in  $\mathcal{C}$ 

// Stage 2: Extracting features at  $C^*$ 
16: Train Lasso on the full  $X_{\text{train}}$  at  $C^*$ 
17:  $\mathcal{S}^* \leftarrow \{j : |\hat{\beta}_j| > 0\}$  ▷ Preliminary Lasso feature set

// Stage 3: Secondary filtering using Information Value
18: for each feature  $j \in \mathcal{S}^*$  do
19:   Discretise  $X_j$  into bins; compute  $\text{IV}_j =$ 
    $\sum_i (\text{Dist}_{\text{event},i} - \text{Dist}_{\text{non-event},i}) \ln\left(\frac{\text{Dist}_{\text{event},i}}{\text{Dist}_{\text{non-event},i}}\right)$ 
20: end for
21:  $\mathcal{F}^* \leftarrow \{j \in \mathcal{S}^* : \text{IV}_j \geq \delta\}$ 
22: return  $\mathcal{F}^*$ 

```

4.2.5 Data Discretisation (K_Pro)

As presented in Section 3.1.3, discretising continuous variables is a prerequisite for the Bayesian network to be able to compute conditional probability tables through frequency statistics, since for a continuous variable with infinitely many values, each parent-child combination would appear almost exactly once in the training data, rendering maximum likelihood estimation meaningless due to the lack of repeated observations. However, applying the K-means algorithm directly according to Equation 3.4 to the original financial variables — which have heavily skewed distributions owing to the nature of credit data (income, outstanding debt, and credit balances are typically densely concentrated at low levels with a long tail towards high values) — would lead to clusters dominated by a few extreme outliers, causing most observations to be crowded into one or two bins while the remaining bins are almost empty.

To overcome this limitation, the experimental procedure adds a distribution-transformation step before feeding the data into K-means according to Algorithm 2 presented in Section 3.1.3. This additional step applies the Yeo-Johnson power transformation, defined by four cases depending on the sign of the original value x and the shape parameter λ :

$$\psi(x; \lambda) = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda}, & \lambda \neq 0, x \geq 0 \\ \ln(x+1), & \lambda = 0, x \geq 0 \\ -\frac{(-x+1)^{2-\lambda} - 1}{2-\lambda}, & \lambda \neq 2, x < 0 \\ -\ln(-x+1), & \lambda = 2, x < 0 \end{cases} \quad (4.4)$$

where the parameter λ is estimated by the maximum likelihood method on the training set, such that the distribution of $\psi(x; \lambda)$ approaches as closely as possible a normal distribution [78]. The first two branches handle non-negative values using the same mechanism as the Box-Cox transformation with an added shift constant, while the latter two branches extend the formula to the negative domain by reflection through the origin, thereby overcoming the inherent limitation of Box-Cox, which is only defined on the positive domain. This transformation was chosen for its ability to handle both negative and zero values, which commonly occur in variables such as debt differences or the number of months since the most recent event, while also flattening skewness and reducing the influence of outliers without losing information, unlike traditional capping techniques.

For the Lending Club dataset, because the scales among the variables differ very widely — for instance, `loan_amnt` is measured in thousands of dollars while `dti` is measured as a percentage — a `StandardScaler` normalisation step was performed before Yeo-Johnson in order to bring the variables onto the same scale, avoiding a situation in which a variable with large variance dominates the estimation of the transformation

parameters for the other variables. After the data distributions were normalised, the K-means algorithm according to Algorithm 2 was applied to the transformed space with a fixed number of bins $k = 4$ for all continuous variables. The number of bins $k = 4$ was chosen to balance two opposing objectives: maintaining sufficient resolution for the bins to reflect differences in credit risk across value levels, while avoiding excessive fragmentation that would cause some parent-child state combinations in the Bayesian network to fall into sparse data regions. The boundaries of each bin were then mapped back to the original value scale by taking the actual minimum and maximum values of the observations falling into that bin in the training set, ensuring that the reported value ranges accurately reflect the data distribution rather than the theoretical cluster boundaries in the transformed space.

The discretisation results for the Australian dataset are presented in Table 4.6. The variable **Debt** was divided into four intervals $[0.00; 1.08)$, $[1.08; 3.29)$, $[3.29; 8.00)$ and $[8.00; 28.00]$, in which the default rate decreases progressively from 54.78% in the lowest bin to 32.41% in the highest bin — a somewhat counterintuitive trend if judged by the ordinary logic that higher debt implies greater risk, though it can be explained by the fact that large debts are typically associated with credit profiles that have already been carefully appraised and backed by corresponding collateral. The variable **YearsEmployed** shows a clearer monotonic relationship, with the default rate falling sharply from 72.16% in the group with less than 0.54 years of employment to just 19.23% in the group with more than 4.33 years, correctly reflecting the intuition that stable job tenure is inversely correlated with credit risk. The variable **CreditScore** shows the strongest separation in event rate among the continuous variables in this dataset, from 71.61% in the lowest bin down to only 9.18% in the highest bin, reinforcing its role as a leading risk indicator.

The discretisation results for the German dataset are presented in Table 4.7. On the German dataset, the variable **month_duration** — the loan term in months — shows a positive relationship with default risk, rising from 14.00% in the short-term loan group (4-9 months) to 70.79% in the long-term loan group (31-60 months), consistent with financial theory that a longer loan term leads to greater accumulated credit risk due to a longer period of exposure to income shocks. The variable **age** shows the opposite trend, with the default rate falling from 57.38% in the 19-25 age group to 32.74% in the group over 43 years old, reflecting increasing financial stability with age. The variable **credit_amount** does not show a monotonic relationship, fluctuating between 43.65%-62.15% across the bins, suggesting that loan size alone is not a strong determinant of default likelihood in this dataset when separated from other variables.

The discretisation results for the Lending Club dataset are presented in Table 4.8. On the Lending Club dataset, the variable **int_rate** — the loan interest rate — shows the clearest and strongest positive relationship among all the continuous variables examined, with the default rate rising from 22.54% in the lowest interest-rate bin (6.00%-10.80%) to 69.09% in the highest interest-rate bin (19.59%-26.06%), consistent with the risk-pricing mechanism of peer-to-peer lending platforms, whereby higher interest rates are assigned to loans assessed as higher risk from the initial appraisal stage. The

variable `dti` (debt-to-income ratio) also shows a consistent positive trend, from 38.02% to 56.03% across the four bins, in line with the expectation that a relatively high debt burden compared with income undermines repayment ability. Conversely, variables reflecting the depth of credit history, such as `mo_sin_rcnt_tl` (number of months since the most recent credit account was opened) and `mths_since_recent_bc`, show an inverse relationship at the highest bin, with the default rate falling to 35.88% and 41.51% respectively — showing that customers who have not opened a new credit account for a long time tend to be more financially stable. Notably, the variable `revol_util` in this dataset produces a bin containing only a single observation with a 100% default rate, a phenomenon typical of K-means-based discretisation when an extreme outlier exists that is clearly separated from the rest of the distribution even after the Yeo-Johnson transformation.

The entire power-transform and K-means discretisation procedure is fitted (parameters estimated) solely on the training set, after which the corresponding transform is applied to the test set, in order to avoid information leakage from the test set into the process of determining the bin boundaries.

Bảng 4.6: Description of the selected features of the Australian dataset after K_Pro

Variable Name	Description	States
CreditHistory	Applicant's prior credit history No, yes	0, 1
CreditScore	Applicant's credit score (0; 1], (1; 2], (2; 4], (4; 40]	0, 1, 2, 3
Employed	Employment status at the time of application No, yes	0, 1
YearsEmployed	Number of years employed at current job (0.00; 0.54], (0.54; 1.96], (1.96; 4.33], (4.33; 28.50]	0, 1, 2, 3
Income	Applicant's income (1; 5], (5; 51], (51; 679], (679; 51101]	0, 1, 2, 3
Debt	Applicant's total current outstanding debt (0.00; 1.08], (1.08; 3.29], (3.29; 8.00], (8.00; 28.00]	0, 1, 2, 3

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Bảng 4.6 (continued)

Variable Name	Description	States
Age	Applicant's age (13.75; 21.67], (21.67; 29.17], (29.17; 41.00], (41.00; 80.25]	0, 1, 2, 3
Citizen	Applicant's citizenship/residency status	1, 2, 3
target	Credit application approval status Default, non-default	0, 1

Bảng 4.7: Description of the selected features of the German Credit dataset after K_Pro

Variable Name	Description	States
status_ account	Customer's current account status at the bank < 0 DM, $0 \leq \dots < 200$ DM, ≥ 200 DM, no checking account	0, 1, 2, 3
month_ duration	Loan term in months (4; 9], (9; 17], (17; 31], (31; 60]	0, 1, 2, 3
credit_ history	Customer's credit history No credits taken/all credits paid back duly, all credits at this bank paid back duly, existing credits paid back duly till now, delay in paying off in the past, critical account/other credits existing	0, 1, 2, 3, 4
purpose	Loan purpose Car (new), car (used), furniture/equipment, radio/television, domestic appliances, repairs, education, retraining, business, others	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
credit_amount	Loan amount (250; 1068], (1068; 2279], (2279; 5433], (5433; 18424]	0, 1, 2, 3

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Bảng 4.7 (continued)

Variable Name	Description	States
status_and_ sex	Marital status and sex Male: divorced/separated, female: divorced/separated/married, male: single, male: married/widowed	0, 1, 2, 3
secondary_ obligor	Guarantor or co-applicant None, co-applicant, guarantor	0, 1, 2
residence_ since	Number of years at current residence	1, 2, 3, 4
collateral	Collateral Real estate, building society savings agreement/life insurance, car or other, unknown/no property	0, 1, 2, 3
age	Customer's age (19; 25], (25; 32], (32; 43], (43; 75]	0, 1, 2, 3
other_ installment_ plans	Other existing instalment plans Bank, stores, none	0, 1, 2
housing	Housing status Rent, own, for free	0, 1, 2
n_credits	Number of existing credits at this bank	1, 2, 3, 4
job	Employment status Unemployed/unskilled non-resident, unskilled resident, skilled employee/official, management/self-employed/highly qualified	0, 1, 2, 3
is_foreign_ worker	Foreign worker status Yes, no	0, 1
target	Customer's credit status Good, bad	0, 1

Bảng 4.8: Description of the selected features of the Lending Club 2007-2014 dataset after K_Pro

Variable Name	Description	States
term	Repayment term of the loan in months 36 months, 60 months	0, 1
grade	Credit grade assigned by Lending Club to the loan (A to G)	0, 1, 2, 3, 4, 5, 6
emp_length	Borrower's number of years employed Unknown, < 1 year, 1 year, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, 8 years, 9 years, 10+ years	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
verification_status	Verification status of the borrower's income source Not Verified, Verified, Source Verified	0, 1, 2
home_ownership	Borrower's home ownership status MORTGAGE, OTHER, OWN, RENT	0, 1, 2, 3
purpose	Loan purpose as declared by the borrower car, credit_card, debt_consolidation, home_improvement, house, major_purchase, medical, moving, other, renewable_energy, small_business, vacation, wedding	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
debt_settlement_flag	Indicates whether the loan is currently under a debt settlement programme	0, 1
loan_amnt	Loan amount requested by the borrower at the time of application (1000.00; 7713.40], (7713.40; 13649.90], (13649.90; 22325.00], (22325.00; 35000.00]	0, 1, 2, 3
annual_inc	Borrower's self-reported annual income (3000.00; 43264.00], (43264.00; 66296.67], (66296.67; 106453.62], (106453.62; 2000000.00]	0, 1, 2, 3

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Bảng 4.8 (continued)

Variable Name	Description	States
int_rate	Interest rate of the loan (6.00; 10.80], (10.80; 15.11], (15.11; 19.59], (19.59; 26.06]	0, 1, 2, 3
dti	Ratio of total monthly debt (excluding this loan and mortgage) to self-reported monthly income (0.00; 11.50], (11.50; 18.53], (18.53; 25.82], (25.82; 39.99]	0, 1, 2, 3
fico_range_ low	Lower bound of the borrower's FICO score range at the time of loan origination (660.00; 675.58], (675.58; 695.41], (695.41; 726.54], (726.54; 845.00]	0, 1, 2, 3
revol_util	Ratio of revolving credit line utilisation to total credit limit (0.00; 45.45], (45.45; 71.26], (71.26; 366.60], = 366.60	0, 1, 2, 3
tot_cur_bal	Total current balance across all credit accounts (0.00; 48110.00], (48110.00; 115417.00], (115417.00; 246217.00], (246217.00; 2179304.00]	0, 1, 2, 3
bc_open_to_ buy	Available credit limit on bankcard accounts (0.00; 1824.00], (1824.00; 4873.00], (4873.00; 11413.00], (11413.00; 168099.00]	0, 1, 2, 3
mo_sin_ rcnt_rev_ tl_op	Number of months since the most recent revolving credit account was opened (0.00; 4.54], (4.54; 9.99], (9.99; 21.11], (21.11; 217.00]	0, 1, 2, 3

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Bảng 4.8 (continued)

Variable Name	Description	States
mo_sin_ rcnt_tl	Number of months since the most recent credit account was opened (0.00; 3.44], (3.44; 7.13], (7.13; 14.05], (14.05; 102.00]	0, 1, 2, 3
mths_since_ recent_bc	Number of months since the most recent bankcard account was opened (0.00; 8.25], (8.25; 18.03], (18.03; 38.10], (38.10; 521.00]	0, 1, 2, 3
mths_since_ recent_inq	Number of months since the most recent credit inquiry (0.00; 2.60], (2.60; 6.34], (6.34; 12.26], (12.26; 24.00]	0, 1, 2, 3
inq_last_ 6mths	Number of credit inquiries in the last 6 months (0.00; 0.41], (0.41; 1.37], (1.37; 2.54], (2.54; 6.00]	0, 1, 2, 3
acc_open_ past_24mths	Number of credit accounts opened in the past 24 months (0.00; 3.33], (3.33; 6.03], (6.03; 9.38], (9.38; 50.00]	0, 1, 2, 3
mort_acc	Number of existing mortgage accounts (0.00; 1.09], (1.09; 3.23], (3.23; 6.03], (6.03; 24.00]	0, 1, 2, 3
num_accts_ ever_120_pd	Number of accounts ever 120 days or more past due (0.00; 0.08], (0.08; 0.30], (0.30; 0.71], (0.71; 21.00]	0, 1, 2, 3
num_actv_ rev_tl	Number of currently active revolving credit accounts (0.00; 3.64], (3.64; 6.21], (6.21; 10.11], (10.11; 31.00]	0, 1, 2, 3
num_bc_tl	Number of current bankcard accounts (1.00; 5.49], (5.49; 9.18], (9.18; 14.42], (14.42; 44.00]	0, 1, 2, 3
num_il_tl	Number of current instalment credit accounts (1.00; 4.29], (4.29; 8.29], (8.29; 15.31], (15.31; 97.00]	0, 1, 2, 3

Continued on next page...

Bảng 4.8 (continued)

Variable Name	Description	States
num_tl_op_ past_12m	Number of credit accounts opened in the past 12 months (0.00; 1.42], (1.42; 3.16], (3.16; 5.46], (5.46; 17.00]	0, 1, 2, 3
target	Default status of the loan No, yes	0, 1

4.3 Building the Bayesian Network and Prediction (BN_Pro)

4.3.1 Building the Network Structure and Prediction

The core library for building and performing inference in the Bayesian network is pgmpy, using the DiscreteBayesianNetwork and HillClimbSearch classes (the Tabu Search algorithm, with the DAG structure learned by Tabu Search optimising the BIC score for discrete data (bic-d)), with 50 independent training runs ($N_RUNS = 50$), each initialised from a different random DAG (random restart) to avoid local optima.

Configuration:

- Number of training runs: $N_RUNS = 50$
- Tabu memory (tabu_length) = 100 acts as a short-term memory storing the edge addition, deletion, and reversal actions that the algorithm has performed in the most recent 100 steps.
- Maximum in-degree per node (max_indegree) = 5 is the maximum number of arrows pointing directly into a node, equivalent to the maximum number of parent nodes for that node, in order to control the size of the CPT.
- Maximum number of search steps per run (max_iter) = 1000.
- Initial edge generation probability (edge_prob) = 0.3 (set to a low probability ≤ 0.5 to avoid generating a very dense graph, with nodes intricately connected to one another, which would increase the complexity of the graph. For a directed acyclic graph with n nodes, the number of pairs of nodes that could possibly be connected is $M_{max} = \binom{n}{2} = \frac{n(n-1)}{2}$, and the average number of edges appearing in the initial generated graph is

$$E(M) = edge_prob \times M_{max} = edge_prob \times \frac{n(n-1)}{2} \quad (4.5)$$

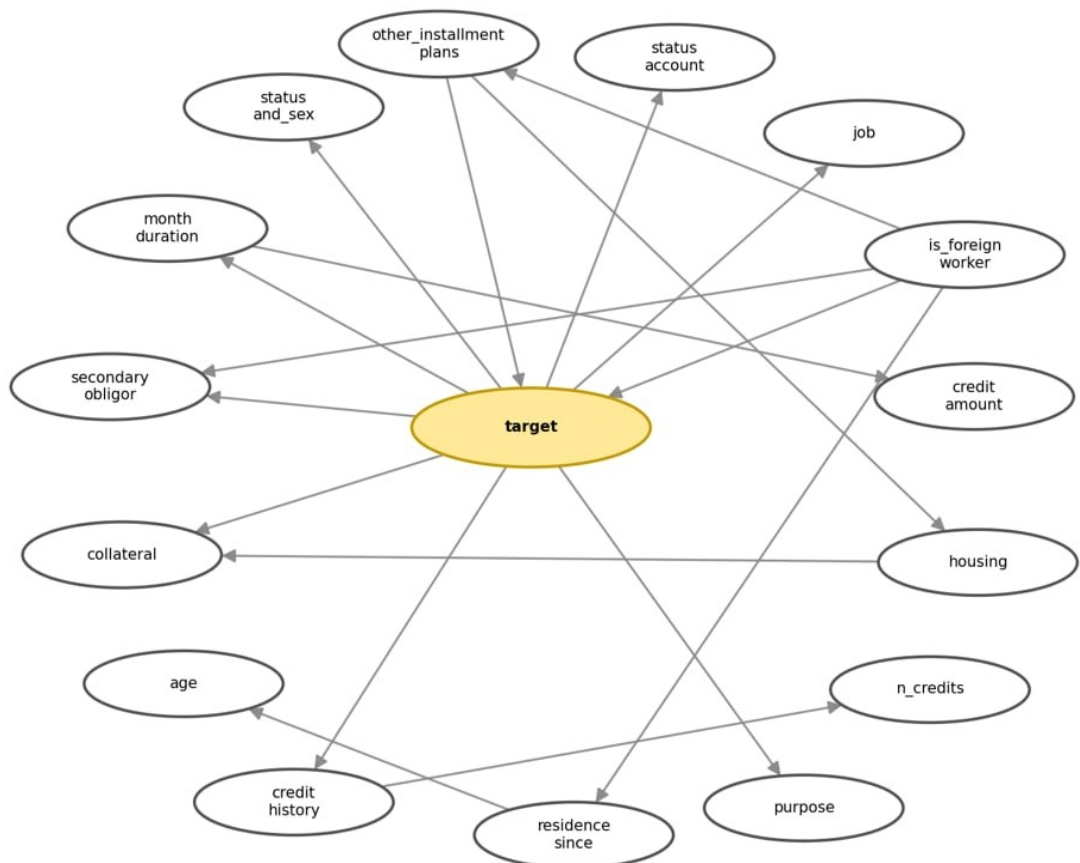
After 50 runs, the DAG with the highest BIC score (least negative) was selected as the final structure. Table 4.9 summarises the structure-learning results on the three datasets.

Bảng 4.9: DAG structure optimisation results using the Tabu search algorithm

Dataset	Best BIC	Best run	No. of nodes	No. of edges	BIC variation across runs
Australian Credit	-3343.32	15	9	10	≈ 44 units (-3387 $\rightarrow$ -3343)
German Credit	-14755.32	20	16	18	≈ 114 units (-14869 $\rightarrow$ -14755)
Lending Club	-725970.15	14	28	58	$\approx 3,500$ units (-729492 $\rightarrow$ -725970)

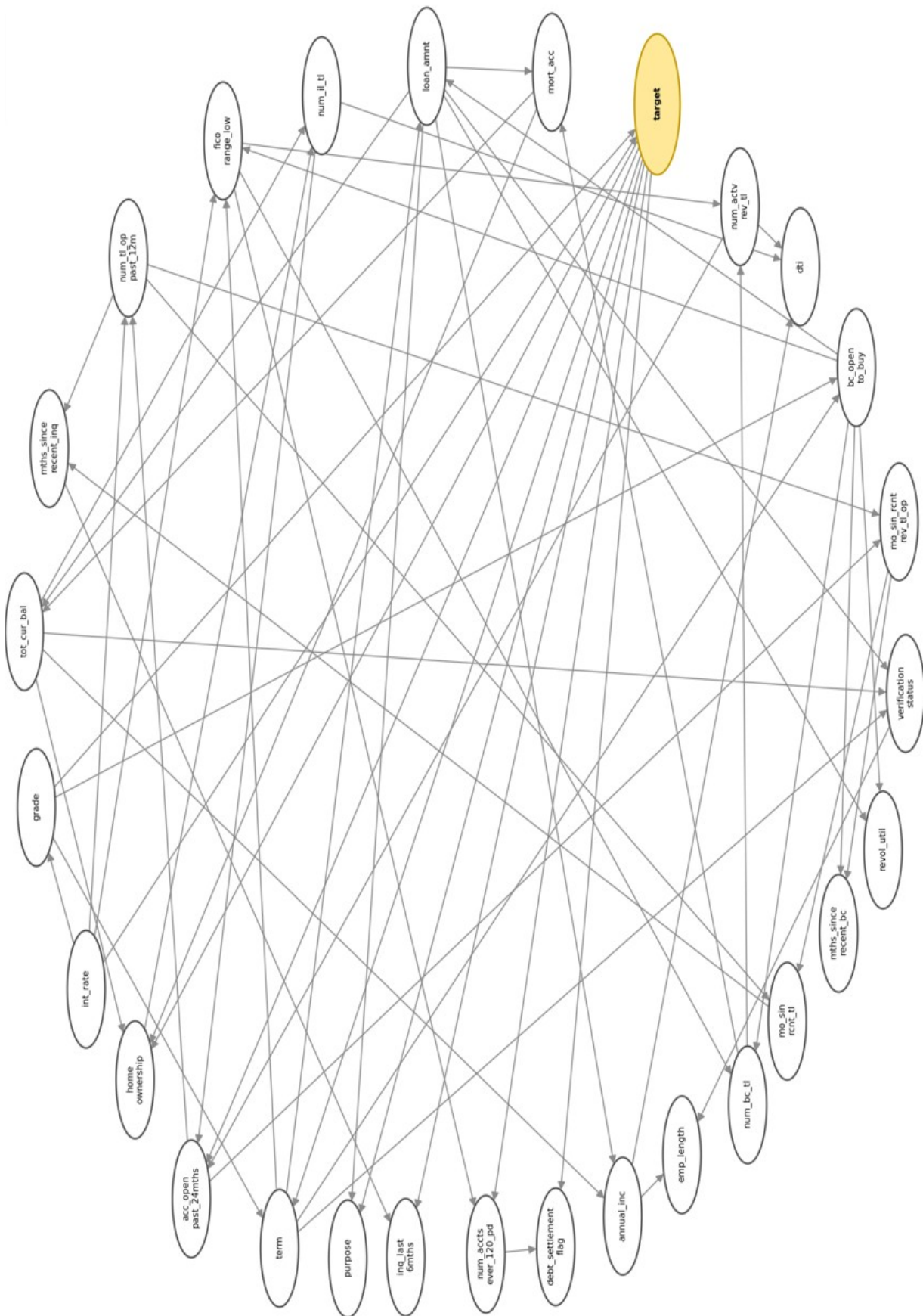


(a) Bayesian network structure diagram for the Australian Credit dataset



(b) Bayesian network structure diagram for the German Credit dataset

Hình 4.14: Bayesian network structure diagram (Part 1)



(c) Bayesian network structure diagram for the Lending Club dataset

Hình 4.14: Bayesian network structure diagram (continued).

Figure 4.14 shows that the complexity of the graph structure is directly proportional to the number of input features: Lending Club, with 27 features, learned a dense DAG with as many as 58 edges, while Australian Credit, with only 8 features, learned a lightweight structure with 10 edges. The stability of the optimisation process (measured by the range of BIC score fluctuation across the 50 runs) also differs markedly: for Australian Credit, the amplitude of BIC fluctuation across runs is very narrow (only about 30 units around the optimal value), indicating a small structure search space and very stable, consistent convergence of the Tabu algorithm across different random initialisations. In contrast, for Lending Club, the amplitude of fluctuation is much larger (thousands of BIC units), reflecting a wide structure space and high sensitivity to the random starting point, an inevitable consequence of the large number of variables causing the space of feasible DAGs to grow exponentially (super-exponentially with the number of nodes).

Across all three learned structures, the target variable has many direct links (edges to or from target) with core credit features, correctly reflecting domain knowledge about credit risk:

- Lending Club: target is directly linked to `int_rate` (interest rate), `grade` (credit grade), `debt_settlement_flag`, `num_accts_ever_120_pd` (number of accounts ever 120 days past due), `inq_last_6mths` (number of credit inquiries in the last 6 months), `purpose`, `term`, `acc_open_past_24mths` and `home_ownership`.
- German Credit: target is directly linked to `status_account` (current account balance), `purpose`, `credit_history`, `job`, `collateral`, `secondary_obligor`, `month_duration` and `status_and_sex`, and is also directly influenced by `is_foreign_worker` and `other_installment_plans`.
- Australian Credit: target is directly and solely linked to three variables, `CreditHistory`, `Employed` and `Income`, reflecting the very compact causal structure of this dataset.

The consistency between the graph structure learned by the purely data-driven algorithm and traditional credit expert knowledge (e.g. credit history, interest rate, and income being decisive factors) is an important qualitative indicator that the BN model is not only capable of prediction but also of providing reasonable causal interpretation.

In addition to the pure graph structure, Figures 4.15, 4.16 and 4.17 present an overview of the Bayesian network for the three datasets together with the prior marginal distribution of each node, i.e. the probability distribution of each variable when no evidence has yet been entered, including the **target** variable at 50%/50% due to having undergone the SMOTE-NC data balancing step at S_{Pro} . This visualisation approach allows both the dependency structure and the distributional shape of each variable to be observed simultaneously on the same diagram, rather than having to look up each CPT table individually.

On Australian Credit (Figure 4.15), the marginal distribution shows that most customers belong to citizenship/residency state 2 (90.8%), while `CreditHistory` and

Employed are relatively balanced between the two states (42.9%/57.1% and 53.9%/46.1% respectively), correctly reflecting the nature of a dataset that has had its label balanced but whose input features retain their natural distribution.

On German Credit (Figure 4.16), notably **is_foreign_worker** is strongly skewed towards state 1 (83.0%) and **secondary_obligor** is strongly skewed towards state 2, "guarantor" (73.2%), showing that these two features have a clearly imbalanced natural distribution across the dataset as a whole — a factor to bear in mind when interpreting the forward/backward inference results related to these two variables in Sections 4.4.1 and 4.4.2, since the minority states (**is_foreign_worker**=0, **secondary_obligor**∈{0,1}) inherently have few supporting observations from the outset.

On Lending Club (Figure 4.17), **debt_settlement_flag** is extremely strongly skewed towards state 0 (98%), with only 2% of customers ever having participated in a debt settlement programme; this is precisely the root cause of the data-sparsity phenomenon whereby this variable repeatedly produces extreme probability values ($\delta = 0.5$, or a saturated 100% posterior probability). Similarly, **num_accts_ever_120_pd** is also strongly skewed towards state 0 (73.1%). Observing the marginal distribution in advance in this way provides a well-founded explanation for the anomalous results appearing throughout the model-interpretation experimental sections that follow, rather than having to work backwards from the results.

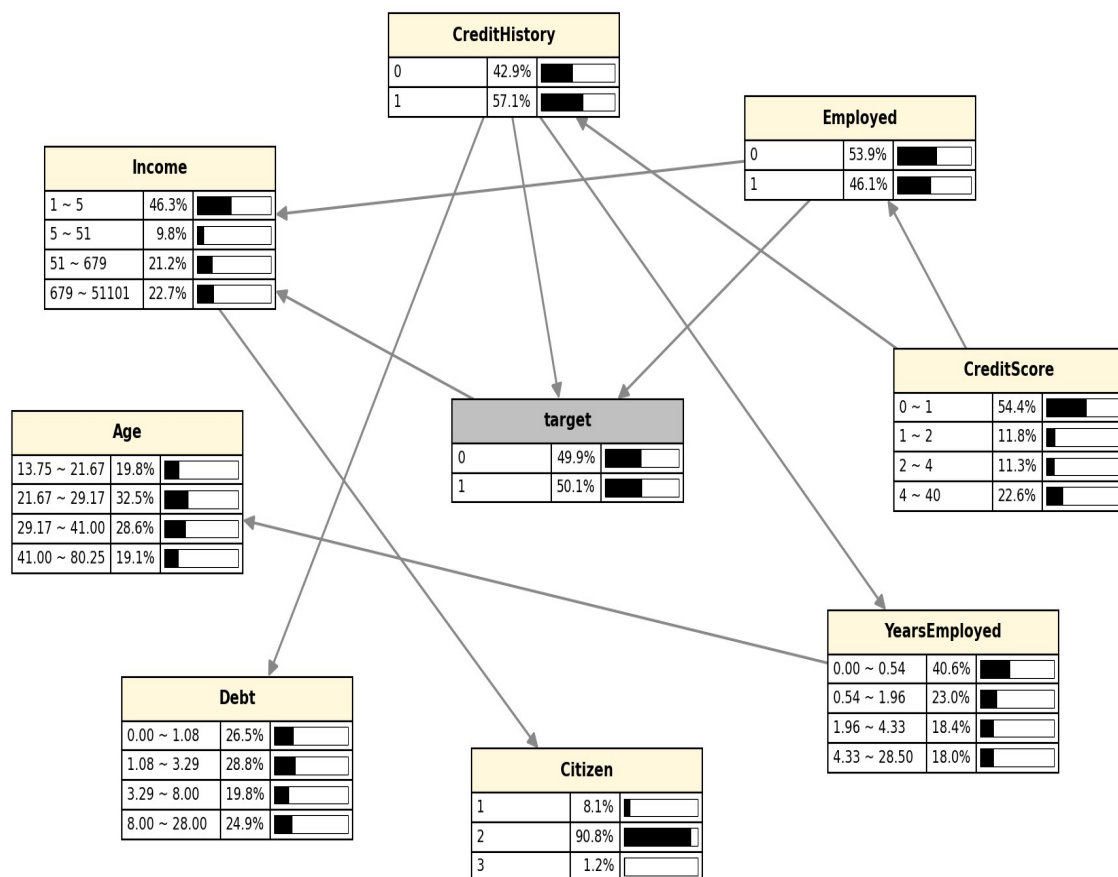
Bảng 4.10: Summary of Bayesian network model evaluation metrics on the training and test sets

Metric	Lending Club		German Credit		Australian Credit	
	Train	Test	Train	Test	Train	Test
Accuracy	0.7085	0.6808	0.7811	0.7300	0.8641	0.8406
Precision	0.7269	0.4610	0.8386	0.6047	0.9247	0.9211
Recall	0.6679	0.4878	0.6962	0.4127	0.7926	0.8140
Specificity	0.7490	0.7615	0.8660	0.8759	0.9355	0.8846
F1-Score	0.7034	0.4740	0.7608	0.4906	0.8536	0.8642
ROC-AUC	0.7944	0.6910	0.8686	0.7667	0.9278	0.9220
PR-AUC	0.8238	0.5224	0.8886	0.5421	0.9070	0.9371

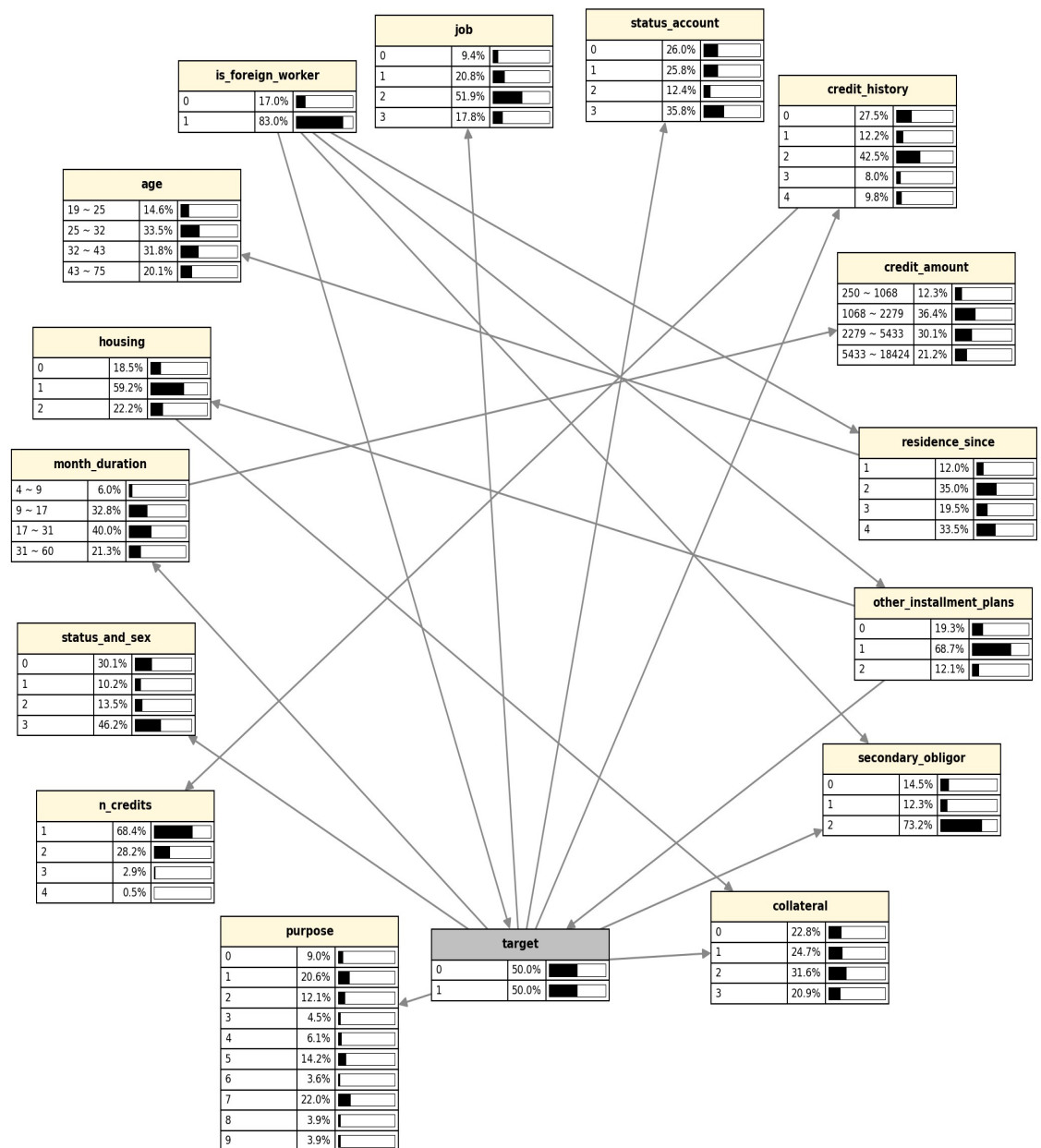
The results in Table 4.10 show a clear trend: the predictive performance of the BN model increases progressively in the direction Lending Club → German Credit → Australian Credit, with ROC-AUC values of 0.6910, 0.7667 and 0.9220 respectively. This can be explained from three perspectives. Regarding the intrinsic noise level of the data, Lending Club is real-world, large-scale consumer lending data in which default behaviour depends on many random factors outside the credit profile (sudden unemployment, macroeconomic fluctuations, etc.), whereas German and Australian

Credit are highly standardised UCI datasets that have already been processed and have clearer feature-label relationships. Regarding dimensional complexity, the greater the number of features (27 compared with 8), the more complex the CPT space of the Bayesian network becomes, which easily leads to data sparsity in the conditional probability tables and reduces the accuracy of parameter estimation. For Australian Credit, because the prevalence of the default label is high ($\approx 58\%$), the classification problem is somewhat easier statistically, since it does not face extreme imbalance on the test set.

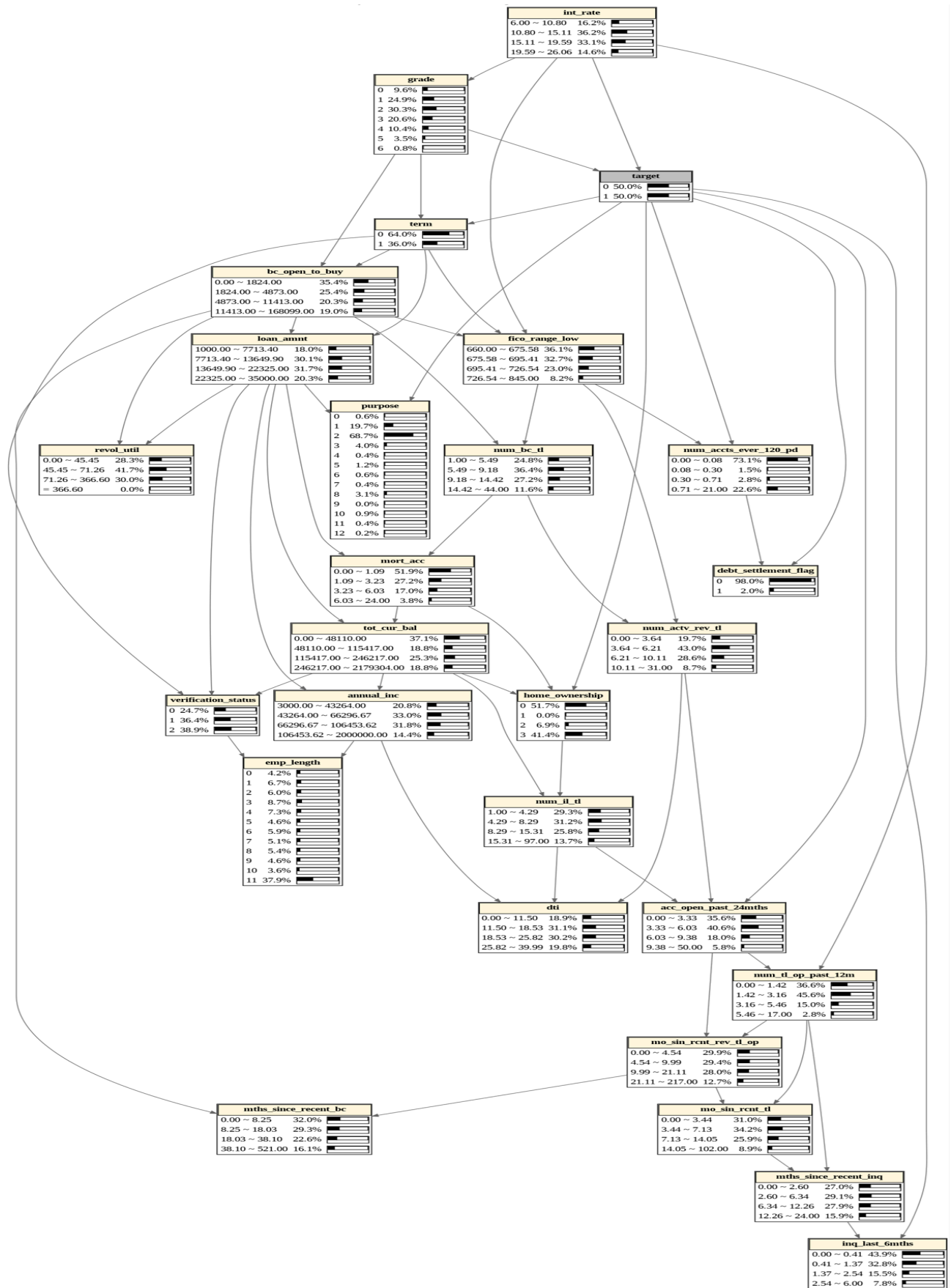
On the Lending Club dataset, the AUC on the training set reached 0.7944 while the AUC on the test set reached only 0.6910, a fairly large gap (≈ 0.1034), indicating that the model shows signs of mild to moderate overfitting with respect to the 58-edge graph structure learned from 27 variables. This is a reasonable consequence of the fact that a discrete Bayesian network with a CPT having a large number of state combinations (due to many variables and many edges) is prone to overfitting the empirical distribution of the training set. For Australian Credit, which has a more compact structure with fewer edges, the train-test gap is expected to be considerably narrower, owing to the model's degrees of freedom being much lower relative to the sample size.



Hình 4.15: Probability distribution of the entire network — Australian Credit



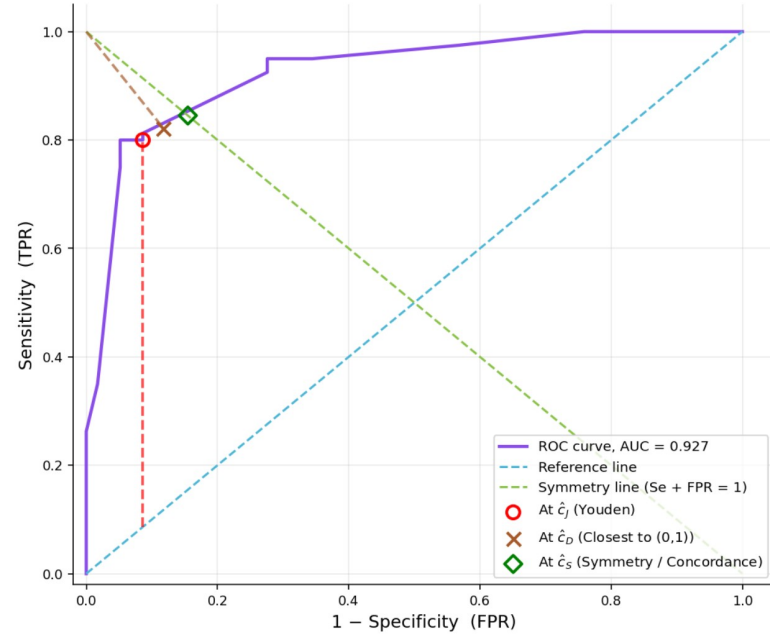
Hinh 4.16: Probability distribution of the entire network — German Credit



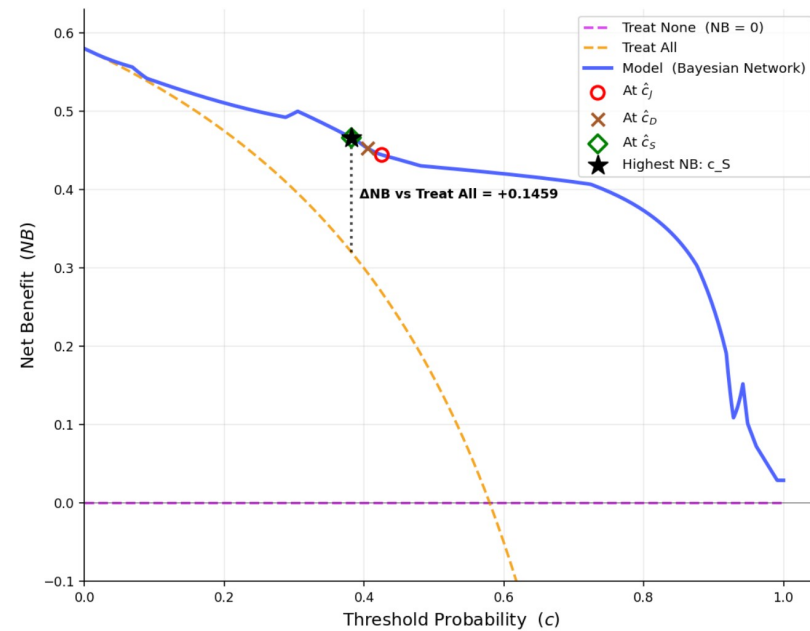
Hình 4.17: Probability distribution of the entire network — Lending Club

4.3.2 Analysis of the ROC Curve, Optimal Threshold, and Decision Curve Analysis on the Validation Set

Australian Credit



Hình 4.18: Plot of the three optimal decision thresholds according to ROC — Australian Credit



Hình 4.19: Plot of the three optimal decision thresholds on DCA — Australian Credit

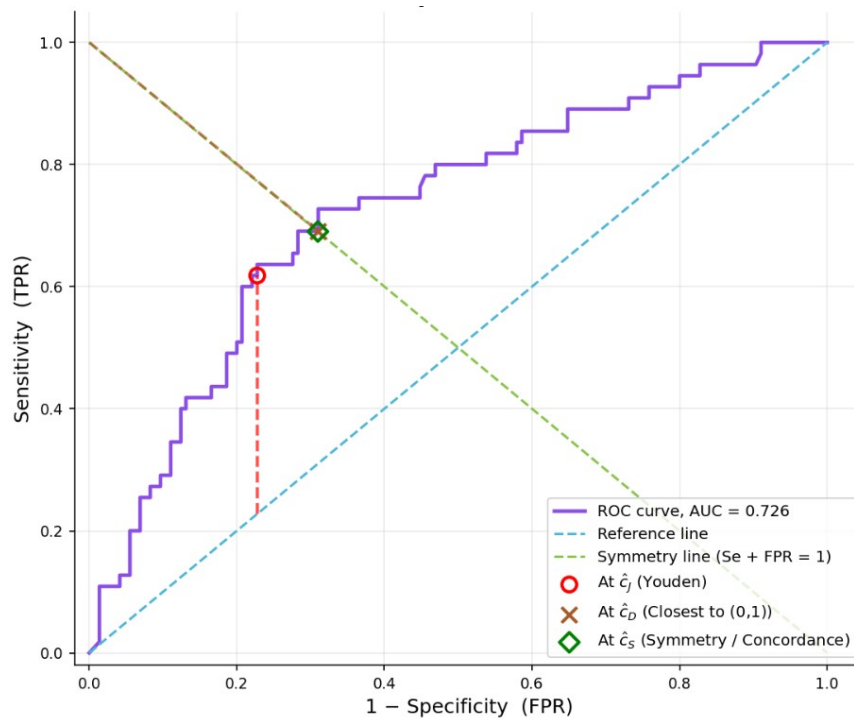
Bảng 4.11: Three optimal decision thresholds according to ROC – Australian Credit (validation set)

Threshold criterion	Threshold (c)	Sensitivity	Specificity	Accuracy	F1-Score
Youden Index	0.4254	0.8125	0.9138	0.8551	0.8667
Closest to (0,1)	0.4052	0.8125	0.9138	0.8551	0.8667
Symmetry Point	0.3818	0.9250	0.7241	0.8406	0.8706

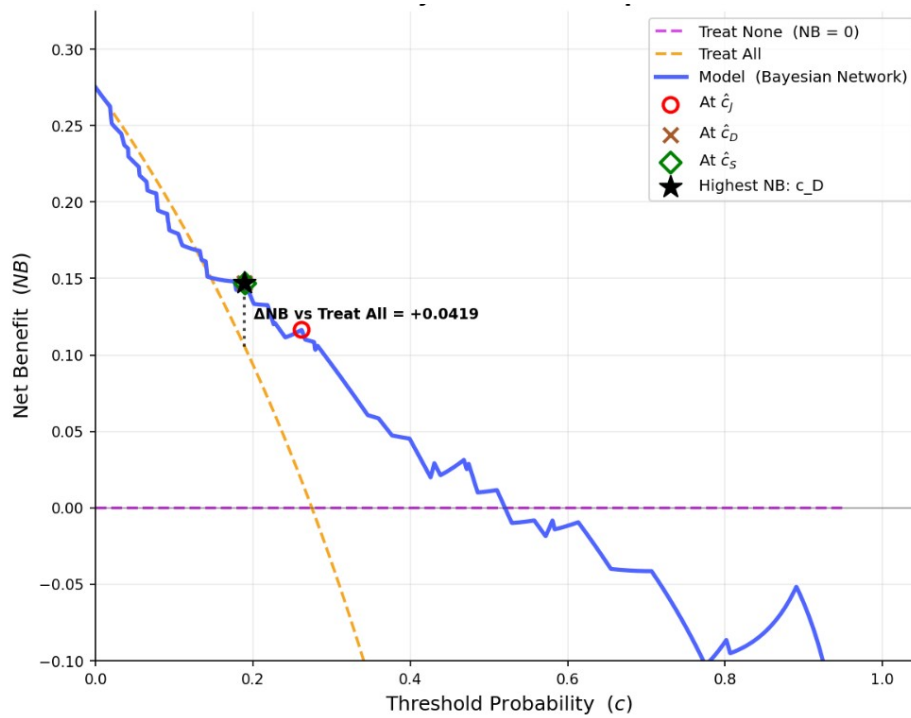
The DCA results based on Figures 4.18, 4.19 and Table 4.11: with prevalence = 0.5797, the Symmetry Point threshold ($c_S = 0.3818$) gives the highest Net Benefit among the three thresholds (NB = 0.4661), exceeding Treat All by 0.1459 points.

German Credit

The DCA results based on Figures 4.20, 4.21 and Table 4.12: prevalence (default rate) on the validation set = 0.2750. Among the three optimal ROC thresholds, the Closest-to (0,1) threshold ($c_D = 0.1895$) gives the highest Net Benefit (NB = 0.1474), exceeding Treat All by 0.0419 Net Benefit points.



Hình 4.20: Plot of the three optimal decision thresholds according to ROC — German Credit



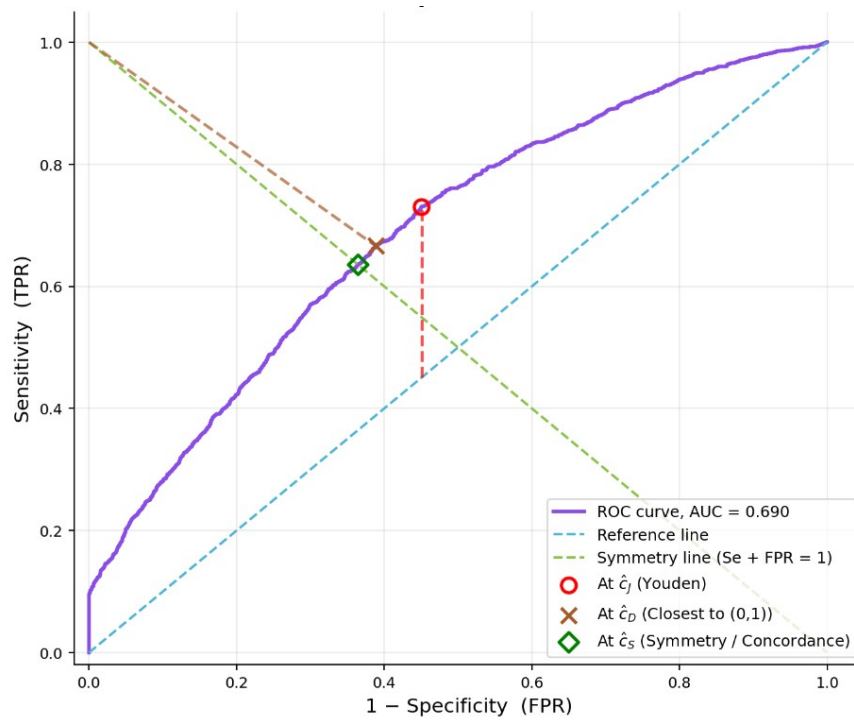
Hình 4.21: Plot of the three optimal decision thresholds on DCA — German Credit

Bảng 4.12: Three optimal decision thresholds according to ROC – German Credit (validation set)

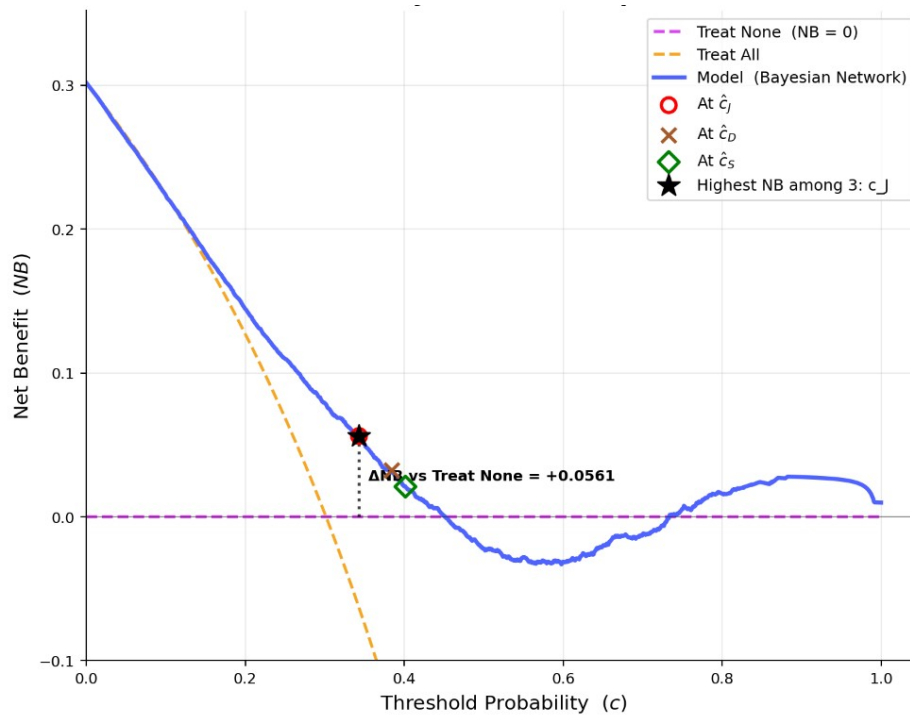
Threshold criterion	Threshold (c)	Sensitivity	Specificity	Accuracy	F1-Score
Youden Index	0.2619	0.6364	0.7724	0.7350	0.5691
Closest to (0,1)	0.1895	0.7091	0.6897	0.6950	0.5612
Symmetry Point	0.1901	0.7091	0.6897	0.6950	0.5612

Lending Club

The DCA results in Figures 4.22, 4.23 and Table 4.13: prevalence = 0.3017. Notably, at all three thresholds the Net Benefit of Treat All is negative, so the comparison baseline is Treat None (NB = 0); the Youden threshold ($c_J = 0.3431$) gives the highest positive Net Benefit among the three thresholds (NB = 0.0561), but at a level considerably lower than for the other two datasets, showing that the practical benefit of the BN model on Lending Club is more limited.



Hình 4.22: Plot of the three optimal decision thresholds according to ROC — Lending Club



Hình 4.23: Plot of the three optimal decision thresholds on DCA — Lending Club

Bảng 4.13: Three optimal decision thresholds according to ROC – Lending Club (validation set)

Threshold criterion	Threshold (c)	Sensitivity	Specificity	Accuracy	F1-Score
Youden Index	0.3593	0.7232	0.5313	0.5892	0.5151
Closest to (0,1)	0.4016	0.6403	0.6098	0.6190	0.5035
Symmetry Point	0.4088	0.6227	0.6227	0.6227	0.4989

On the Lending Club dataset, lowering the decision threshold from the default 0.5 to the Youden-optimal level (≈ 0.3593) considerably increases sensitivity (from 0.4878 to 0.7232) but simultaneously reduces specificity and overall accuracy — a classic trade-off between detecting more defaulting customers (reducing the risk of missing them, FN) at the cost of wrongly rejecting more good customers (increasing FP). On Australian Credit, since the model already achieves high performance at the default threshold, at the Youden-optimal and Closest-to (0,1) thresholds the model's performance coincides completely ($0.4254 \approx 0.4052$, both giving an Accuracy of 0.8551), indicating that the ROC curve of this dataset is shaped very close to the top-left corner, a characteristic of a strong classifier.

The BN model provides a positive net benefit on all three datasets in the low-to-moderate probability threshold range, outperforming both boundary strategies, demonstrating the practical deployment value of the model beyond purely statistical value. Notably, the absolute Net Benefit for Australian Credit (0.4661) is much higher than for German Credit (0.1474) and Lending Club (0.0561), directly reflecting the model's superior classification performance on this dataset (ROC-AUC = 0.9220), while also reflecting the higher prevalence (0.58), which increases the absolute scale of the net benefit achievable. The changes in the metrics when applying the optimal thresholds on the three datasets are reflected in Table 4.14.

Bảng 4.14: Comparison of Bayesian network model evaluation metrics with the optimal threshold on the test set

Metric	Australian Credit		German Credit		Lending Club	
	$c = 0.5$	$c^* = 0.3818$	$c = 0.5$	$c^* = 0.1895$	$c = 0.5$	$c^* = 0.3593$
Accuracy	0.8406	0.8551	0.7300	0.6750	0.6808	0.5897
Precision	0.9211	0.8367	0.6047	0.4891	0.4610	0.3926
Recall	0.8140	0.9535	0.4127	0.7143	0.4878	0.7157
Specificity	0.8846	0.6923	0.8759	0.6569	0.7615	0.5370
F1-Score	0.8642	0.8913	0.4906	0.5806	0.4740	0.5070

The Australian economy operates under the Comprehensive Credit Reporting (CCR) system [1]. Previously, Australia used a negative reporting system (recording only bad debts), but the current CCR system requires banks to share positive data as well (24-month on-time payment history, credit limits, actual payment behaviour). Thanks to a rich two-way database (positive + negative) from the CCR system, the machine learning model has enough high-quality features to smoothly and accurately separate out the risky customer group [67]. This information transparency helps the model's DCA curve to widen the gap with the Treat All line to its maximum extent.

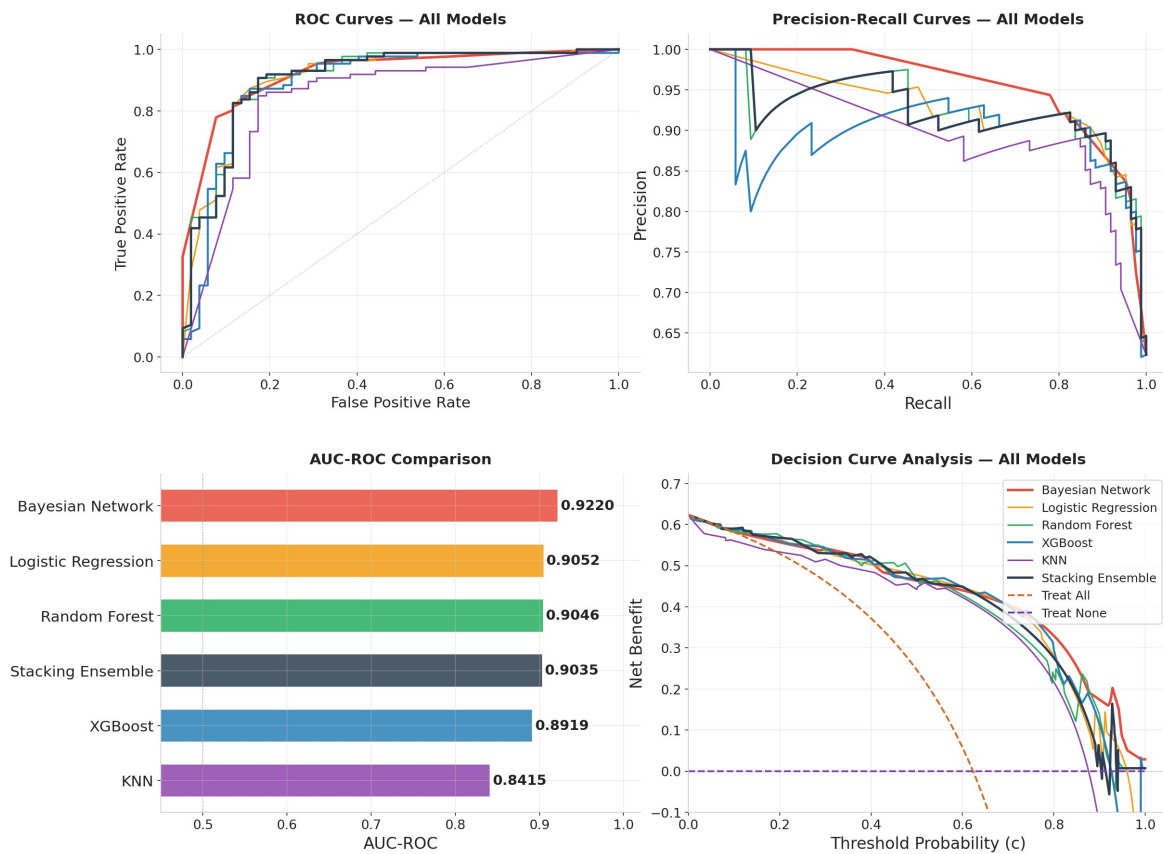
German financial culture is famous worldwide for its prudence, discipline, and extreme risk-aversion [19]. German people have a high rate of asset accumulation, prefer using cash or debit cards, and make very limited use of consumer borrowing through credit cards [19]. Germany's national credit scoring system is dominated by the SCHUFA group. SCHUFA's scoring criteria place extremely heavy weight on negative factors (late-payment history, debt collection records, personal bankruptcy) [15]. Those without a "black mark" are by default considered to be highly creditworthy. Because the preliminary screening process of German banks combined with the SCHUFA system is so strict, the pool of customers that passes through this initial gate to reach the in-depth scoring model consists mostly of extremely clean profiles, resulting in an ultra-low natural default base rate in the dataset [74]. The German market leans heavily towards a Recall-maximising strategy, with the core mindset: "Better to wrongly reject a good customer than to let a single bad debt slip through." Their optimal point lies at

a very low probability threshold ($\hat{c}_D \approx 0.19$), meaning that as soon as the model detects even a small hint of risk, they immediately reject the loan. If the probability threshold is deliberately raised in an attempt to save more customers (increasing Precision), the net benefit immediately drops sharply and falls into negative territory, violating the strict safety standards of the German financial system.

Lending Club is not a traditional commercial bank but originated as a peer-to-peer lending / Fintech platform in the United States, specialising in providing unsecured personal loans [38]. In a high-risk unsecured lending environment such as P2P, if a "Treat All" strategy (lending indiscriminately) were applied, the resulting wave of defaults would immediately wipe out the entire capital pool, causing heavy losses for investors (the Treat All line plunges deeply negative) [59]. Therefore, their minimum safe anchor point must be "Treat None" (holding cash, doing nothing, in order to preserve capital). They need to ensure that profiles labelled "Good" and approved have high accuracy, in order to protect profitability and maintain sustained trust among the investors contributing capital on the platform.

4.3.3 Comparison of the Bayesian Network with Benchmark Machine Learning Models

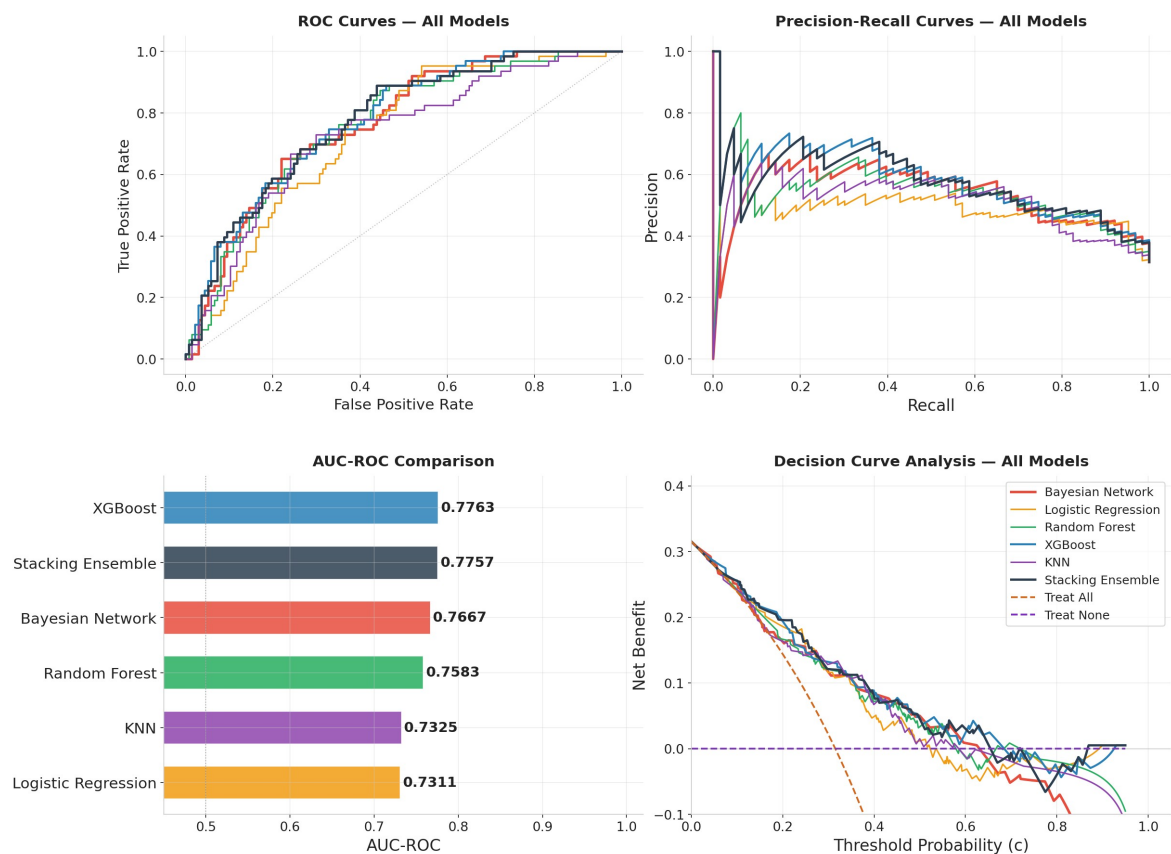
On the Australian Credit dataset (Figure 4.24 and Table 4.15), BN achieves the highest ROC-AUC (0.9220) and PR-AUC (0.9371) among all the models tested, including the Stacking Ensemble. This is the most outstanding result for BN across all three datasets, showing that for a small dataset with few features and a relatively clear causal structure, the Bayesian network has a decisive advantage even over more complex ensemble models.



Hình 4.24: Comparison of the performance of the Bayesian network model with machine learning models on the Australian Credit dataset

Bảng 4.15: Comparison of model performance on Australian Credit

Model	ROC-AUC	PR-AUC
Bayesian Network	0.9220	0.9371
Logistic Regression	0.9052	0.9218
Random Forest	0.9046	0.9238
XGBoost	0.8919	0.8997
KNN	0.8415	0.8633
Stacking Ensemble	0.9055	0.9233

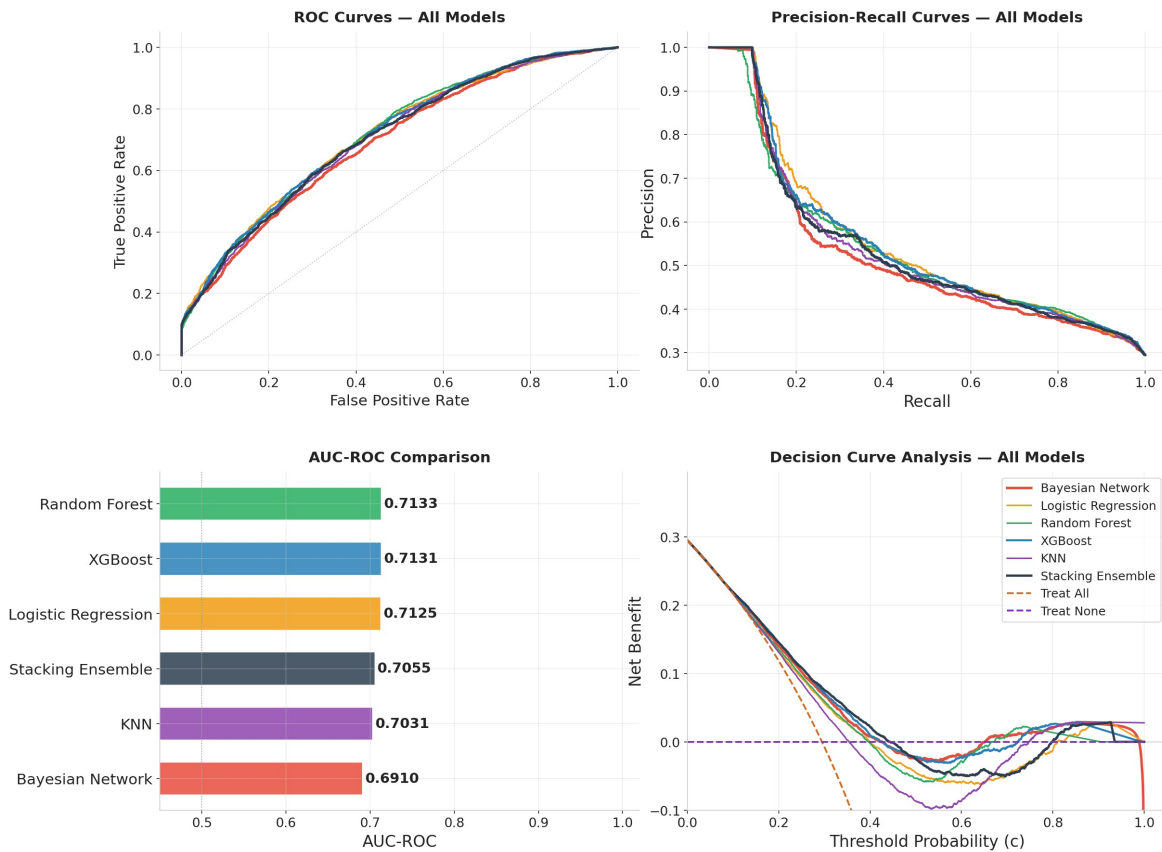


Hình 4.25: Comparison of the performance of the Bayesian network model with machine learning models on the German Credit dataset

Bảng 4.16: Comparison of model performance on German Credit

Model	ROC-AUC	PR-AUC
Bayesian Network	0.7667	0.5421
Logistic Regression	0.7311	0.5015
Random Forest	0.7583	0.5433
XGBoost	0.7763	0.5888
KNN	0.7325	0.5172
Stacking Ensemble	0.7757	0.5785

On German Credit (Figure 4.25 and Table 4.16), the Bayesian network's ROC-AUC = 0.7667 outperforms Logistic Regression, Random Forest and KNN, but is slightly lower than XGBoost (0.7763) and the Stacking Ensemble (0.7757).



Hình 4.26: Comparison of the performance of the Bayesian network model with machine learning models on the Lending Club dataset

Bảng 4.17: Comparison of model performance on Lending Club

Model	ROC-AUC	PR-AUC
Bayesian Network	0.6910	0.5224
Logistic Regression	0.7125	0.5528
Random Forest	0.7133	0.5395
XGBoost	0.7131	0.5483
KNN	0.7031	0.5352
Stacking Ensemble	0.7055	0.5371

In contrast, on the large-scale Lending Club dataset (Figure 4.26 and Table 4.17), BN has the lowest ROC-AUC (0.6910) among the six models, ranking behind all four individual machine learning models and the Stacking Ensemble. This relative decline is consistent with the observation that as the feature dimensionality and DAG structure complexity increase (28 nodes, 58 edges), exact probabilistic inference in BN becomes

more difficult compared with boosting/ensemble models, which are more flexible in capturing non-linear interactions among many features.

The comparison results reveal an empirical finding of considerable significance for the thesis’s argument: the relative standing of the Bayesian network compared with modern machine learning models (XGBoost, Random Forest) is not fixed but depends heavily on the characteristics of the data. On a large, high-feature, noisy dataset (Lending Club), BN ranks last (6th out of 6), reflecting an inherent limitation of the discrete Bayesian network when handling a large feature space: discretising continuous variables causes information loss, and a CPT with many state combinations easily suffers from inaccurate parameter estimation due to data sparsity in each combination cell. On German Credit (medium scale), BN rises to 3rd out of 6, surpassing Random Forest, KNN and Logistic Regression, and trailing only the Stacking Ensemble and XGBoost by a very small margin (less than one percentage point). Notably, on Australian Credit (small scale, few features), BN achieves the absolute top position (1st out of 6) on both the AUC-ROC (0.9220) and AUC-PR (0.9371) metrics, surpassing all benchmark models including the Stacking Ensemble, providing clear evidence that the BN’s explicit probabilistic graph structure delivers maximum advantage when the number of variables is small and the causal relationships among the variables are simple.

4.4 Causal Inference (In_Pro)

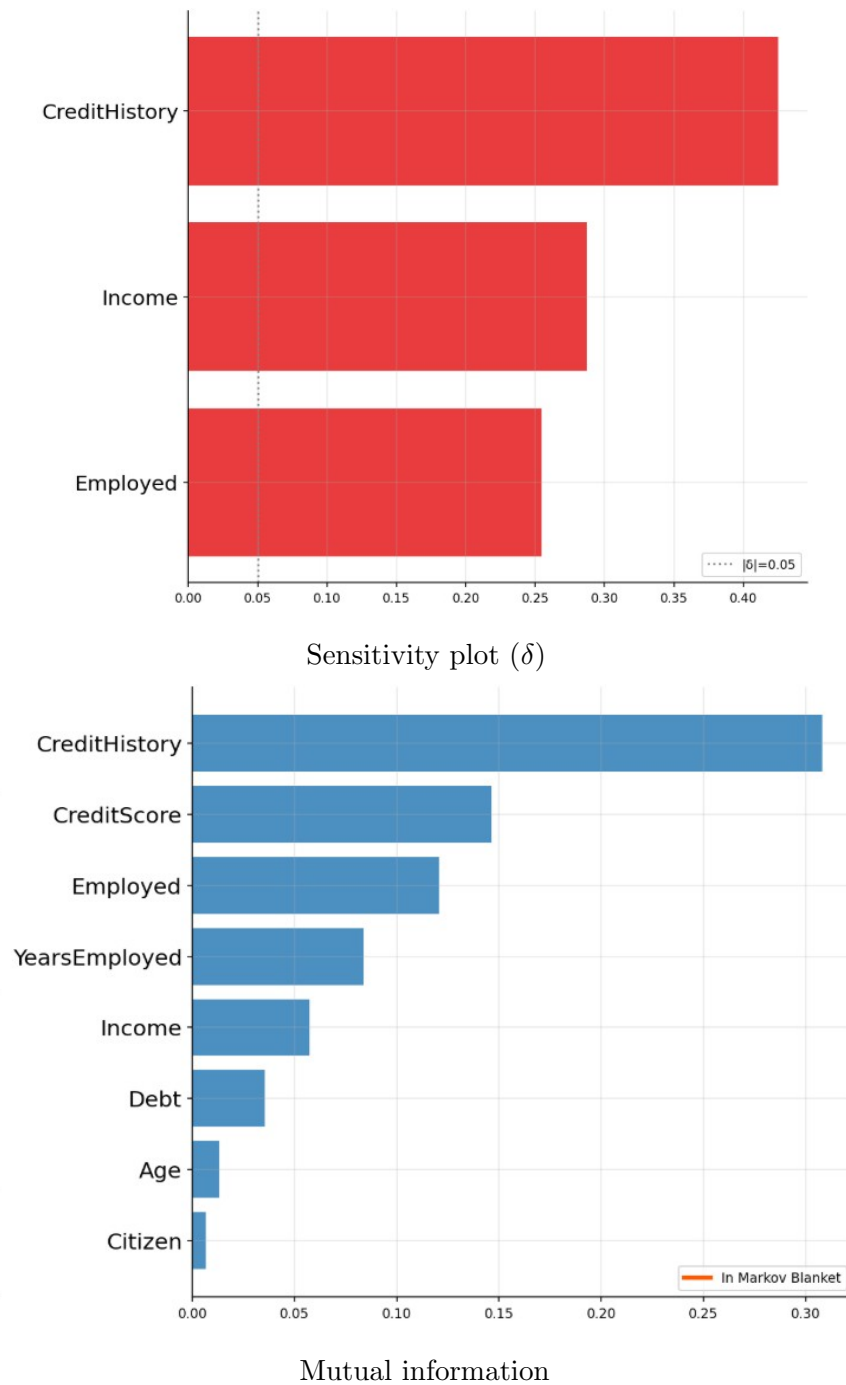
Four interpretation techniques are applied sequentially to all three datasets, Australian Credit, German Credit and Lending Club, comprising parameter sensitivity analysis combined with mutual information, forward inference, backward inference combined with what-if scenario analysis, and finally comparison with two popular post-hoc interpretation methods, LIME and SHAP. This approach closely follows the theoretical framework of Liu et al. [42], whereby the four techniques above are regarded as four mutually complementary perspectives for fully exploiting the intrinsic interpretability of the Bayesian network, rather than relying on a single isolated measure.

4.4.1 Sensitivity Analysis

Applying the method presented in Section 3.3.3, the study computes δ and mutual information experimentally on all three datasets using `pgmpy` and `scikit-learn`, restricting the scope of the δ analysis to the set of variables belonging to the Markov Blanket of `target`.

For Australian Credit, the Markov Blanket of `target` consists of only three variables, `CreditHistory`, `Income` and `Employed`, exactly as recorded for the compact graph structure in Section 4.3.1, with $\max |\delta|$ values of 0.4249, 0.2871 and 0.2545 respectively, shown in Figure 4.27. `CreditHistory` shows a clear monotonic relationship in Figure 4.30: customers with no credit history carry a default probability of 92.5%, while customers with an existing credit history fall sharply to 18.3%, correctly reflecting the role of this variable as a fundamental trustworthiness indicator. In contrast, `Income` does

not show a monotonic relationship across the four bins: the default probability rises from 60.1% in the lowest bin to a peak of 69.3% in the second bin, before falling progressively to 50.2% and 21.3% in the two highest bins. This counterintuitive phenomenon may arise because the posterior probability of **Income** has been integrated through the entire dependency structure with **CreditHistory** and **Employed** in the Markov Blanket, causing a particular income bin to overlap heavily with the group of customers who lack a credit history or stable employment; the small sample size of this dataset also makes some bins after discretisation more susceptible to statistical noise.



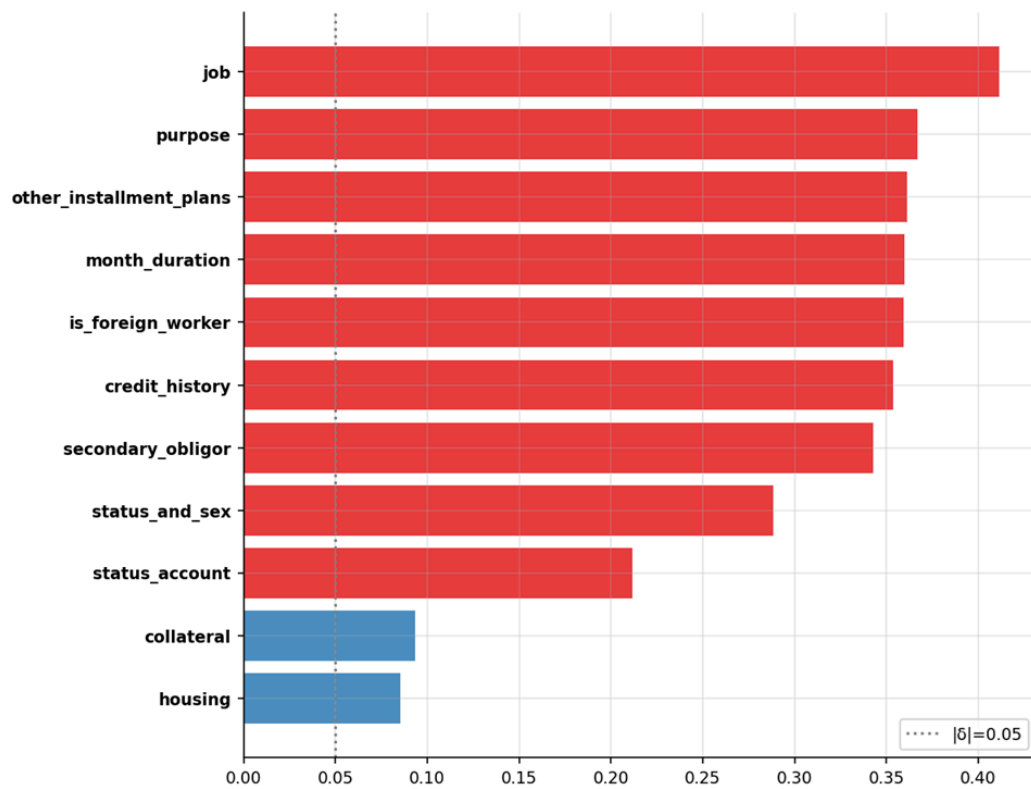
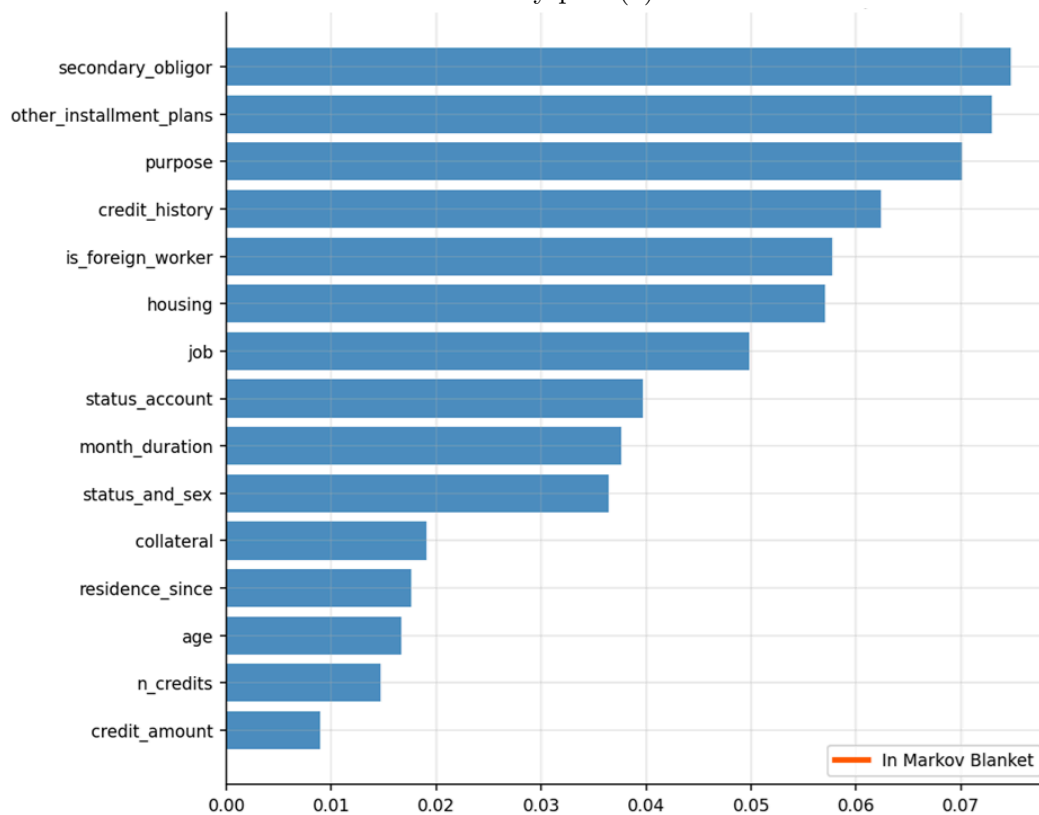
Hình 4.27: Sensitivity plot (δ) and mutual information on the set of variables belonging to the Markov Blanket — Australian Credit

For German Credit, the Markov Blanket expands to 11 variables, led in terms of δ respectively by `job` (0.4114), `purpose` (0.3667), `other_installment_plans` (0.3614), `month_duration` (0.3600), `is_foreign_worker` (0.3592), `credit_history` (0.3537), `secondary_obligor` (0.3430), `status_and_sex` (0.2882), `status_account` (0.2115), `collateral` (0.0936) and `housing` (0.0838), shown in Figure 4.28. Notably, the rank-

ing by δ does not coincide with the ranking by mutual information: the variable with the highest mutual information with **target** is **secondary_obligor**, followed by **other_installment_plans** and **purpose**, whereas **job**, which leads by δ , ranks only seventh on the mutual information axis, showing that these two metrics reflect two different aspects of the relationship between a variable and **target** on the same network. Looking in detail at the two most sensitive variables in Figure 4.31, **is_foreign_worker** shows that foreign-worker customers carry a default probability of 85.9%, considerably higher than the 42.7% of the non-foreign-worker group. **other_installment_plans** shows a non-linear result similar to **Income** on Australian Credit: the group with installment plans at another bank carries a probability of 70.8%, the group with instalment plans at stores falls to 37.8%, but the group with no other instalment plans at all reaches the highest probability, 86.1%, most likely due to overlap with the group having an unfavourable **status_account** or **credit_history** in the Markov Blanket, rather than the absence of a secondary loan itself carrying independent risk meaning.

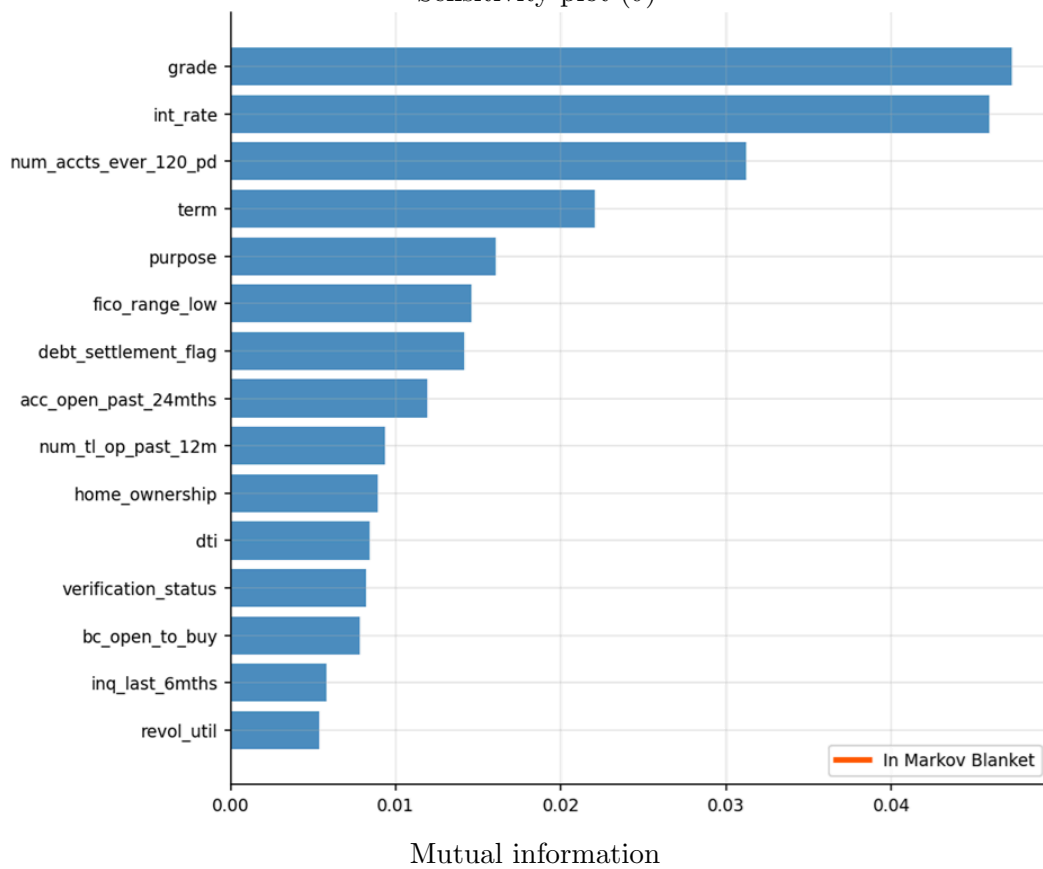
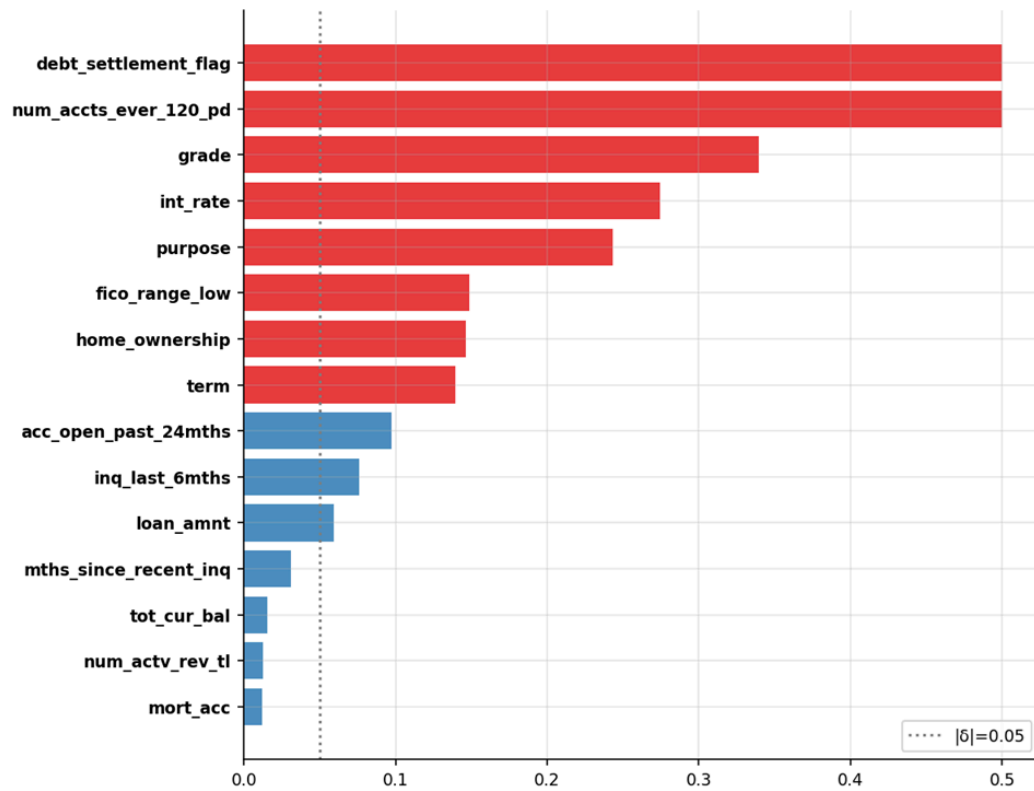
For Lending Club, the Markov Blanket comprises 15 variables, corresponding to the 58-edge DAG recorded in Table 4.9. In Figure 4.29, **debt_settlement_flag** and **num_accts_ever_120_pd** reach $\delta = 0.5000$, likely reflecting a data-sparsity phenomenon similar to the **revol_util** case recorded in Section 4.2.5, rather than an absolute causal effect, and should be interpreted with caution. These are followed by **grade** (0.3395), **int_rate** (0.2746), **purpose** (0.2435), **fico_range_low** (0.1486), **home_ownership** (0.1465) and **term** (0.1392); of these, **grade** and **int_rate** are also simultaneously the two leading variables in terms of mutual information on the same chart. The remaining variables, such as **acc_open_past_24mths**, **inq_last_6mths**, **loan_amnt**, **mths_since_recent_inq**, **tot_cur_bal**, **num_actv_rev_tl** and **mort_acc**, all have $\max|\delta|$ below the 0.1 threshold, showing a negligible impact on default probability compared with the group of variables mentioned above. Looking in detail at the two leading variables in Figure 4.32, the default probability by **grade** rises from 16.1% at the highest credit grade to 67.4% at the second-lowest grade, before fluctuating slightly around 65%–67% at the two lowest grades, showing a diminishing marginal effect once customers have already fallen into the high-risk group. For **int_rate**, the default probability rises monotonically and almost linearly from 22.5% in the lowest interest-rate bin to 69.1% in the highest bin.

Summarising across all three datasets, features that directly reflect credit history and the lender’s risk pricing (**CreditHistory**, **credit_history**, **grade**, **int_rate**) are consistently among the leading factors influencing default probability, consistent with traditional credit expert intuition. Comparing δ against mutual information also helps identify cases where a large probability shift does not stem from a robust causal relationship but rather from data sparsity or a noise effect propagated through the network structure — a diagnostic capability that closed-form post-hoc interpretation methods can rarely provide directly.

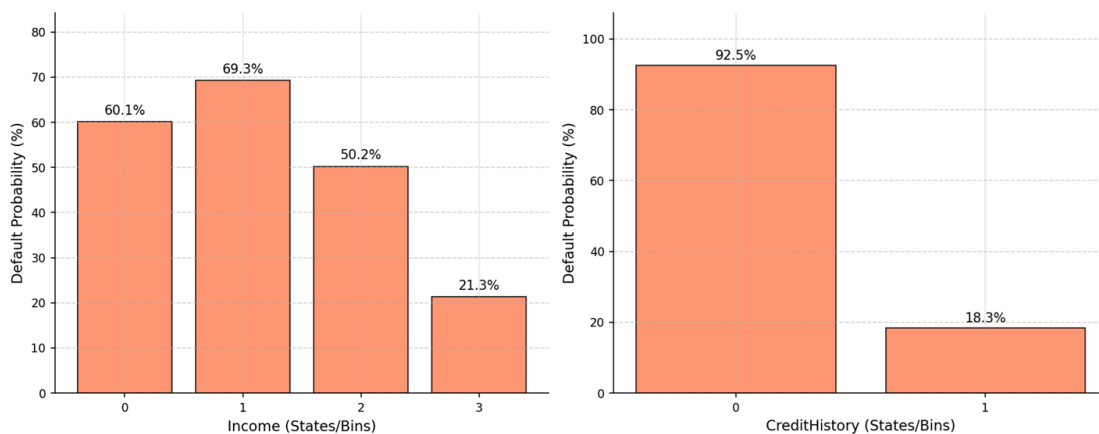
Sensitivity plot (δ)

Mutual information

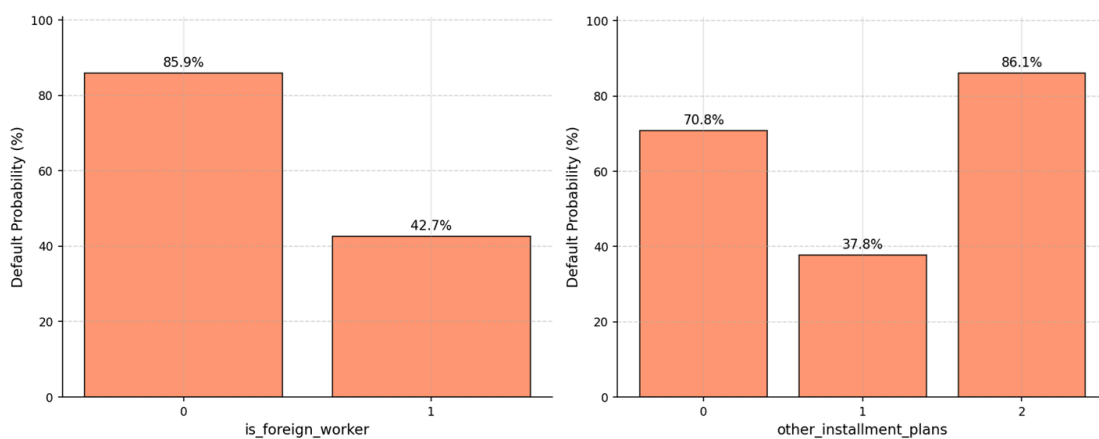
Hình 4.28: Sensitivity plot (δ) and mutual information on the set of variables belonging to the Markov Blanket — German Credit



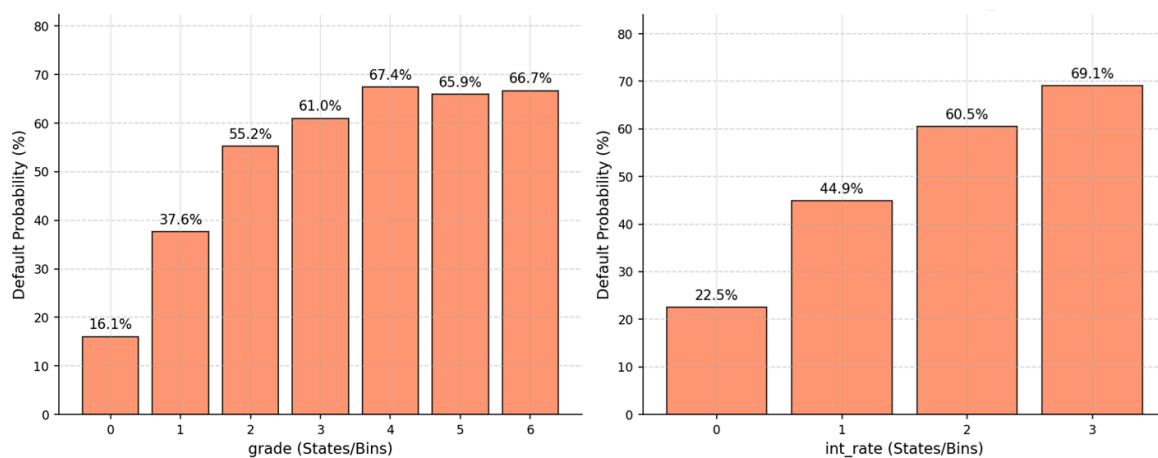
Hình 4.29: Sensitivity plot (δ) and mutual information on the set of variables belonging to the Markov Blanket — Lending Club



Hình 4.30: Marginal default probability under intervention on the key features — Australian Credit



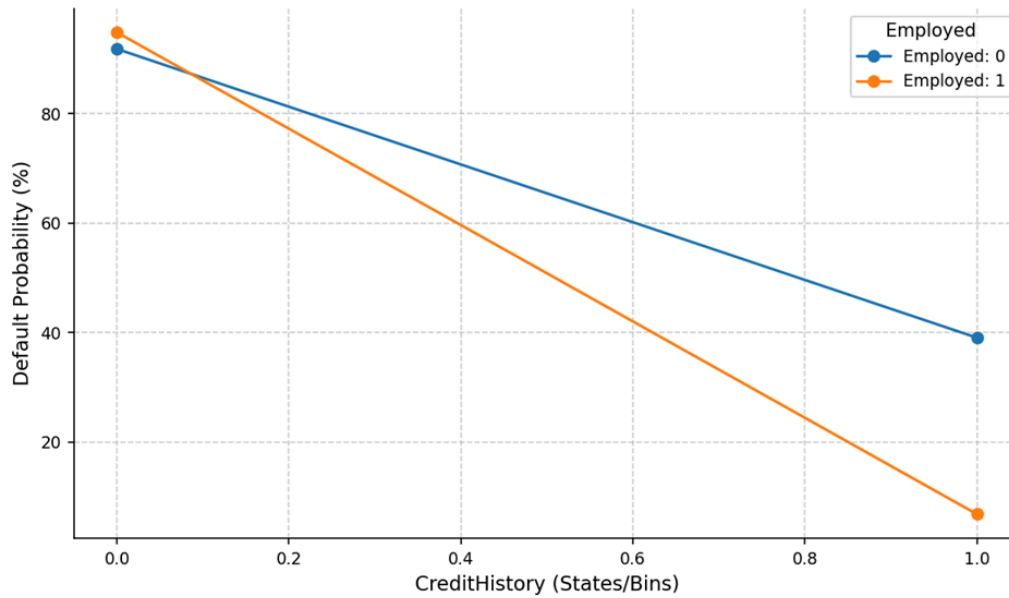
Hình 4.31: Marginal default probability under intervention on the key features — German Credit



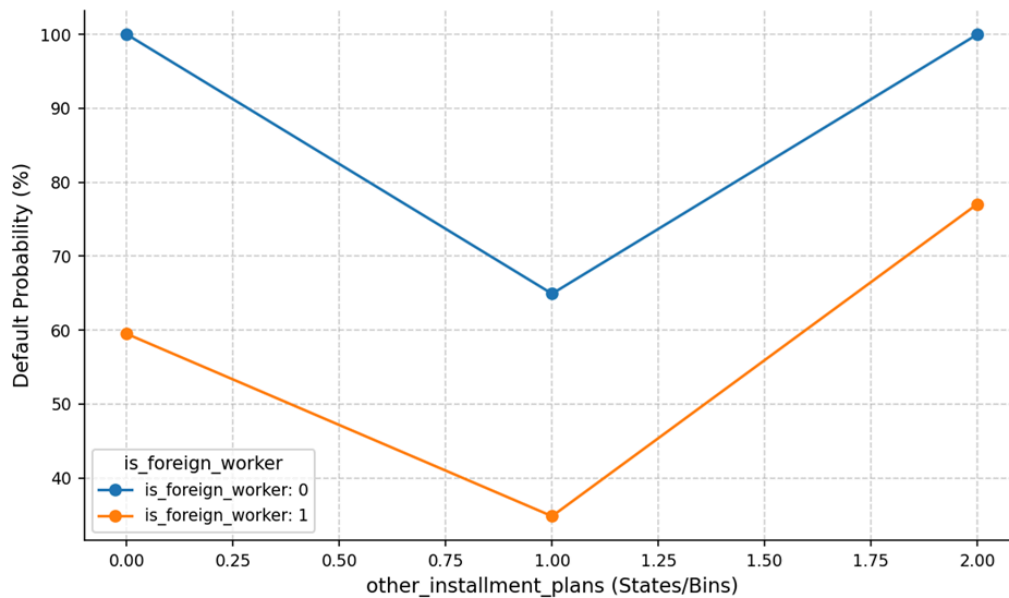
Hình 4.32: Marginal default probability under intervention on the key features — Lending Club

4.4.2 Forward Inference

Forward inference is performed simultaneously on the pair of direct parent variables of `target` in the DAG of each dataset, in order to observe the interaction effect between the direct causes on default probability.

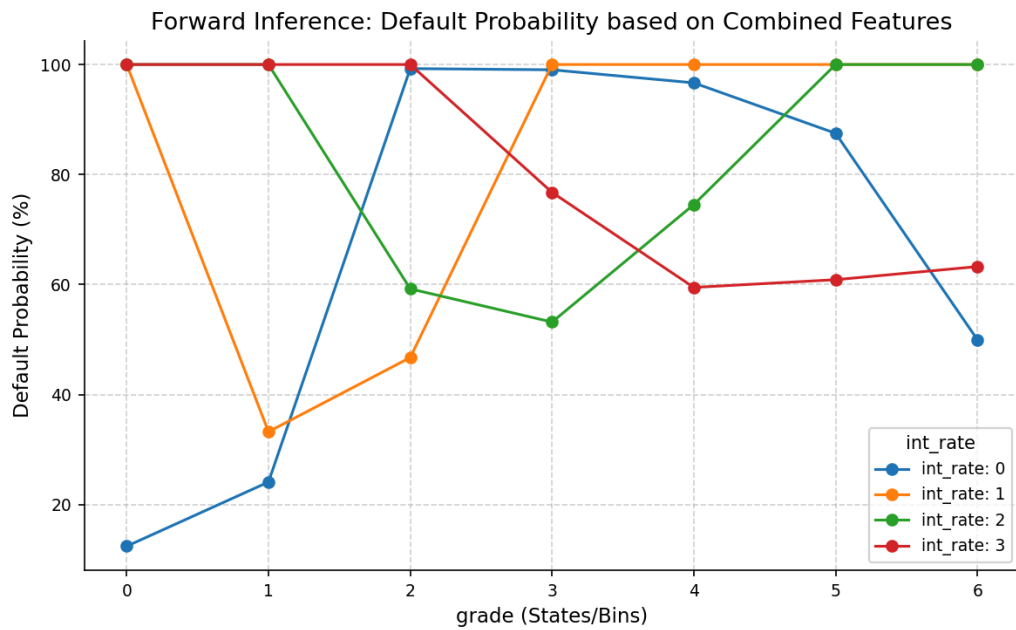


(a) Australian Credit



(b) German Credit

Hình 4.33: Simultaneous forward inference on the pair of direct parent variables of `target`



(c) Lending Club

Hình 4.33: Simultaneous forward inference on the pair of direct parent variables of **target** (continued)

On Australian Credit, the two parent variables **CreditHistory** and **Employed** are set as evidence simultaneously. The results in Figure 4.33a show that when a customer has no credit history, the default probability remains very high regardless of employment status, fluctuating around 93% for those without a job and rising slightly to about 95% for those with a job. Once a credit history exists, the default probability falls sharply in both groups, but the decrease is markedly greater in the employed group, dropping to only about 7%, compared with about 39% in the unemployed group. In other words, **Employed** does not offer a meaningful protective role when **CreditHistory** is in an unfavourable state, but it does markedly amplify the positive effect of **CreditHistory** when the latter is in a favourable state.

On German Credit, the two parent variables **is_foreign_worker** and **other_installment_plans** are examined simultaneously. Figure 4.33b shows that both lines corresponding to **is_foreign_worker** retain the same V-shape already recorded in Section 4.4.1 for **other_installment_plans**, but shifted by two different amplitudes: the foreign-worker group reaches a maximum default probability of 100% at both the "Bank" and "none" states, while the non-foreign-worker group, although also showing the V-shape, has a markedly lower trough and peak, fluctuating only within 35%–77%. This result shows that **is_foreign_worker** acts as an amplifying variable that intensifies the pre-existing non-linear effect of **other_installment_plans**.

On Lending Club, the two parent variables **grade** and **int_rate** are examined simultaneously. Quite unlike the two datasets above, Figure 4.33c shows that the relationship between these two variables is no longer monotonic or clearly patterned,

but fluctuates erratically: at the lowest **grade** (grade A, state 0), the default probability reaches only 13% at the lowest **int_rate** but jumps to nearly 100% at all three remaining **int_rate** levels; meanwhile, at the highest **grade** (grade G, state 6), the trend is almost reversed, with the lowest **int_rate** giving the highest default probability in the group (50%) while the remaining three **int_rate** levels are all approximately 60%–100%. This irregular fluctuation phenomenon likely stems from the very high correlation between **grade** and **int_rate** in Lending Club’s actual lending practice, since the interest rate is largely set based on the credit grade right from the initial appraisal stage; the consequence is that many (**grade**, **int_rate**) combinations are rare or almost non-existent in the training set, causing the corresponding CPT to be estimated from very few observations and to produce unstable extreme probabilities, similar to the data-sparsity phenomenon already recorded in Section 4.4.1 for **debt_settlement_flag** and **num_accts_ever_120_pd**. This result is clear evidence of the limitation of simultaneous inference on two highly intrinsically correlated variables, and should be interpreted with great caution rather than drawing a direct causal conclusion.

Overall, forward inference on simultaneous parent-variable pairs shows that the degree of interaction between features is not uniform: Australian Credit and German Credit exhibit clearly patterned amplifying interactions, while Lending Club reveals the limitation of the method when the two chosen parent variables have excessively high intrinsic correlation, leading to a data-sparsity phenomenon in the combination space. This finding confirms the added value of multivariate forward inference over univariate sensitivity analysis, while also showing that the choice of parent-variable pairs for simultaneous inference needs to take into account the intrinsic correlation between them, rather than relying solely on their direct causal relationship with **target**.

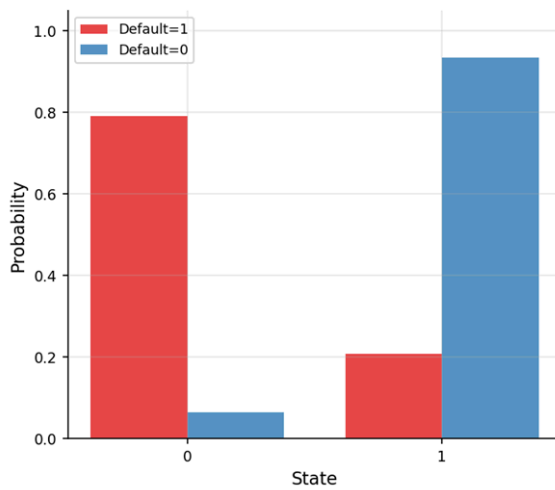
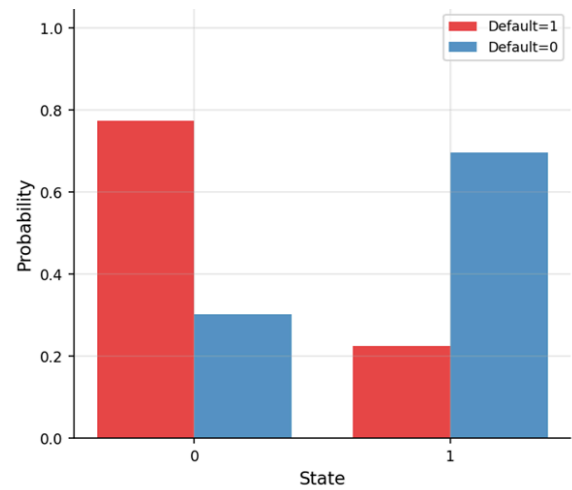
4.4.3 Backward Inference

Backward inference is performed on each direct parent variable of **target** in the DAG of each dataset, querying $P(X_i \mid \text{target} = 1)$ against $P(X_i \mid \text{target} = 0)$ in order to trace the common characteristics of the group of defaulted profiles compared with the group of non-defaulted profiles.

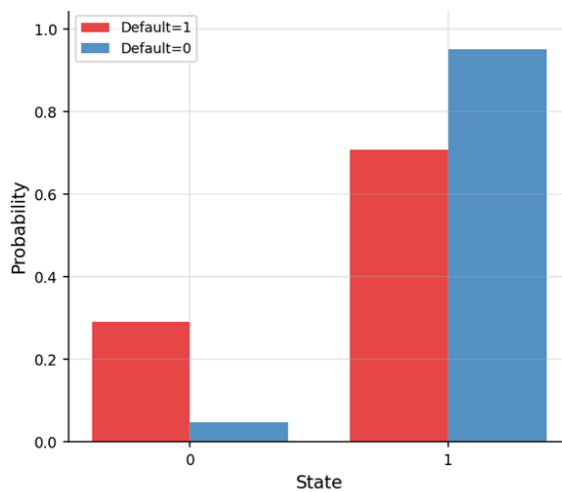
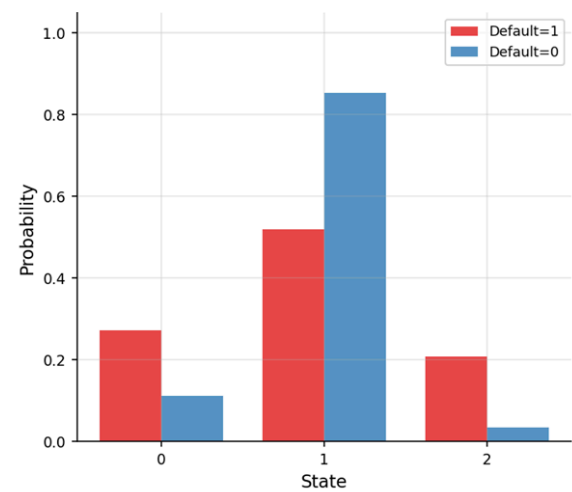
On Australian Credit, **CreditHistory** and **Employed** are the two parent variables queried. The results in Figure 4.34 show that within the group of defaulted customers, 79.1% have no credit history, whereas this proportion in the non-defaulted group is only 6.5% — the largest gap in the entire backward inference results across all three datasets, confirming that **CreditHistory** is the strongest diagnostic signal for tracing a defaulted profile. Similarly, 77.4% of the defaulted group were not employed at the time of application, compared with 30.3% in the non-defaulted group, a considerably smaller gap than that of **CreditHistory** but still showing clear discriminative ability.

On German Credit, **is_foreign_worker** and **other_installment_plans** are the two parent variables queried in Figure 4.35. For **is_foreign_worker**, 29.19% of the defaulted group are foreign workers compared with only 4.78% in the non-defaulted group — the second-largest gap in the entire backward inference results across the three datasets,

after only `CreditHistory` for Australian Credit. For `other_installment_plans`, the "none" state accounts for 20.81% of the defaulted group compared with only 3.35% in the non-defaulted group, while the "stores" state is considerably more common in the non-defaulted group (85.41%) than in the defaulted group (51.91%). This result shows that the discriminative direction of `other_installment_plans` is not concentrated in a single state but spread across the two ends "Bank" and "none", consistent with the non-linear V-shape already recorded in Sections 4.4.1 and 4.4.2.

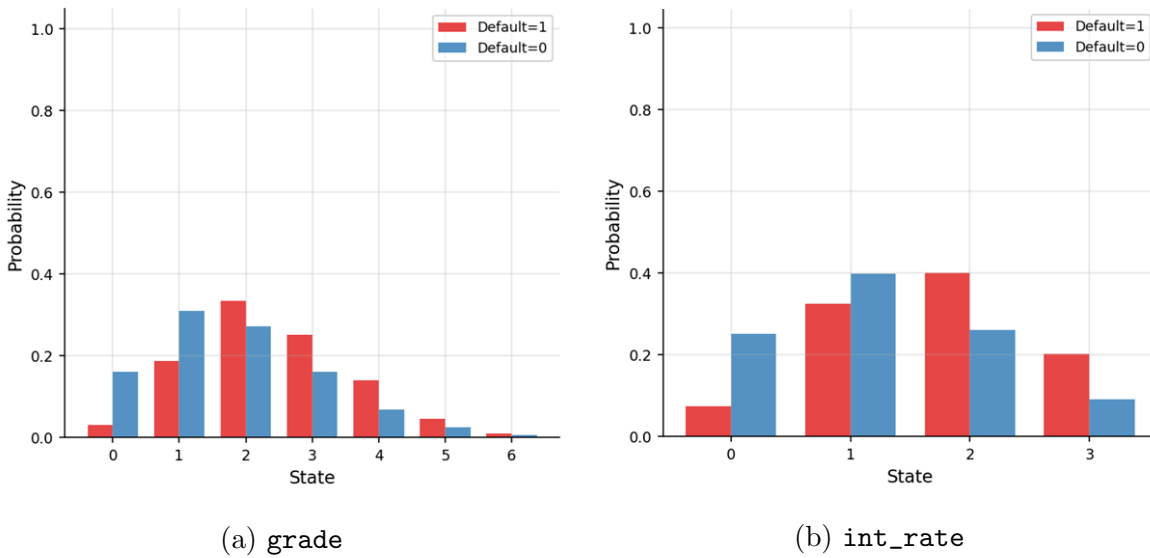
(a) `CreditHistory`(b) `Employed`

Hình 4.34: Backward inference on the direct parent variables of `target` — Australian Credit

(a) `is_foreign_worker`(b) `other_installment_plans`

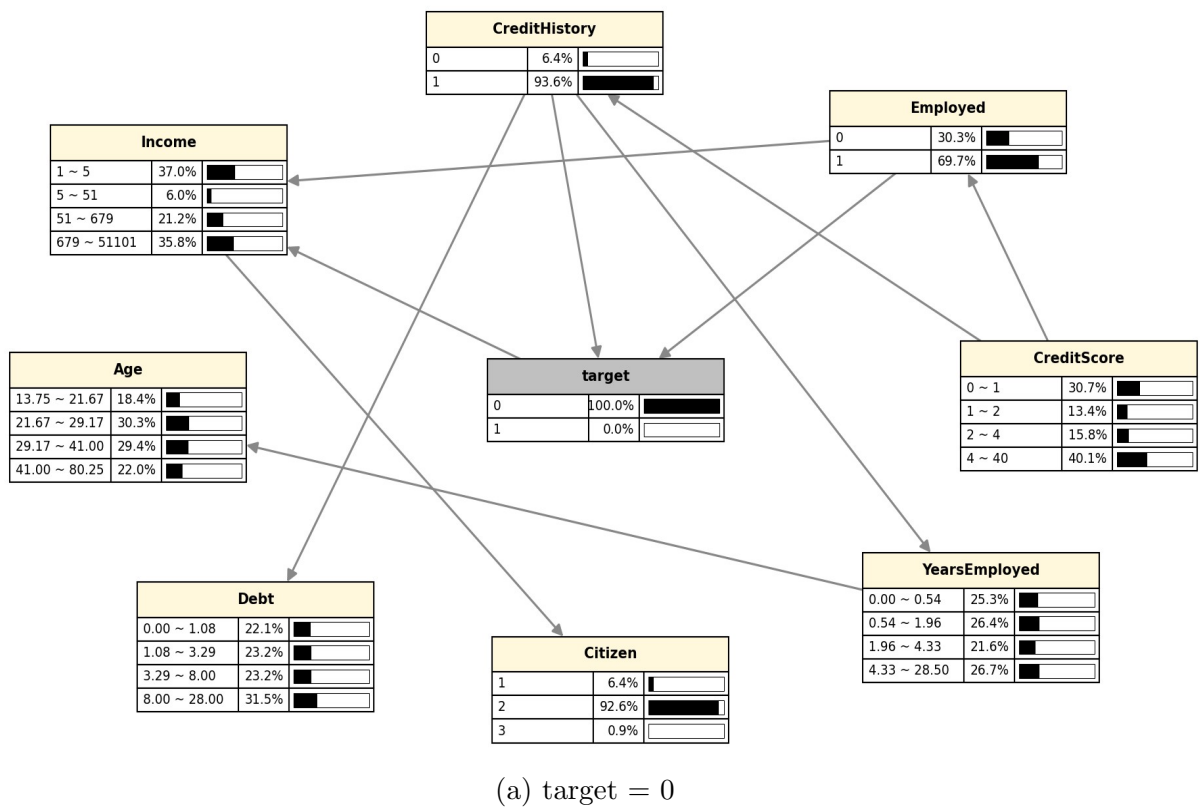
Hình 4.35: Backward inference on the direct parent variables of `target` — German Credit

On Lending Club, `int_rate` and `grade` are the two parent variables queried in Figure 4.36. For `int_rate`, the highest interest-rate bin accounts for 20.13% of the defaulted group compared with only 9.01% in the non-defaulted group, while the lowest interest-rate bin is considerably more common in the non-defaulted group (25.05%) than in the defaulted group (7.29%), consistent with the positive direction of effect already recorded in Sections 4.4.1 and 4.4.2. For `grade`, the proportion in the defaulted group rises steadily with worsening credit grade, from 3.08% at grade A up to 33.47% at grade C, before falling at lower grades due to the small sample size, while the proportion in the non-defaulted group is concentrated most heavily at grade B (31.04%) and declines thereafter, showing that the `grade` distribution of the defaulted group is markedly skewed towards poorer credit grades compared with the non-defaulted group.



Hình 4.36: Backward inference on the direct parent variables of `target` — Lending Club

To obtain a more comprehensive view rather than being limited to the parent variables alone, backward inference is also extended to all nodes of the network: `target=0` and `target=1` are set as evidence in turn, after which the posterior distribution of all remaining variables is queried again, shown in Figures 4.37, 4.38 and 4.39, each figure comprising two plots corresponding to `target = 0` and `target = 1`.

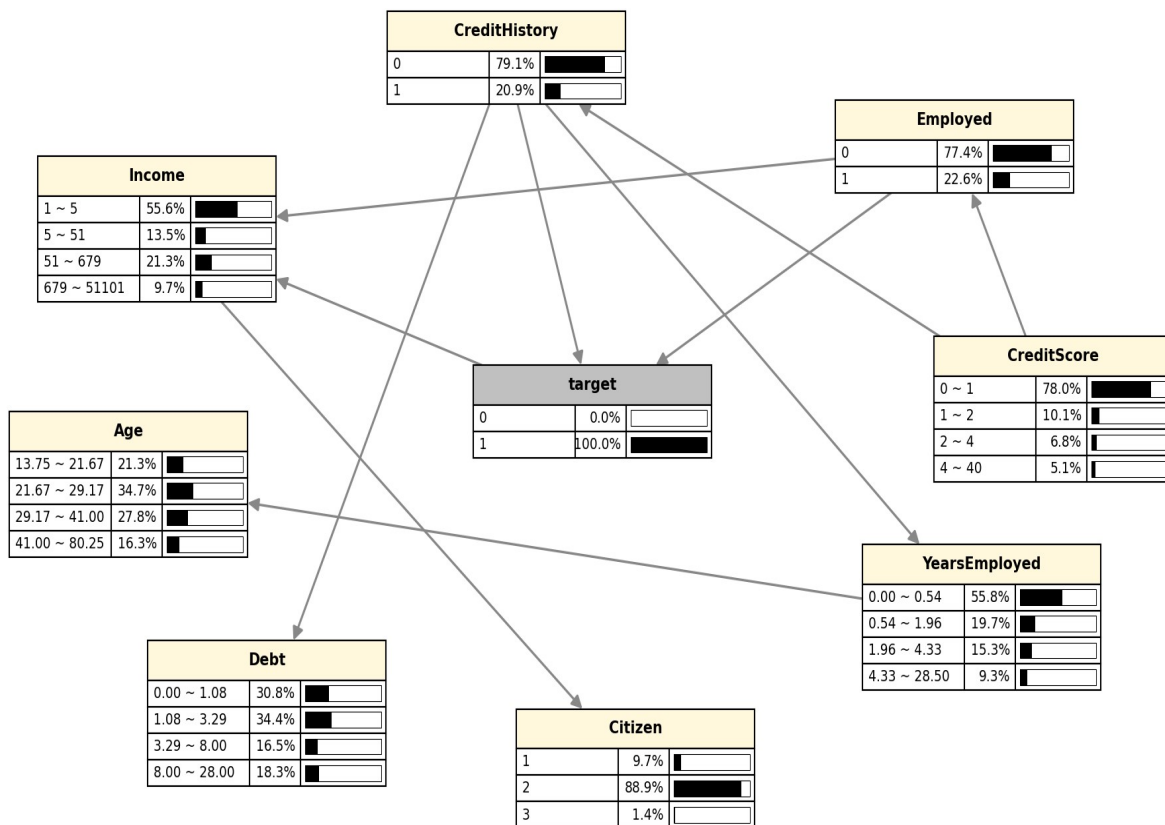


Hình 4.37: Global backward inference conditioned on target evidence — Australian Credit

On Australian Credit, **CreditHistory** shifts the most across the entire network (Figure 4.37): from a prior of 42.9%/57.1% (no/yes history) it skews sharply towards 79.1%/20.9% when **target**=1 and, conversely, towards 6.4%/93.6% when **target**=0. **Employed** also shifts considerably (53.9%/46.1% prior → 77.4%/22.6% when defaulted → 30.3%/69.7% when not defaulted), while **Citizen** remains almost unchanged across all three scenarios (fluctuating within about 1 percentage point), reinforcing the assessment that this is a variable with little relevance to credit risk, even though it remains in the selected feature set.

On German Credit, the three variables that shift most markedly (Figure 4.38) are **credit_history**, **other_installment_plans** and **secondary_obligor**. Most notable is **secondary_obligor**: the "guarantor" state accounts for 73.2% in the prior distribution, rising to 89.7% in the non-defaulted group but falling to 56.7% in the defaulted group, showing that the presence of a guarantor has a clear protective role at the level of the whole network, not only within the direct relationship with **is_foreign_worker** analysed earlier. Similarly, **other_installment_plans** shifts from 68.7% (the "stores" state in the prior) up to 85.4% in the non-defaulted group but down to only 51.9% in the defaulted group.

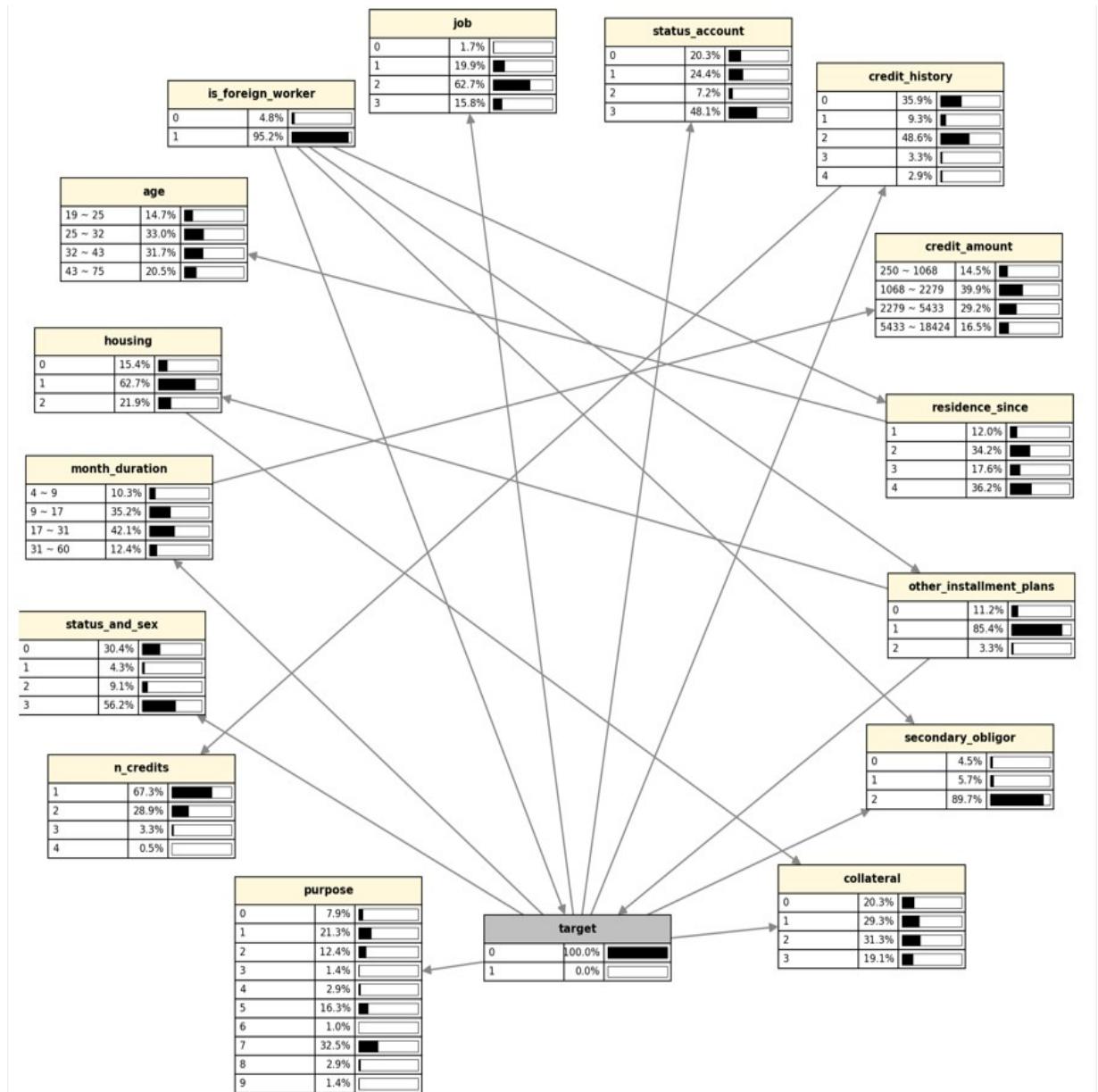
On Lending Club, the most notable and most diagnostic shift across the entire network (Figure 4.39) is **debt_settlement_flag**: in the non-defaulted group, this



(b) target = 1

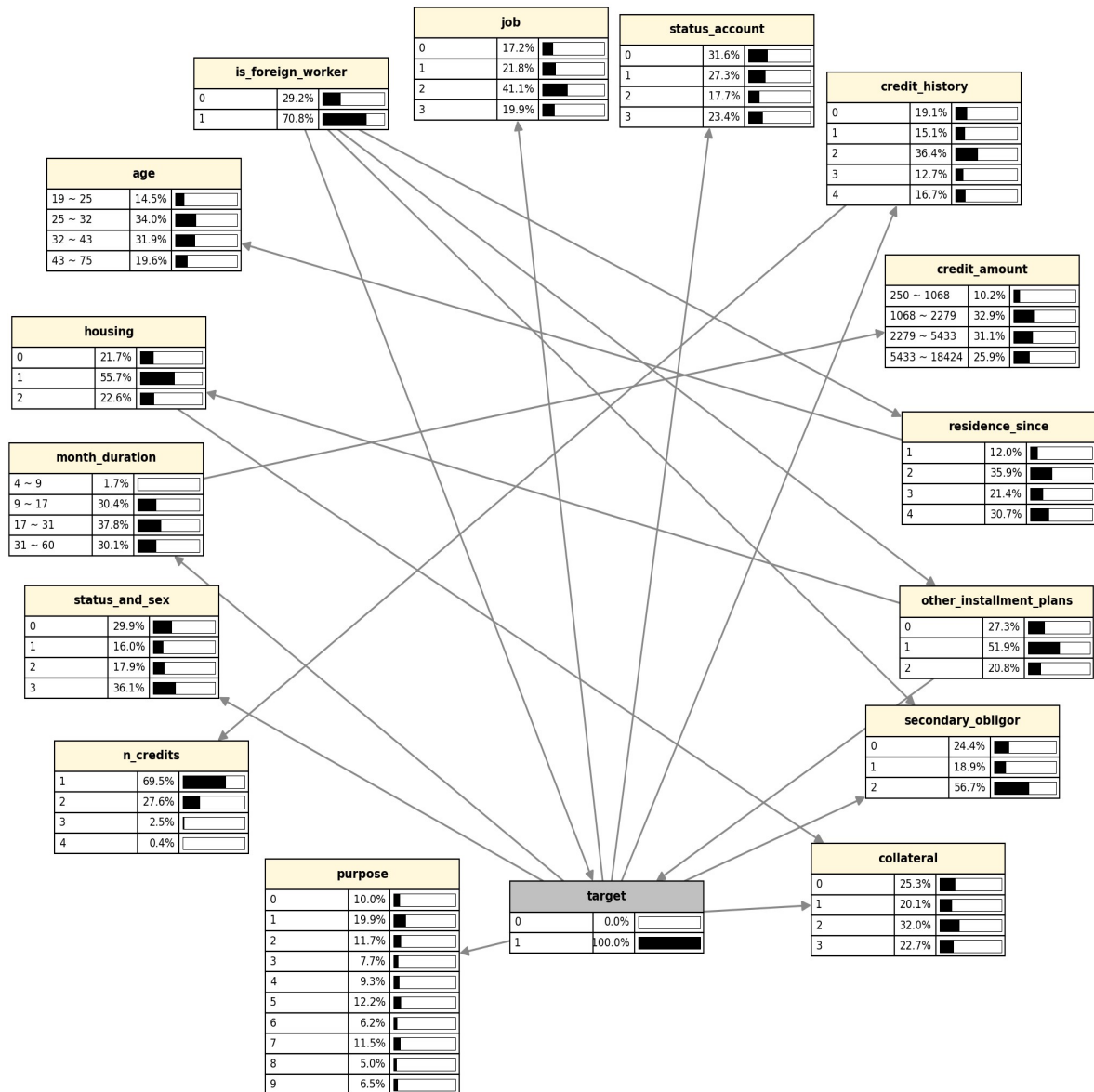
Hình 4.37: Global backward inference conditioned on target evidence — Australian Credit (continued)

variable reaches 100% at the "no" state, i.e. in the data, not a single non-defaulted customer has ever participated in a debt settlement programme, whereas in the defaulted group the proportion of "yes" rises to 4.0% compared with 2.0% in the prior. **grade** and **int_rate** also shift consistently with the earlier analyses: the defaulted group is markedly skewed towards lower grades and higher interest rates (for example, grade G rises from 0.8% in the prior to 1.0%, while grade A falls from 9.6% to 3.1%), while the non-defaulted group is skewed in the opposite direction towards grade A and low interest rates (grade A rises to 16.1%, the lowest interest-rate bin rises to 25.0%).



(a) target = 0

Hình 4.38: Global backward inference conditioned on target evidence — German Credit



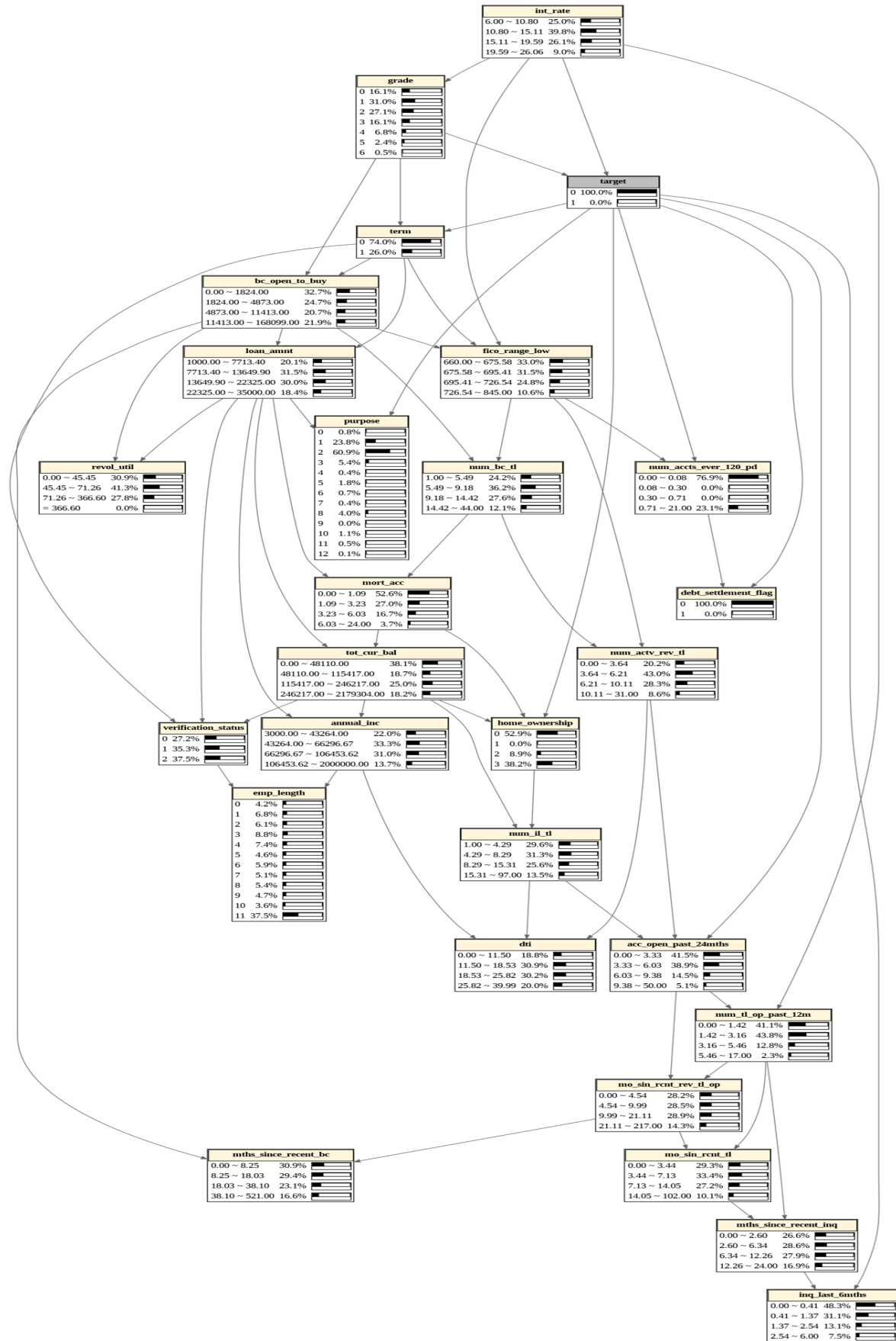
(b) target = 1

Hình 4.38: Global backward inference conditioned on target evidence — German Credit (continued)

This global backward inference result is entirely consistent with the local findings presented above, while also more clearly showing the extent to which the **target** evidence propagates through the entire network structure.

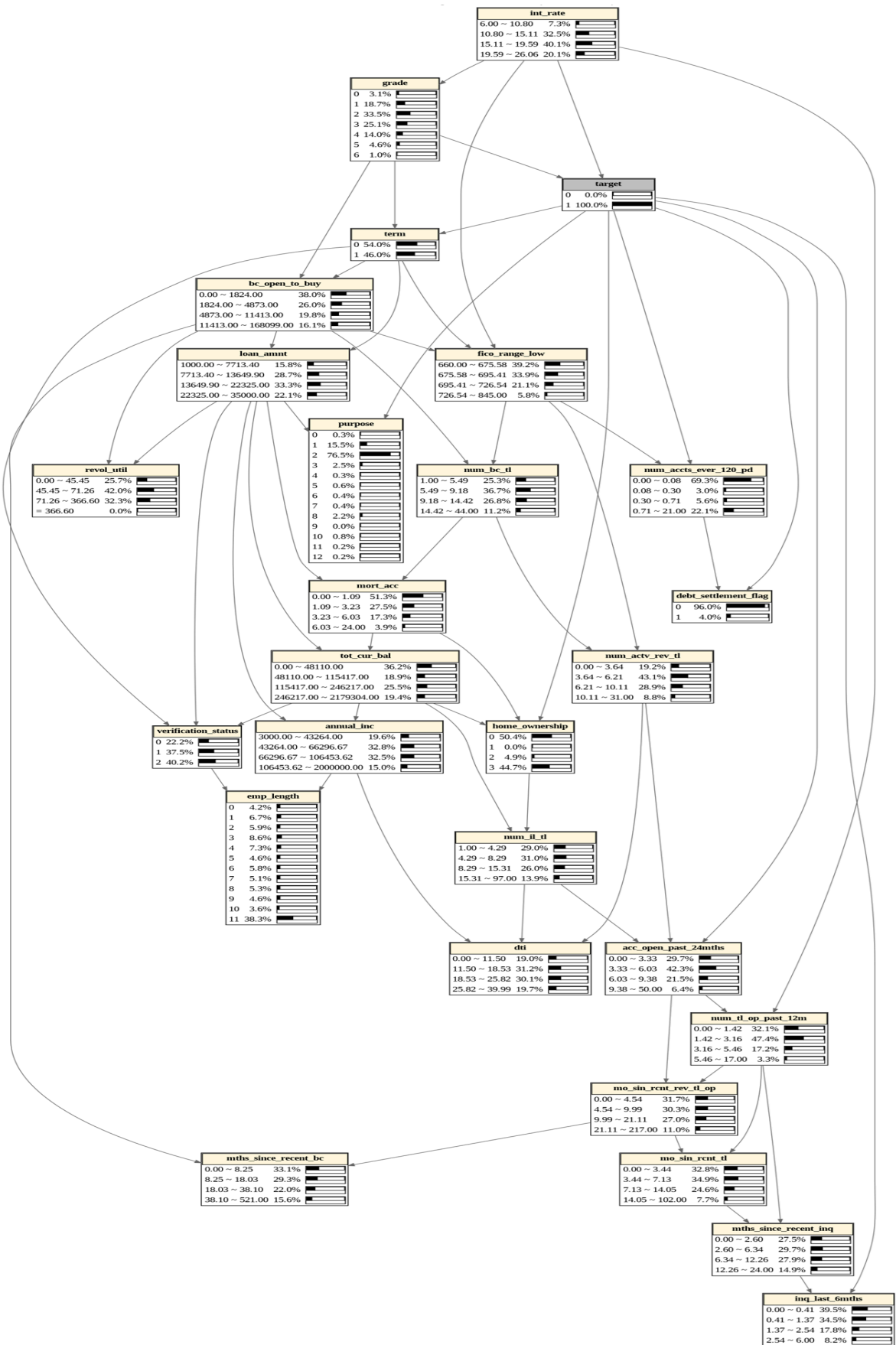
Summarising across all three datasets, backward inference provides an important complementary perspective compared with the two previous techniques: instead of answering "how much risk does a specific profile carry", it answers "what characteristics does a group of profiles with a known outcome typically have", serving directly the task

of profile tracing and building a risk profile after the default outcome has already been observed in practice.



(a) target = 0

Hình 4.39: Global backward inference conditioned on target evidence — Lending Club



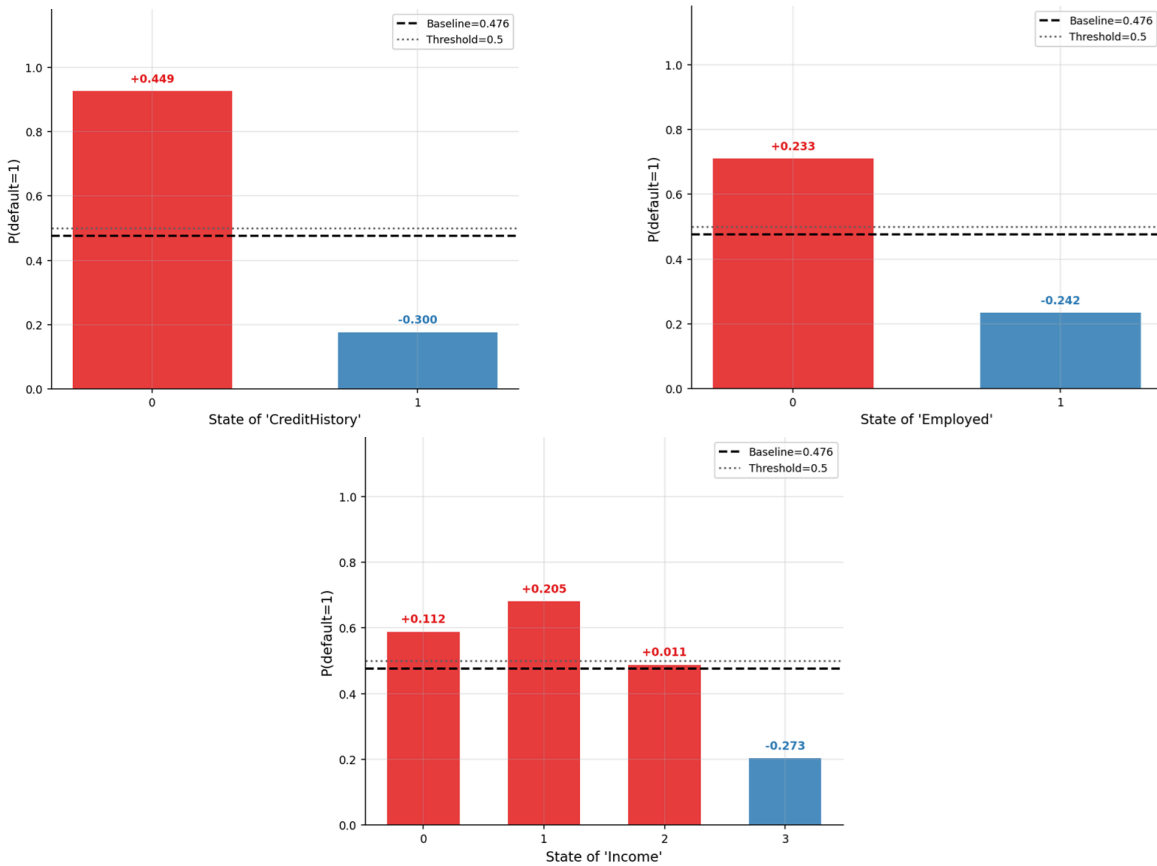
(b) target = 1

Hình 4.39: Global backward inference conditioned on target evidence — Lending Club
(continued)

4.4.4 What-If Scenario Analysis

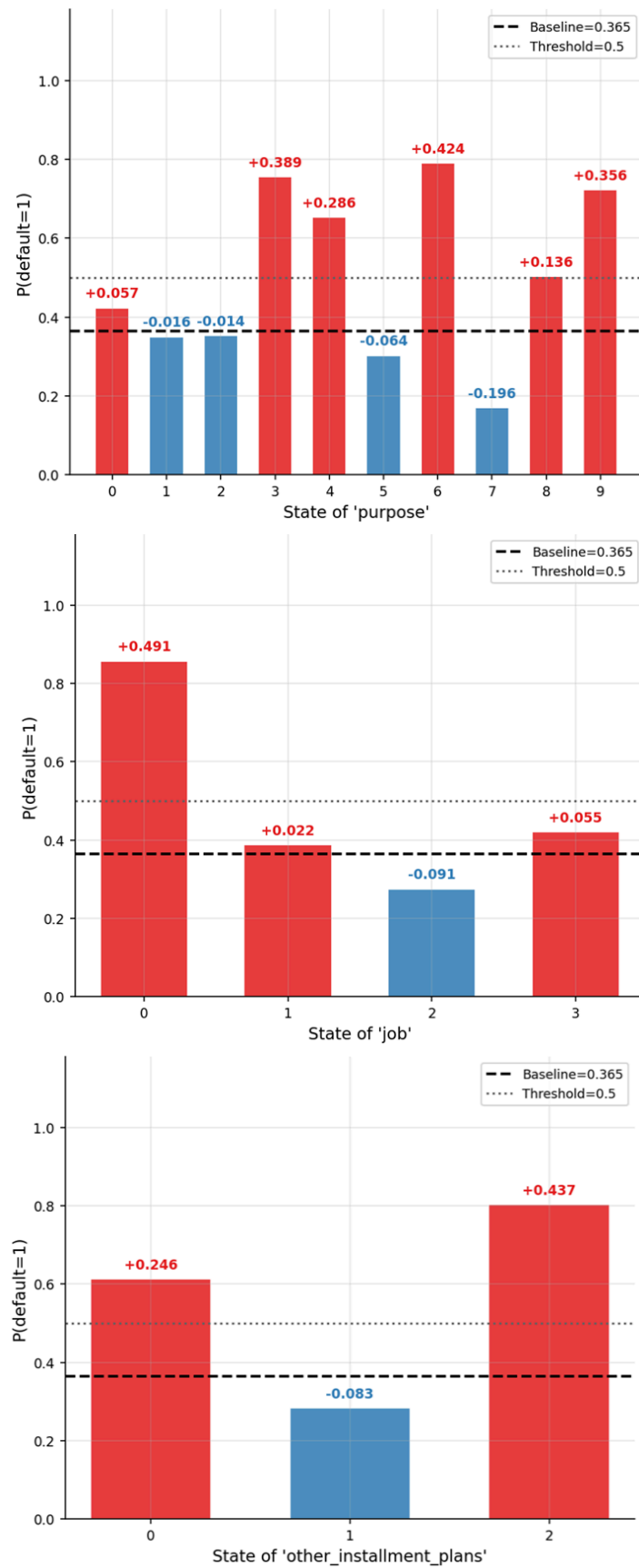
Two approaches are implemented in parallel for what-if scenario analysis: one examined on a specific individual profile, and one examined on a subnetwork structure.

Persona-based Scenarios For each dataset, a few demographic variables that cannot be intervened upon are fixed as an anchor profile (Australian: **Age**, **Citizen**; German: **age**, **status_and_sex**, **housing**; Lending Club: **annual_inc**, **fico_range_low**, **mort_acc**), after which each variable in the group of the three most sensitive variables is changed in turn and $\Delta P(\text{target} = 1)$ is observed relative to the baseline of that anchor profile.

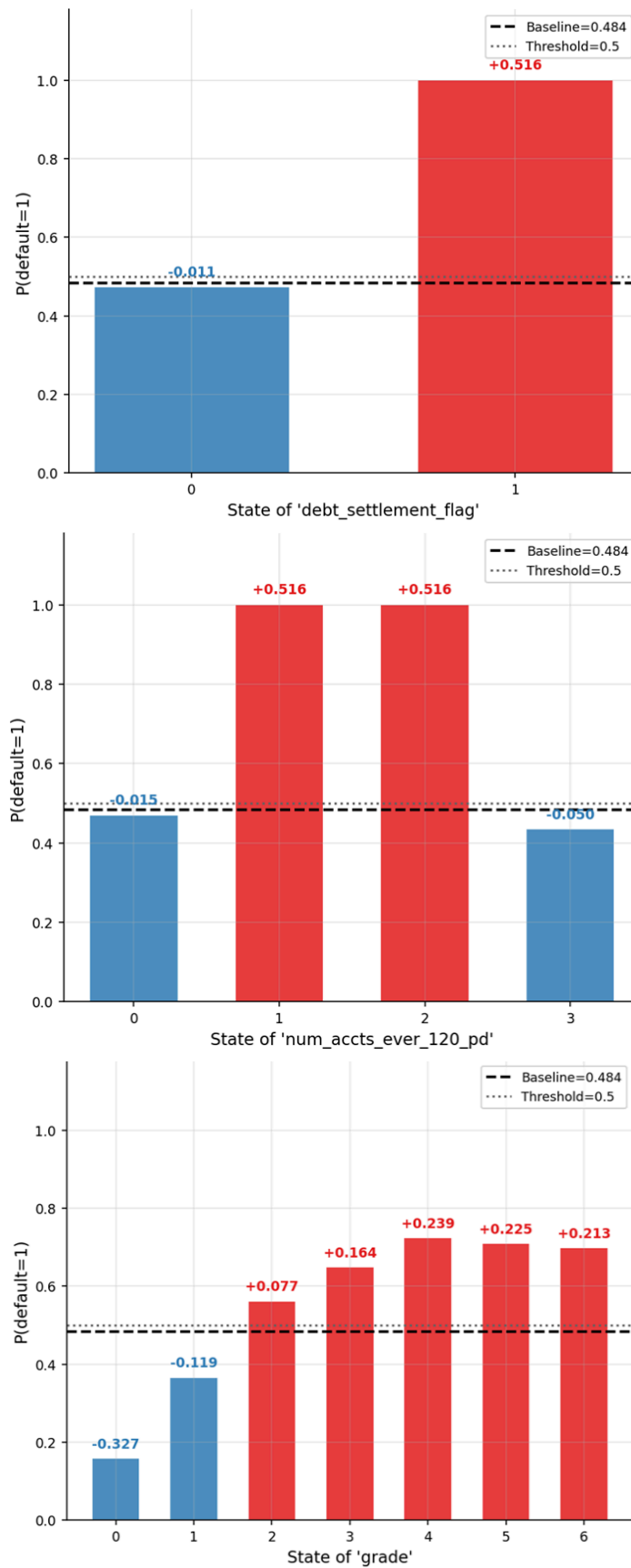


Hình 4.40: Persona-based what-if scenario analysis — Australian Credit
(Where blue: target = 0 and red: target = 1)

On Australian Credit, the baseline of the anchor profile is 47.64%. When **CreditHistory** changes from "no" to "yes", the default probability falls from 92.54% to 17.68% ($\Delta P = -0.300$). **Employed** gives a smaller amplitude, falling from 70.95% to 23.45% ($\Delta P = -0.242$) when changing to employed. **Income** is again not monotonic, as already recorded in Section 4.4.1: the probability rises slightly from bin 0 (58.80%) to a peak at bin 1 (68.12%) before falling progressively to bin 3 (20.31%) (Figure 4.40).



Hình 4.41: Persona-based what-if scenario analysis — German Credit
(Where blue: target = 0 and red: target = 1)

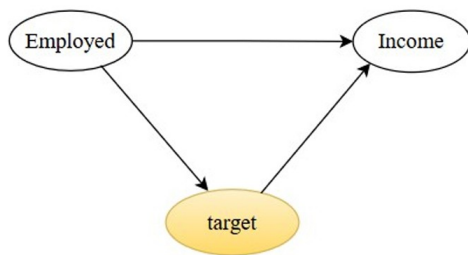


Hình 4.42: Persona-based what-if scenario analysis — Lending Club
(Where blue: target = 0 and red: target = 1)

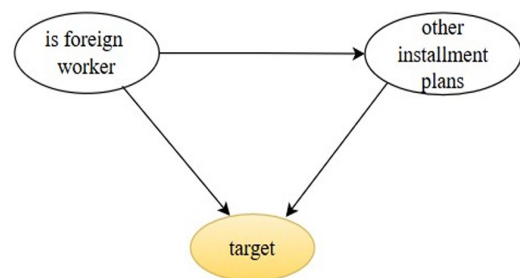
On German Credit, the baseline is 36.45%. **Purpose** is the variable with the largest amplitude among the three examined, ranging from 16.84% (bin 7) to 78.85% (bin 6), reflecting a clear differentiation of risk according to specific loan purpose. **Job** ranges from 27.35% to 85.51%, with bin 0 (unemployed/unskilled) carrying the highest risk. **Other_installment_plans** ranges from 28.13% (the "stores" bin) to 80.10% (the "none" bin), further exhibiting the non-linear V-shape already recorded throughout the previous sections (Figure 4.41).

On Lending Club, the baseline is 48.39%; **num_accts_ever_120_pd** and **Debt_settlement_flag** both show an extreme amplitude identical to the $\delta = 0.5$ result already recorded in Section 4.4.1: switching to the "yes" state causes the default probability to jump straight to 100%, once again confirming that this is a consequence of data sparsity rather than a stable causal relationship. **grade** gives a broader spread with clearer interpretive meaning, falling to a low of 15.73% at grade A and rising to a high of 72.32% at grade E (Figure 4.42).

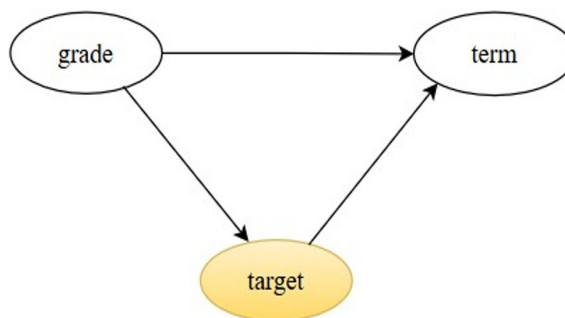
Subnetwork and CPT Analysis Based on the DAG structure learned in Section 4.3.1, for each dataset a closed triangular subnetwork structure around target is selected to build the full CPT.



(a) Subgraph of Australian Credit



(b) Subgraph of German Credit



(c) Subgraph of Lending Club

Hình 4.43: Subgraphs of the three Bayesian networks

Bảng 4.18: CPT of target by Employed $\times$ Income — Australian Credit

Employed	0	0	0	0	1	1	1	1
Income	0	1	2	3	0	1	2	3
$P(\text{target} = 1)$	0.7190	0.7474	0.7871	0.6174	0.2455	0.6606	0.3026	0.0441
$P(\text{target} = 0)$	0.2810	0.2526	0.2129	0.3826	0.7545	0.3394	0.6974	0.9559

On Australian Credit, the selected subnetwork structure is **Employed** (parent) - target - **Income** (child), closed via the direct edge **Employed** $\rightarrow$ **Income** (Figure 4.43a, Table 4.18). The highest-risk rule: unemployed and **Income** at bin 2 gives the highest default probability (78.71%). The lowest-risk rule: employed and **Income** at bin 3 gives the lowest probability (4.41%), a gap of more than 74 percentage points between the two extremes.

Bảng 4.19: CPT of target by is_foreign_worker $\times$ other_installment_plans — German Credit

is_foreign_worker	0	0	0	1	1	1
other_installment_plans	0	1	2	0	1	2
$P(\text{target} = 1)$	1.0000	0.6491	1.0000	0.5948	0.3482	0.7705
$P(\text{target} = 0)$	0.0000	0.3509	0.0000	0.4052	0.6518	0.2295

Bảng 4.20: CPT of target by grade $\times$ term — Lending Club

grade	0	0	1	1	2	2	3	3
term	0	1	0	1	0	1	0	1
$P(\text{target} = 1)$	0.1515	0.3977	0.3528	0.5214	0.5109	0.6186	0.5645	0.6579
$P(\text{target} = 0)$	0.8485	0.6023	0.6472	0.4786	0.4891	0.3814	0.4355	0.3421
grade	4	4	5	5	6	6		
term	0	1	0	1	0	1		
$P(\text{target} = 1)$	0.6255	0.6961	0.5829	0.6802	0.6111	0.6723		
$P(\text{target} = 0)$	0.3745	0.3039	0.4171	0.3198	0.3889	0.3277		

On German Credit, the selected subnetwork structure is **is_foreign_worker** and **other_installment_plans**, two parent variables directly linked to each other, both

closing the loop through target (Figure 4.43b, Table 4.19). The two combinations reaching an absolute default probability of 100% both fall into the foreign-worker group (`is_foreign_worker=0`, coded as "Yes"), albeit at two opposite states of `other_installment_plans` ("Bank" and "none"). The lowest-risk rule is non-foreign-worker combined with "stores" (34.82%).

On Lending Club, the selected subnetwork structure is `grade` (parent) - target - `term` (child), closed via the direct edge `grade` $\rightarrow$ `term` (Figure 4.43c, Table 4.20). At every credit grade, a 60-month loan term always carries a higher default probability than a 36-month term, and this gap is fairly stable across grades (ranging 6–25 percentage points), showing that `term` exerts a relatively uniform additive effect on `grade`, rather than creating an amplifying interaction like the variable pair in German Credit.

Summarising both approaches, the persona-based method is suitable when the goal is to advise on intervention for a specific profile, giving intuitive results that are easy to interpret via ΔP relative to a fixed baseline, but only accurately reflects the specific anchor profile chosen. Conversely, the subnetwork/CPT method gives a comprehensive picture of every possible state combination between two variables with a direct structural relationship, suitable for deriving risk-management rules at the policy level rather than for a specific individual. Combining both approaches, as done in this thesis, allows the study to serve both personalised advisory needs and to provide a basis for credit policy planning at the overall scale.

4.4.5 Comparison with LIME/SHAP

For each dataset, a randomly selected customer is explained simultaneously by three methods: LIME, SHAP, and the Bayesian network's interpretation chart, assigning each variable as evidence in the exact order of the nodes in the graph and recording the change in posterior probability after each step.

On Australian Credit, the selected customer has a default probability of 87.6%, has no credit history, is unemployed, and has *Income* at the highest bin. All three interpretation methods agree that *CreditHistory* is the main dominant factor increasing default risk: LIME assigns +65.4%, the Bayesian network assigns +42.4%. With SHAP, *CreditHistory* = 0.0 and *Income* = 3.0 are the two variables on the side pushing the prediction upwards, with *CreditHistory* lying right at the decision boundary $f(x) = 0.88$ — a position reflecting the dominant role of this variable on the final prediction outcome. The agreement of all three methods on the dominant variable, despite each method relying on a different mathematical mechanism, reinforces confidence in the conclusion that the lack of credit history is the main cause leading to a high-risk assessment for this profile.

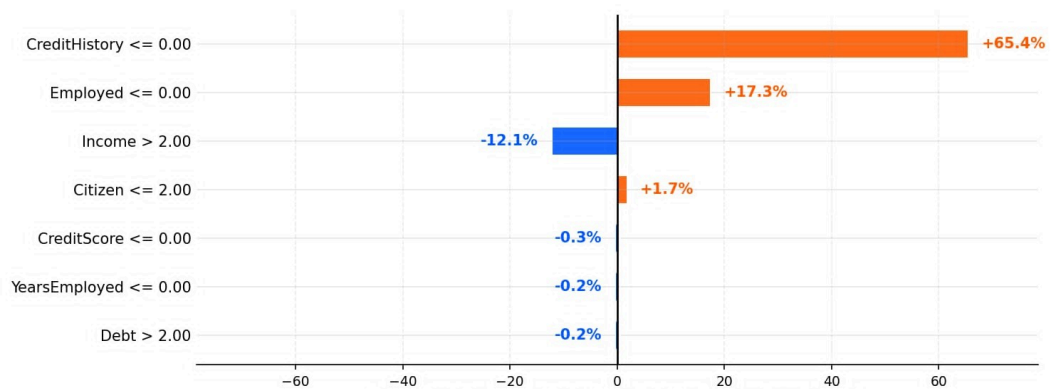
On German Credit, the selected customer has a default probability of 5.3% (Figure 4.45). All three methods agree on the overall direction of effect for `other_installment_plans`, `secondary_obligor`, `job` and `status_and_sex` (all reducing risk). However, there are two notable points of disagreement. First,

credit_history: LIME does not place this variable in the top 7 most important variables, whereas the Bayesian network assigns +10.6% (the largest risk-increasing shift for this profile) and SHAP also confirms **credit_history** = 1.0 as a red (risk-increasing) segment. Second, **housing**: LIME (+8.0%) and SHAP (a clear red segment for **housing** = 0.0) both regard this as a risk-increasing factor, but this variable disappears entirely from the Bayesian network's interpretation chart, meaning its contribution fell below the recording threshold precisely at the point it was added into the sequential evidence chain.

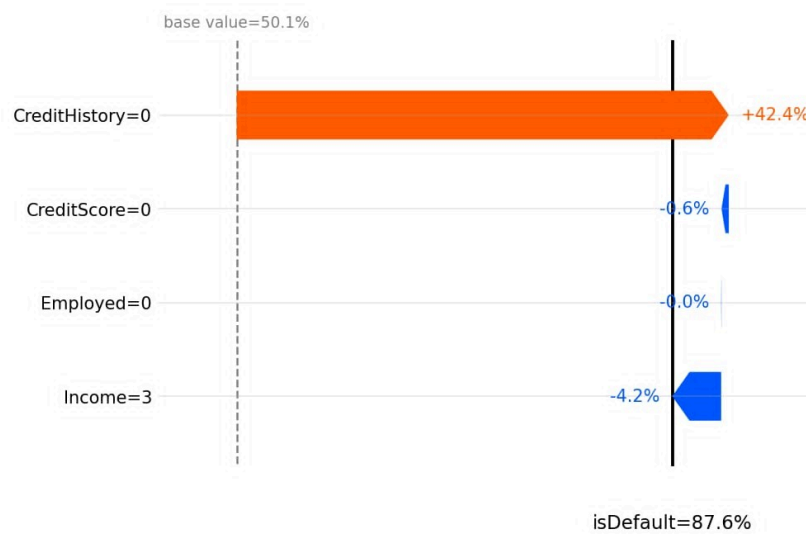
On Lending Club, the selected customer has a very low default probability, only 3.4%, with **debt_settlement_flag** in the "no" (0) state, in contrast to the extreme-risk profile analysed earlier. LIME still ranks **debt_settlement_flag** ≤ 0.00 as the strongest risk-reducing factor (−21.4%), whereas the Bayesian network shows that the role of this variable is almost negligible (−0.4%) and instead identifies **int_rate** = 0 as the true dominant factor (−28.1%), followed by **grade** = 0 (−11.1%); SHAP also agrees with the Bayesian network, with **int_rate** = 0.0 accounting for most of the blue (risk-reducing) segment, while **debt_settlement_flag** does not meet the threshold to appear in the chart. This asymmetry between the two states of **debt_settlement_flag** is quite notable: when in the rare state (= 1, currently in a debt settlement programme), this variable almost absolutely determines the outcome, as already recorded for the previous profile (Bayesian network +54.6%); but when in the common state (= 0, no settlement), it carries very little additional information beyond what **int_rate** and **grade** have already shown, since most customers are inherently already in this state. LIME appears unable to distinguish this asymmetry, since the local linear approximation method assigns weights based on the level of perturbation around the data point rather than accurately reflecting the true distribution of the variable across the entire dataset.

Summarising across all three datasets, all three methods are broadly consistent in the direction of effect and the main dominant variable for each profile, showing that the causal structure learned by the Bayesian network does not contradict common post-hoc approximation methods. However, the specific magnitude of contribution (in the case of German Credit, even the direction of effect of a few variables) is not consistent across the three methods, particularly when the relevant CPT involves sparse data or when the influence of a variable depends heavily on other variables that have already been fixed earlier in the evidence propagation chain. In addition, the order in which variables are presented in the Bayesian network depends on the order of the nodes in the graph rather than automatically being sorted by contribution magnitude as with LIME or SHAP, so some important variables may appear in the middle of the chart rather than at the top. This is a difference in presentation to bear in mind when directly comparing the three methods, rather than a difference in the underlying nature of the inference.

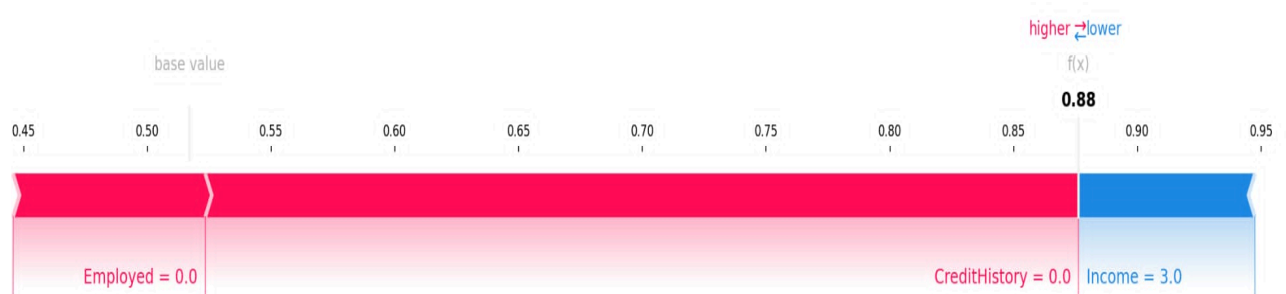
Predicted probability: Is default = 0.8762, No default = 0.1238



(a) LIME



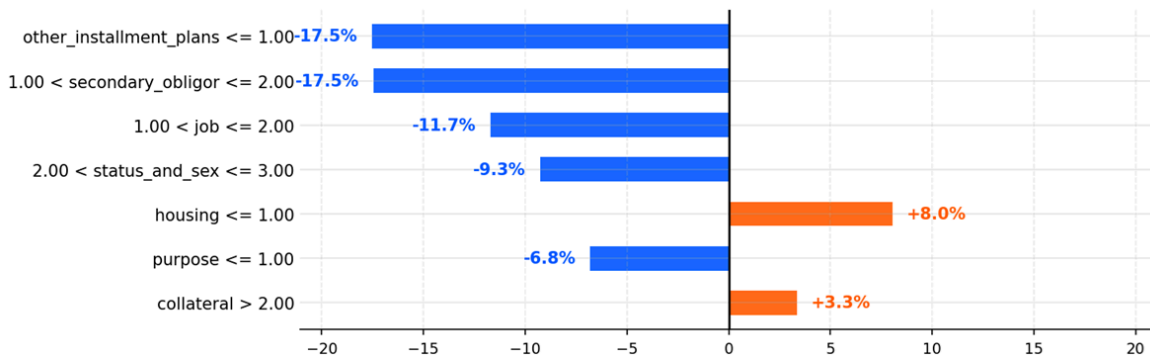
(b) Bayesian Network



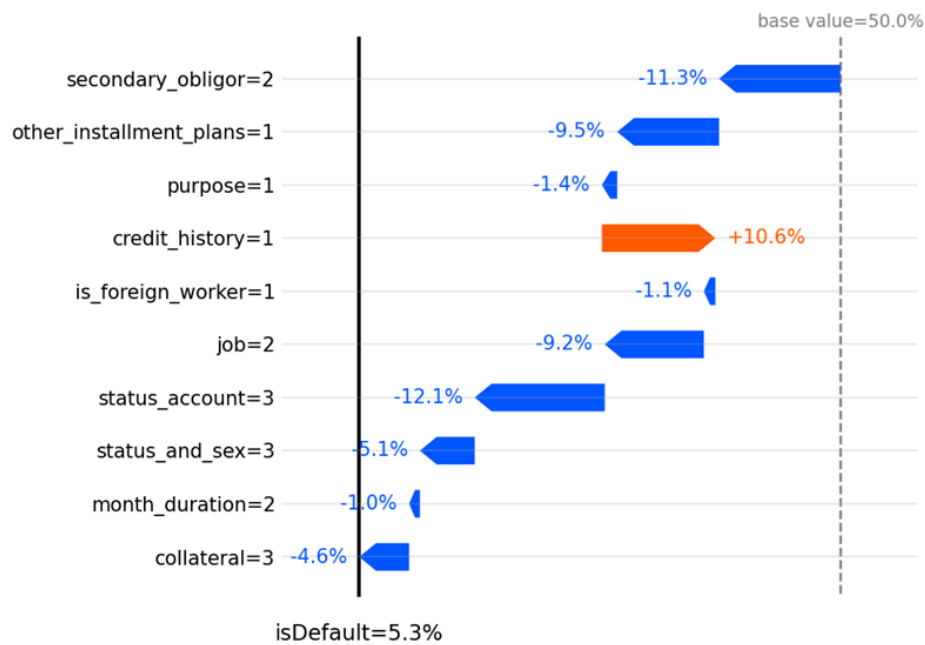
(c) SHAP

Hình 4.44: Comparison of three local explanation methods for a randomly selected customer profile — Australian Credit

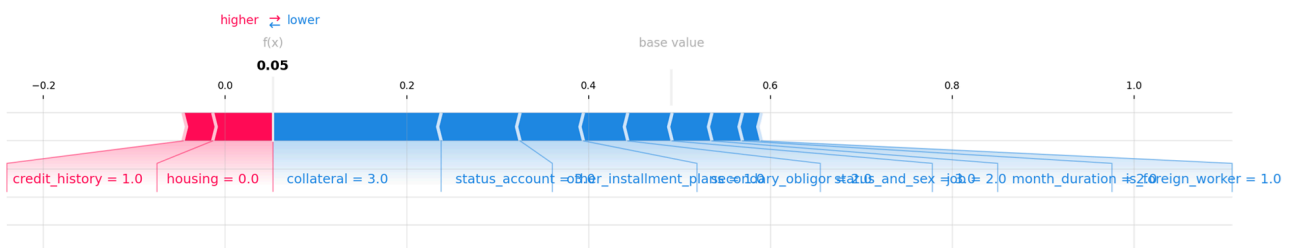
Predicted probability: Is default = 0.0526, No default = 0.9474



(a) LIME



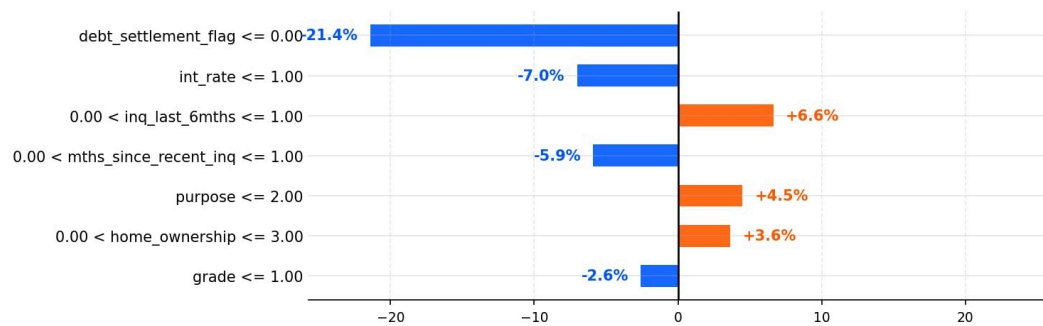
(b) Bayesian Network



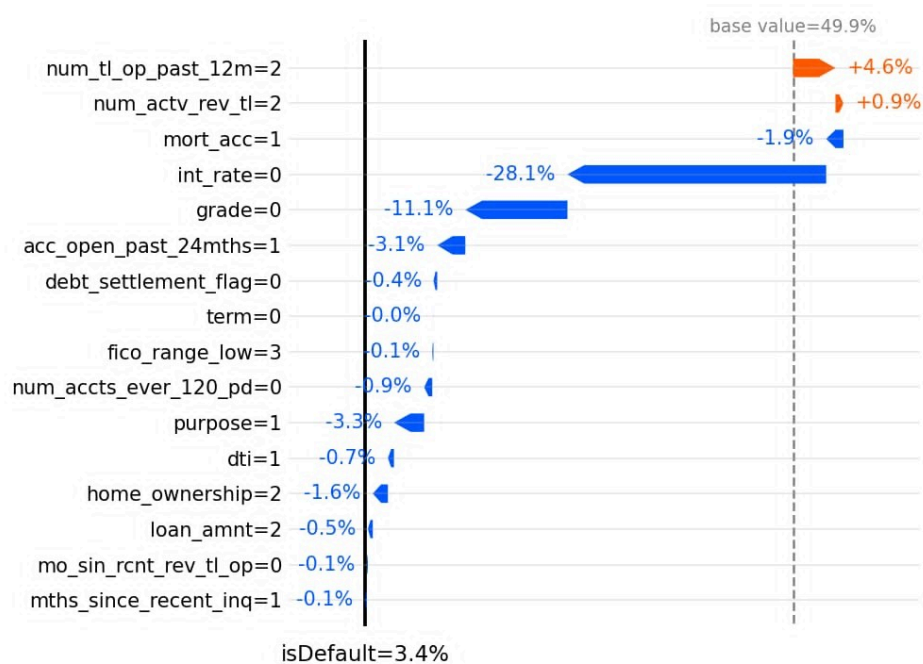
(c) SHAP

Hình 4.45: Comparison of three local explanation methods for a randomly selected customer profile — German Credit

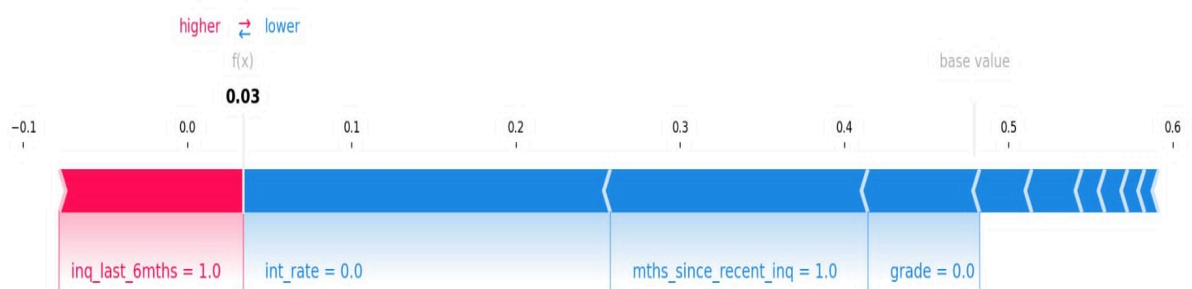
Predicted probability: Is default = 0.0338, No default = 0.9662



(a) LIME



(b) Bayesian Network



(c) SHAP

Hình 4.46: Comparison of three local explanation methods for a randomly selected customer profile — Lending Club

Chapter 5

Conclusion

5.1 Results of the Thesis

The experimental results on the three datasets confirm that the BN_Pro model — combining structure learning via BIC-Tabu Search with expert-knowledge constraints, parameter learning via MLE, exact inference via VE, and a full suite of interpretation tools (sensitivity, forward/backward inference, Individual Analysis, what-if) — not only competes in predictive accuracy with modern machine learning models on small- to medium-scale datasets (with an absolute advantage on Australian Credit), but also provides explicit causal interpretability that closed-form models such as XGBoost or Stacking Ensemble do not inherently offer.

The predictive performance of BN varies considerably with data scale and complexity: ROC-AUC rises from 0.6910 (Lending Club, 27 variables) to 0.7667 (German Credit, 15 variables) and peaks at 0.9220 (Australian Credit, 8 variables), trending inversely with the number of input features.

Compared with modern machine learning models, BN is not always the optimal choice in terms of pure accuracy: on large, high-dimensional data, tree-ensemble models (XGBoost, Random Forest) are superior; on small, low-dimensional data, BN can outperform all benchmark models, including the Stacking Ensemble.

DCA analysis confirms that BN yields a positive net benefit and outperforms the boundary strategies (Treat All / Treat None) within a reasonable probability threshold range across all three datasets, confirming its practical application value even though its absolute AUC may be lower than benchmark models in some cases.

The outstanding and distinguishing advantage of BN over the benchmark models is its explicit interpretability through the causal graph structure and posterior probability sensitivity analysis — a capability that closed-form models (Random Forest, XGBoost, neural networks) do not inherently possess, requiring the addition of separate post-hoc interpretation tools (such as SHAP, LIME).

The structure optimisation process using Tabu Search with 50 independent runs shows stability that is inversely related to the number of variables: the smaller the dataset, the narrower the structure search space, and the more consistent the convergence of results across runs.

5.2 Limitations and Future Directions

The greatest limitation observed for the BN model on the large-scale dataset (Lending Club) is the performance gap between train and test (0.7944 versus 0.6910), reflecting the risk of structural overfitting as the number of variables and graph edges increases. Discretising continuous variables using K-means is also a potential source of information loss, particularly affecting financial variables with diverse continuous distributions such as `annual_inc`, `dti`, and `revol_util`.

The results on Lending Club also reveal limitations in the scalability of exact inference as the number of features and DAG structure complexity increase, as well as the phenomenon of sparse CPTs leading to extreme probabilities when forward inference is based only on narrow evidence. This provides an empirical basis for proposing directions for improvement: imposing a tighter limit on `max_indegree`, applying approximate inference instead of variable elimination on large-scale datasets, or exploiting BN as an interpretive component within the architecture of ensemble models rather than as a fully independent predictive model.

Future development directions include: (i) applying stronger regularisation techniques in the structure-learning step (for example, reducing `max_indegree` or imposing stricter `expert_knowledge/forbidden_edges` constraints); (ii) experimenting with adaptive binning methods instead of fixed K-means discretisation; (iii) extending the comparison to continuous Bayesian network variants (Gaussian Bayesian Network, Hybrid BN) to minimise information loss due to discretisation, particularly suited to large-scale datasets such as Lending Club.

5.3 Conclusion

This study arose from a practical question of high urgency in the modern banking and finance sector: can a credit risk prediction model be built that achieves both competitive classification performance and sufficiently deep causal explanation to serve practical credit appraisal and risk management, rather than having to choose between the two? The answer this study provides, verified by empirical evidence across three independent datasets, is yes — but conditionally. That condition is the quality of the entire data-preparation pipeline before feeding data into the generative model: appropriate feature filtering, control of information leakage, and a discretisation strategy compatible with the distribution of each variable.

Evidence for this argument is drawn directly from the experimental results. On the Australian Credit dataset, where the data-preparation pipeline achieves the most uniformly high quality with only 8 carefully filtered and consistently discretised features, BN achieves an AUC-ROC of 0.9220 on the test set, ranking 1st out of 6, surpassing the Stacking Ensemble (0.9035), Logistic Regression (0.9052), Random Forest (0.9046), XGBoost (0.8919) and KNN (0.8415). More notably, BN achieves this with the smallest overfitting gap in the group: the training AUC is 0.9278, i.e. a train–test gap of only

0.0058, compared with 0.0852 for XGBoost (train 0.9771 versus test 0.8919), as much as 0.1555 for KNN (train 0.9970 versus test 0.8415), and even 0.0570 for the Stacking Ensemble (train 0.9605 versus test 0.9035). This shows that when supplied with input data compatible with its discrete assumptions, BN is not inferior to ensemble models in terms of AUC, and is considerably superior in terms of generalisation stability — a key characteristic in the context of practical credit deployment, where data distributions are constantly shifting over time.

The core contribution extending beyond the AUC figure is the four-dimensional In_Pro inference framework, which delivers a kind of value that no discriminative model can provide from a single unified model: quantifying the sensitivity of each variable within the Markov Blanket, answering the reverse diagnostic question about the characteristics of already-defaulted customers, explaining default probability at the level of each specific profile, and quantifying the intervention effect of changing each risk factor. Three independent analyses — sensitivity, likelihood ratio, and individual contribution — converge consistently on the same set of risk-dominant variables, confirming that the learned graph structure correctly reflects genuinely domain-meaningful dependency relationships in each credit context.

The genuine limitation of the generative model is not in classification performance when the data and pipeline are appropriate, but in scalability to a large feature space and unstructured data. On Lending Club, with 27 features, BN achieves $AUC = 0.6910$. This is not a research failure but a reasonable outcome of a generative model that is forced to discretise the entire continuous feature space and learn a dependency structure from data with many local non-linear relationships — precisely the condition under which discriminative decision-tree-based models are inherently more effective. This trade-off does not negate the value of BN, but rather clarifies its appropriate application domain: BN delivers maximum advantage in the context of a moderate feature space, a relatively clear dependency structure, and where the requirement for causal explanation is the top priority — precisely the context of most practical credit appraisal problems at small- and medium-sized credit institutions.

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